From the Quadratic Sequence to the Conical Helix

An Inductive Derivation:
Algebraic Geometry of the Congruence Classes $1, 5 \pmod 6$

Dogan Balban

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Abstract

This work develops a detailed analytical and geometric framework for the quadratic integer sequence

$$k(n) = \tfrac{1}{6}(2n^2 + 3n + 1),$$

which takes integer values precisely for the congruence classes $n \equiv 1, 5 \pmod 6$.

A fine-grained analysis of the first differences reveals an alternating pattern of *minor* and *major steps* whose pairwise sums—as proved in this work—form the minimal even block size for which the difference sequence yields an arithmetic progression of circle circumferences. This minimality, combined with the circumference interpretation, determines a constant radial increment $\Delta R = 12/\pi$ and leads, with no remaining free parameter once one revolution is assigned to each pair, to an explicit closed parametrisation of an Archimedean spiral with slope $a = 6/\pi^2 = 1/\zeta(2)$, whose uniform radial growth is obtained directly from the discrete arithmetic data of the sequence.

Extending the planar spiral by a linear height function produces a three-dimensional conical helix on a right circular cone with half-angle $\alpha = \arctan(2/\pi) \approx 32.48°$. The algebraic core of the entire construction is the fundamental identity $(4n+3)^2 = 48\,k(n)+1$, which defines a universal quantity $Q(k) = \sqrt{48k+1}$ from which radius, azimuth, and height of the helix can be expressed as exact closed functions of a single parameter.

As a corollary of the congruence-class structure, all primes $p > 3$—which by a classical result of number theory satisfy $p \equiv 1, 5 \pmod 6$—appear as a distinguished subset of the integer-indexed points on the helix. The helix furthermore contains all composite numbers of these congruence classes; the primes form a infinite subset of relative density zero therein — one that the geometry does not separate from the composites.

This investigation demonstrates how a simple quadratic form generates a coherent geometric representation encompassing Archimedean spirals, conical helices, uniform discretisation, and closed analytical expressions. It illustrates how discrete modular patterns can be embedded into continuous geometric structures, thereby opening a unified perspective on the interplay between integer sequences, modular arithmetic, and parametric geometry. The contribution lies not in new number theory but in a geometric synthesis: known arithmetic building blocks are connected along a single derivation path.

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Keywords: Quadratic integer sequence, Archimedean spiral, conical helix, congruence classes, prime numbers, parametric geometry, arc-length parametrisation, Basel problem $\zeta(2)$.

Preface

This work does not begin with a theory to be tested, nor with a question posed by a textbook. It begins with a table.

Specifically: with the values

$$1, \quad 11, \quad 20, \quad 46, \quad 63, \quad 105, \quad 130, \quad 188, \quad \ldots$$

These numbers appeared in a computation. They did not seem to form any known sequence—at least not one that I was aware of at the time.

What drew my attention was the pattern. The differences between consecutive values—$10, 9, 26, 17, 42, 25, \ldots$—exhibited a regularity that could not be ignored. Two subsequences alternate, each arithmetic in its own right—and their pairwise sums $19, 43, 67, \ldots$ again form an arithmetic progression. The question of *why* this is so would not let me go.

From this question came another, and then yet another. Each answer opened the view to something I had not anticipated: a spiral, a cone, and finally a space curve on which—as a corollary of the algebraic construction—all prime numbers greater than 3 appear as discrete points.

This work follows the order in which these discoveries were made, not the order of a retrospective deductive presentation. This is a deliberate choice. In mathematics, one often distinguishes between the *path of discovery* and the *path of exposition*—the former is inductive and tentative, the latter deductive and orderly. Textbooks and dissertations almost always choose the second path: they begin with axioms and definitions and proceed to theorems as though someone already knew the answer before posing the question.

This work begins with observation, develops conjectures from patterns, and only then seeks the proof. At no point is the reader confronted with a result whose origin remains unclear. Every concept appears when it is needed—not before.

Formally, the sequence under investigation is the arithmetic mean of the first n perfect squares:

$$k(n) = \frac{1}{n}\sum_{j=1}^{n} j^2 = \frac{2n^2 + 3n + 1}{6}.$$

That this formula carries this name and has this form was not known at the outset. It was reconstructed from the table values—through differencing, pattern recognition, and elementary algebra. The result of this reconstruction is none other than the classical result found in every textbook on analysis. The path to it, however, was a different one.

What carries this work beyond the mere derivation is the question: *What does this sequence mean geometrically?* The answer to this question—an Archimedean spiral, a cone, a helix, a universal quantity $Q(k) = \sqrt{48k + 1}$ that unites all geometric coordinates in a single expression—could not have been foreseen. It emerged step by step, as a consequence of consistently pressing forward.

Beyond its mathematical content, this work is also a testament to what curiosity is capable of. Anyone willing to look closely can follow it—just as it was created: step by step, number by number.

Dogan Balban
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Acknowledgements

This work arose from many hours of intensive engagement with a subject that revealed unexpected new dimensions with each insight. What began as a manageable analysis turned into a journey whose length and depth could not have been foreseen.

My deepest gratitude goes to my family. They bore with me patiently when I was lost in thought, and were understanding when I was less present than I would have wished.

I also thank those who challenged me with critical questions and thereby compelled me to sharpen my thinking—some of those remarks led to the most productive hours of this work.

Part I

The Parabola — Starting Point

Chapter 1

Introduction

Quadratic integer sequences play a central role in number theory, particularly in the study of congruence properties, polygonal numbers, and prime number distributions. The present work considers the special quadratic sequence

$$k(n) = \frac{1}{6}(2n^2 + 3n + 1), \tag{1.1}$$

which possesses a Diophantine property: it takes integer values if and only if n lies in the congruence classes

$$n \equiv 1, 5 \pmod 6. \tag{1.2}$$

These two classes are of fundamental number-theoretic significance, as they contain all prime numbers greater than 3 (Theorem 2.4 in Chapter 2).

Motivation and Objectives

The geometric visualisation of number-theoretic structures has a long tradition and has led to significant insights. Well-known examples include:

- The *Ulam spiral* [SUW64]
- The *Sacks spiral* [Sac03]
- Quadratic forms in prime number theory

While the Ulam and Sacks spirals are primarily empirically motivated — patterns observed first, explanations sought afterwards — the theory of quadratic forms represents the complementary algebraic tradition; indeed, the diagonal patterns of the former are governed by quadratic polynomials. The present work pursues this second direction: starting from the quadratic sequence (1.1), a geometric structure is *derived*, not merely visualised. The contribution is accordingly a geometric synthesis of known arithmetic building blocks along a single derivation path, rather than new number theory.

Main Results

The central results of this work are:

1. Alternating minor and major steps with closed-form formulae
2. Circle circumferences with constant radius difference $\Delta R = \frac{12}{\pi}$
3. Archimedean spiral $r(\theta) = \frac{6}{\pi^2}\theta$
4. Parametrisation via p: $r(p) = \frac{6}{\pi}\sqrt{p/3}$, $\theta(p) = \pi\sqrt{p/3}$
5. Exact arc-length integral
6. Conical helix as a 3D embedding
7. All primes $p > 3$ appear as a distinguished discrete subset of the helix (corollary of the congruence class structure)

Structure of the Work

The work is organised into five parts, mirroring the derivation path. **Part I**, *The Parabola* (Chapters 2–4), develops the arithmetic foundations: congruence condition, the closed-form step formulae, and Euclidean distances. **Part II**, *The Archimedean Spiral* (Chapters 5–10), derives the spiral from the minimality of the pairing and treats parametrisation, arc length, and the Taylor approximation. **Part III**, *The Cone* (Chapters 11 and 12), constructs the enveloping cone and its ray geometry. **Part IV**, *The Conical Helix* (Chapters 13–15), assembles the helix, derives the prime number corollary, and verifies the arc lengths. **Part V**, *The Direct Solution* (Chapter 16 onwards), presents the exact Q-parametrisation, the direct parabola–helix construction (Chapter 17), the summary (Chapter 18), and the formula collection.

Notation

- $k(n)$: quadratic mean-value sequence
- δ_k: difference sequence of the integer values
- Δ_{Major}, Δ_{Minor}: major and minor step sizes
- a_k: closed-form generator of (δ_k)
- $Q(k) = \sqrt{48k+1}$: universal quantity
- $a = 6/\pi^2$: spiral slope; $c_1 = 3/\pi$: height rate
- θ: angular parameter
- p: function-value parameter

- $r(\theta)$: radius function
- $\gamma(\theta)$: helix parametrisation

Chapter 2

Arithmetic Foundations

In this chapter we investigate the number-theoretic properties of the sequence (1.1) and derive the structure of its differences.

2.1 Derivation via Means of Sums of Squares

The sequence $k(n)$ arises naturally from the classical summation formula for square numbers. For the sum of the first n squares one has

$$S_n = \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}. \tag{2.1}$$

The arithmetic mean of these squares is

$$M_n = \frac{S_n}{n} = \frac{(n+1)(2n+1)}{6} = \frac{2n^2+3n+1}{6} = k(n). \tag{2.2}$$

Remark 2.1. The sequence $k(n)$ thus gives the *average value* of the first n square numbers. It is the parabolic interpolation form of the arithmetic mean. This interpretation is important for understanding the later geometric construction, as it establishes the connection between discrete summation and continuous geometry.

The following table illustrates the structure of the sums of squares and their means for the first fifteen values:

n	**1**	2	3	4	**5**	6	**7**	8	9	10	**11**	12	**13**	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
$\sum_{k=1}^{n} k^2$	1	5	14	30	55	91	140	204	285	385	506	650	819	1015	1240
M_n	**1**	2.5	$4.\overline{6}$	7.5	**11**	$15.1\overline{6}$	**20**	25.5	$31.\overline{6}$	38.5	**46**	$54.1\overline{6}$	**63**	72.5	$82.\overline{6}$

The bold values indicate the integer means, which occur precisely for $n \equiv 1, 5 \pmod 6$.

Remark 2.2 (Connection to the OEIS: A164576). The sequence of integer means

$$1,\ 11,\ 20,\ 46,\ 63,\ 105,\ 130,\ 188,\ \ldots$$

is listed in the OEIS under A164576 [SO24] as *"Integer averages of the set of the first positive squares up to some n^2"* — precisely the sequence $k(n)$ whose discovery stands at the beginning of this work.

The OEIS formula (Colin Barker, 2015) reads:

$$a(j) = \frac{1}{4}\big(12j^2 - 6j + (-1)^j(4j-1) + 1\big), \qquad j = 1, 2, 3, \ldots$$

It uses a *running* index j that enumerates the integer positions $n \equiv 1, 5 \pmod 6$ consecutively. The factor $(-1)^j$ is a *quasi-polynomial of period* 2: it encodes both congruence classes in a single expression — for odd j (i.e. $n \equiv 1 \pmod 6$) and even j (i.e. $n \equiv 5 \pmod 6$) it alternately takes the values $+1$ and -1.

The connection to $k(n)$ is obtained by substituting $A007310(j)$ — the sequence of positive integers coprime to 6 [SO24]:

$$a(j) = k\big(A007310(j)\big),$$

where the two branches yield precisely:

$$k(A007310(j)) = \begin{cases} \dfrac{(2j-1)(3j-1)}{2} & j \text{ odd } (n \equiv 1 \pmod 6), \\ \dfrac{j(6j-1)}{2} & j \text{ even } (n \equiv 5 \pmod 6). \end{cases}$$

The present formula $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ is simpler and more direct: it works with the actual n-values rather than a running index and therefore requires no $(-1)^j$ correction term. The sequence itself is thus not new; what is new in the present work is its geometric interpretation.

2.2 Congruence Condition for Integer Values

Theorem 2.3. *The sequence $k(n)$ takes integer values if and only if*

$$n \equiv 1, 5 \pmod{6}. \tag{2.3}$$

Proof. We examine $2n^2 + 3n + 1 \pmod{6}$ for all residue classes:

$n \pmod 6$	$n^2 \pmod 6$	$2n^2 \pmod 6$	$3n \pmod 6$	Sum $\pmod 6$
0	0	0	0	1
1	1	2	3	0
2	4	2	0	3
3	3	0	3	4
4	4	2	0	3
5	1	2	3	0

The final column includes the constant term $+1$ of the numerator. The sum $2n^2 + 3n + 1$ is divisible by 6 precisely in the rows $n \equiv 1$ and $n \equiv 5$, which proves the claim. □

Theorem 2.4 (Prime Number Property)**.** *All primes $p > 3$ lie in the classes $n \equiv 1, 5 \pmod{6}$.*

Proof. Every prime $p > 3$ is coprime to 6, since it is divisible neither by 2 nor by 3. The residue classes coprime to 6 are precisely 1 and 5 modulo 6 [IR90, Ch. 1]. □

2.3 Value Table

Table 2.1: Integer points of the curve

n	1	5	7	11	13	17	19	23	25	29	31	35	37	41	43
$k(n)$	1	11	20	46	63	105	130	188	221	295	336	426	475	581	638

2.4 Differences: Minor and Major Steps

Table 2.2: Differences of the integer values

Δk	10	9	26	17	42	25	58	33	74	41	90	49	106	57

The indices of the integer values follow a periodic pattern:

$$\ldots, \quad 6m-5, \quad 6m-1, \quad 6m+1, \quad 6m+5, \quad 6m+7, \quad \ldots$$

Here $6m - 5 \equiv 1 \pmod{6}$ and $6m - 1 \equiv 5 \pmod{6}$. The period m $(m = 1, 2, 3, \ldots)$ consists of:

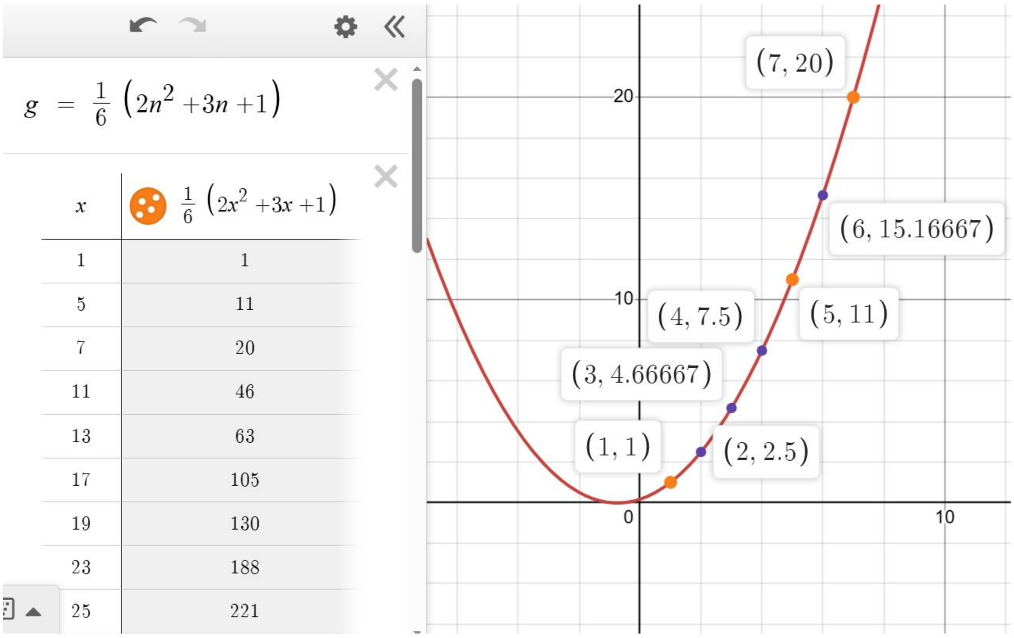


Figure 2.1: The quadratic curve $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ as a continuous parabola with its numerical value table. The orange dots mark the integer values at $n \equiv 1, 5 \pmod 6$; the purple dots correspond to the non-integer intermediate values. The growth of the sequence is already clearly visible from the first values (1, 1), (5, 11), (7, 20).

- a **major step** from $n = 6m - 5$ to $n = 6m - 1$ (index difference $\Delta n = 4$),
- a **minor step** from $n = 6m - 1$ to $n = 6m + 1$ (index difference $\Delta n = 2$).

Theorem 2.5 (Explicit formulae for major and minor steps)**.** *For the m-th period ($m \geq 1$):*

$$\Delta_{\text{Major}}(m) = k(6m-1) - k(6m-5) = 16m - 6, \tag{2.4}$$

$$\Delta_{\text{Minor}}(m) = k(6m+1) - k(6m-1) = 8m + 1. \tag{2.5}$$

Proof. **Major step:** We compute $k(6m-1) - k(6m-5)$ directly. With $k(n) = \frac{2n^2+3n+1}{6}$:

$$\begin{aligned} k(6m-1) - k(6m-5) &= \frac{2(6m-1)^2 + 3(6m-1) + 1}{6} - \frac{2(6m-5)^2 + 3(6m-5) + 1}{6} \\ &= \frac{1}{6}\Big[2\big((6m-1)^2 - (6m-5)^2\big) + 3\big((6m-1) - (6m-5)\big)\Big]. \end{aligned}$$

We use the difference-of-squares identity $a^2 - b^2 = (a+b)(a-b)$ with $a = 6m - 1$ and $b = 6m - 5$:

$$\begin{aligned} (6m-1)^2 - (6m-5)^2 &= \big[(6m-1) + (6m-5)\big]\big[(6m-1) - (6m-5)\big] \\ &= (12m - 6) \cdot 4 = 48m - 24. \end{aligned}$$

Moreover: $(6m-1)-(6m-5)=4$. Substituting:

$$\begin{aligned}
k(6m-1)-k(6m-5) &= \frac{1}{6}\big[2(48m-24)+3\cdot 4\big] \\
&= \frac{1}{6}\big[96m-48+12\big] \\
&= \frac{96m-36}{6} = 16m-6. \checkmark
\end{aligned}$$

Minor step: We compute $k(6m+1)-k(6m-1)$:

$$k(6m+1)-k(6m-1) = \frac{1}{6}\big[2\big((6m+1)^2-(6m-1)^2\big)+3\big((6m+1)-(6m-1)\big)\big].$$

With $a=6m+1$, $b=6m-1$:

$$(6m+1)^2-(6m-1)^2 = (12m)(2) = 24m.$$

Moreover: $(6m+1)-(6m-1)=2$. Substituting:

$$k(6m+1)-k(6m-1) = \frac{1}{6}\big[2\cdot 24m+3\cdot 2\big] = \frac{48m+6}{6} = 8m+1. \checkmark$$

□

Definition 2.6 (Minor and Major Steps)**.** The step sizes of the m-th period are denoted as follows:

$$\Delta_{\text{Minor},m} = 8m+1 \quad (\text{step } \Delta n = 2, \text{ from } n\equiv 5 \text{ to } n \equiv 1), \tag{2.6}$$

$$\Delta_{\text{Major},m} = 16m-6 \quad (\text{step } \Delta n = 4, \text{ from } n\equiv 1 \text{ to } n \equiv 5). \tag{2.7}$$

2.5 Generating Function of the Sequence $k(n)$

We consider the sequence of quadratic means

$$k(n) := \frac{1}{n}\sum_{j=1}^{n} j^2 = \frac{2n^2+3n+1}{6}, \qquad n \geq 1.$$

For this sequence we define the ordinary generating function

$$K(x) := \sum_{n=1}^{\infty} k(n)\, x^n.$$

Theorem 2.7. *The generating function of $k(n)$ is a rational function of the form*

$$K(x) = \frac{1}{6}\left(2\,\frac{x(1+x)}{(1-x)^3} + 3\,\frac{x}{(1-x)^2} + \frac{x}{1-x}\right),$$

which simplifies to the closed form

$$K(x) = \frac{x\,(x^2-3x+6)}{6\,(1-x)^3}. \tag{2.8}$$

Proof. From the explicit representation

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

it follows that

$$K(x) = \sum_{n=1}^{\infty} k(n)\, x^n = \frac{1}{6} \sum_{n=1}^{\infty} (2n^2 + 3n + 1)\, x^n.$$

We decompose the sum into three standard series:

$$K(x) = \frac{1}{6} \left(2 \sum_{n=1}^{\infty} n^2 x^n + 3 \sum_{n=1}^{\infty} n x^n + \sum_{n=1}^{\infty} x^n \right).$$

The closed forms for these series are well known:

$$\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}, \qquad \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}, \qquad \sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3},$$

where all series converge for $|x| < 1$. Substituting yields

$$K(x) = \frac{1}{6} \left(2\, \frac{x(1+x)}{(1-x)^3} + 3\, \frac{x}{(1-x)^2} + \frac{x}{1-x} \right).$$

This establishes the stated representation. □

Remark 2.8. The rational structure of $K(x)$ reflects the fact that $k(n)$ is described by a polynomial of degree two in n. From the representation of $K(x)$ one can, for instance, obtain a linear recurrence for $k(n)$ by considering the common denominator $(1-x)^3$ and comparing coefficients of x^n.

At the same time, $K(x)$ arises as a linear combination of the generating functions of n, n^2, and the constant sequence 1. Thus $k(n)$ fits into the context of simple linear and quadratic sequences. The generating function provides an alternative, analytic perspective on the same structure that is already visible discretely in the congruence-theoretic considerations (Theorem 2.4).

2.5.1 Linear Recurrence from the Generating Function

From the rational representation of $K(x)$ a linear recurrence for the sequence $k(n)$ can be derived immediately. To this end we write

$$K(x) = \sum_{n=1}^{\infty} k(n)\, x^n$$

and consider the factor $(1-x)^3$:

$$(1-x)^3 K(x) = (1 - 3x + 3x^2 - x^3) \sum_{n=1}^{\infty} k(n)\, x^n.$$

Expanding the product and comparing the coefficients of x^n for sufficiently large n yields

$$k(n) - 3k(n-1) + 3k(n-2) - k(n-3) = 0 \qquad \text{for } n \geq 4.$$

Thus the linear recurrence

$$k(n) = 3k(n-1) - 3k(n-2) + k(n-3), \qquad n \geq 4,$$

holds, with initial values

$$k(1) = 1, \quad k(2) = \frac{5}{2}, \quad k(3) = \frac{14}{3}.$$

Remark 2.9. The recurrence reflects the fact that $k(n)$ is given by a polynomial of degree two in n: a sequence described by a polynomial of degree d always satisfies a linear recurrence of constant order $d+1$ with constant coefficients. In the present case $d = 2$, and the order of the recurrence is 3.

Together with the explicit representation

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

and the congruence-theoretic condition for integer values (Theorem 2.3), the generating function thus provides an alternative, analytic perspective on the same structure. It connects the discrete summation formula, the quadratic form of $k(n)$, and the arithmetic patterns of the sequence within a common framework.

Chapter 3

Euclidean Distances Between Function Values

This chapter investigates the geometric distances between consecutive integer points on the curve $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$.

3.1 Motivation

The function takes integer values for all $n \equiv 1, 5 \pmod 6$. This property allows us to consider discrete points $(n, k(n))$ on the curve. A natural question is: **What are the Euclidean distances between consecutive integer points?**

3.2 Point Distances in the Plane

3.2.1 Definition

For two integer arguments n_1, n_2 with $n_1, n_2 \equiv 1, 5 \pmod 6$, the points $(n_1, k(n_1))$ and $(n_2, k(n_2))$ lie on the curve.

Definition 3.1 (Euclidean Distance Between Curve Points)**.** The Euclidean distance between two points on the curve is:

$$d(n_1, n_2) = \sqrt{(n_2 - n_1)^2 + (k(n_2) - k(n_1))^2} \tag{3.1}$$

3.2.2 Numerical Examples

Example 3.2 (Distances between integer points)**.** The following table shows the Euclidean distances for the first integer-valued pairs:

Table 3.1: Euclidean distances between consecutive points of the sequence $n \equiv 1, 5 \pmod 6$

(n_1, n_2)	Δn	$k(n_1)$	$k(n_2)$	Distance d	Exact value
$(1, 5)$	4	1	11	10.7703296143	$\sqrt{116}$
$(5, 7)$	2	11	20	9.21954445729	$\sqrt{85}$
$(7, 11)$	4	20	46	26.3058928759	$\sqrt{692}$
$(11, 13)$	2	46	63	17.1172427686	$\sqrt{293}$
$(13, 17)$	4	63	105	42.1900462195	$\sqrt{1780}$
$(17, 19)$	2	105	130	25.0798724080	$\sqrt{629}$
$(19, 23)$	4	130	188	58.1377674150	$\sqrt{3380}$
$(23, 25)$	2	188	221	33.0605505096	$\sqrt{1093}$
$(25, 29)$	4	221	295	74.1080292546	$\sqrt{5492}$
$(29, 31)$	2	295	336	41.0487515035	$\sqrt{1685}$

3.3 Closed-Form Formulae

3.3.1 Exact Distance Formula

Proposition 3.3 (Closed-form formula for the distance)**.** *With* $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ *and* $\Delta n = n_2 - n_1$*:*

$$k(n_2) - k(n_1) = \frac{\Delta n}{6}(4n_1 + 2\Delta n + 3) \tag{3.2}$$

Hence:

$$d(n_1, n_1 + \Delta n) = \Delta n \cdot \sqrt{1 + \frac{1}{36}(4n_1 + 2\Delta n + 3)^2} \tag{3.3}$$

Proof. Direct computation:

$$\begin{aligned}
k(n_2) - k(n_1) &= \frac{1}{6}[2(n_2^2 - n_1^2) + 3(n_2 - n_1)] && (3.4)\\
&= \frac{1}{6}[2(n_2 - n_1)(n_2 + n_1) + 3(n_2 - n_1)] && (3.5)\\
&= \frac{\Delta n}{6}[2(2n_1 + \Delta n) + 3] && (3.6)\\
&= \frac{\Delta n}{6}(4n_1 + 2\Delta n + 3) && (3.7)
\end{aligned}$$

Substitution into the distance formula yields the claim. □

3.3.2 Special Cases

Case $\Delta n = 2$

$$d(n, n+2) = 2\sqrt{1 + \frac{1}{36}(4n+7)^2} \tag{3.8}$$

n_1	Distance d
5	9.2195444573
11	17.1172427686
17	25.0798724080
23	33.0605505096

Example 3.4 (Numerical values for $\Delta n = 2$)**.**

Case $\Delta n = 4$

$$d(n, n+4) = 4\sqrt{1 + \frac{1}{36}(4n+11)^2} \tag{3.9}$$

n_1	Distance d
1	10.7703296143
7	26.3058928759
13	42.1900462195
19	58.1377674150

Example 3.5 (Numerical values for $\Delta n = 4$)**.**

Case $\Delta n = 6$

$$d(n, n+6) = 6\sqrt{1 + \frac{1}{36}(4n+15)^2} \tag{3.10}$$

Remark 3.6. The case $\Delta n = 6$ corresponds to a full period in the congruence structure modulo 6. Geometrically, this means a jump across both congruence classes.

3.4 Comparison: Distance vs. Arc Length

The Euclidean distance d between two points is always less than or equal to the arc length L along the curve, since the straight line represents the shortest possible path. This is a fundamental metric property.

Proposition 3.7 (Distance as a lower bound)**.** *For $n_1 < n_2$ it always holds that:*

$$d(n_1, n_2) \leq L[n_1, n_2] \tag{3.11}$$

where $L[n_1, n_2] = \int_{n_1}^{n_2} \sqrt{1 + [k'(x)]^2}\, dx$ is the arc length.

Table 3.2: Euclidean distance compared with arc length

Interval	Distance d	Arc length L	Ratio d/L	$L - d$
$[1, 5]$	10.7703296143	10.8394091489	0.9936270018	$6.908 \cdot 10^{-2}$
$[5, 7]$	9.2195444573	9.2210749306	0.9998340244	$1.530 \cdot 10^{-3}$
$[7, 11]$	26.3058928759	26.3101623759	0.9998377243	$4.270 \cdot 10^{-3}$
$[11, 13]$	17.1172427686	17.1174799293	0.9999861451	$2.372 \cdot 10^{-4}$
$[13, 17]$	42.1900462195	42.1910659377	0.9999758309	$1.020 \cdot 10^{-3}$

Example 3.8 (Comparison: distance vs. arc length)**. Observation:** The ratio d/L is in all cases ≤ 1, as guaranteed by the proposition. For short intervals ($\Delta n = 2$), distance and

arc length are nearly identical, since the curve is locally almost rectilinear. The deviation is somewhat larger for small n (more strongly curved region) and approaches the value 1 for large n.

3.5 Distance Sequence

Consider the sequence of all consecutive distances:

Definition 3.9 (Distance Sequence)**.** Let $\{n_k\}$ be the ascending sequence of all integer arguments of $k(n)$ (i.e. $n_k \equiv 1, 5 \pmod 6$, cf. Chapter 2). Then we define:

$$\{d_k\} = \{d(n_k, n_{k+1})\}_{k=1}^{\infty} \tag{3.12}$$

Example 3.10 (The first terms)**.** With $n_1 = 1, n_2 = 5, n_3 = 7, n_4 = 11, \ldots$ we obtain:

$$\{d_k\} = \{10.7703,\ 9.2195,\ 26.3059,\ 17.1172,\ 42.1900,\ 25.0799,\ \ldots\}$$

Remark 3.11 (Irregularity)*.* The sequence $\{d_k\}$ is not monotone, since the spacings $\Delta n_k = n_{k+1} - n_k$ alternate between 2 and 4. However, the sequence exhibits a clear growth pattern: both the local maxima and the local minima increase monotonically with k.

3.6 Summary

Table 3.3: Distance properties by Δn

Δn	Type	Distance formula
2	Adjacent values (minor)	$2\sqrt{1 + \frac{(4n+7)^2}{36}}$
4	Congruence class change (major)	$4\sqrt{1 + \frac{(4n+11)^2}{36}}$
6	Full period	$6\sqrt{1 + \frac{(4n+15)^2}{36}}$

Remark 3.12 (Significance)*.* The distance analysis shows that the geometric spacings between integer function values exhibit a clear pattern directly linked to the congruence structure modulo 6. For large n, all distances grow proportionally to n, with the proportionality factor depending on Δn.

Chapter 4

Euclidean Distance Sequences

The Euclidean distances between consecutive integer points $(n_i,\, k(n_i))$ on the curve $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ can be completely described by closed-form formulae. Since Δn alternates between 2 (twin zone) and 4 (gap zone), two subsequences with explicit expressions in the period index m arise.

4.0.1 Twin-Zone Distance Sequence ($\Delta n = 2$)

Theorem 4.1 (Closed-form formula for twin distances)**.** *The Euclidean distance between the m-th twin pair $(n_1, n_2) = (6m-1,\ 6m+1)$ is:*

$$\boxed{d_{\mathrm{Twin}}(m) = \sqrt{4 + (8m+1)^2} = \sqrt{64m^2 + 16m + 5}.} \tag{4.1}$$

Proof. We have $\Delta n = 2$ and $\Delta k = k(6m+1) - k(6m-1) = 8m + 1$ (minor step from Theorem 2.5). Thus:

$$d^2 = (\Delta n)^2 + (\Delta k)^2 = 4 + (8m+1)^2 = 64m^2 + 16m + 5.$$

□

Table 4.1: Twin-zone distances $d_{\mathrm{Twin}}(m) = \sqrt{4 + (8m+1)^2}$.

m	Pair (n_1, n_2)	$\Delta k = 8m+1$	d^2	$d_{\mathrm{Twin}}(m)$
1	$(5, 7)$	9	85	$\sqrt{85}$
2	$(11, 13)$	17	293	$\sqrt{293}$
3	$(17, 19)$	25	629	$\sqrt{629}$
4	$(23, 25)$	33	1093	$\sqrt{1093}$
5	$(29, 31)$	41	1685	$\sqrt{1685}$
6	$(35, 37)$	49	2405	$\sqrt{2405}$
7	$(41, 43)$	57	3253	$\sqrt{3253}$

4.0.2 Gap-Zone Distance Sequence ($\Delta n = 4$)

Theorem 4.2 (Closed-form formula for gap distances)**.** *The Euclidean distance between the m-th gap pair* $(n_1, n_2) = (6m - 5,\ 6m - 1)$ *is:*

$$\boxed{d_{\text{Gap}}(m) = \sqrt{16 + (16m - 6)^2} = 2\sqrt{64m^2 - 48m + 13}.} \tag{4.2}$$

Proof. We have $\Delta n = 4$ and $\Delta k = k(6m - 1) - k(6m - 5) = 16m - 6$ (major step from Theorem 2.5). Thus:

$$d^2 = 16 + (16m - 6)^2 = 16 + 256m^2 - 192m + 36 = 4(64m^2 - 48m + 13).$$

□

Table 4.2: Gap-zone distances $d_{\text{Gap}}(m) = 2\sqrt{64m^2 - 48m + 13}$.

m	Pair (n_1, n_2)	$\Delta k = 16m - 6$	d^2	$d_{\text{Gap}}(m)$
1	$(1, 5)$	10	116	$\sqrt{116}$
2	$(7, 11)$	26	692	$\sqrt{692}$
3	$(13, 17)$	42	1780	$\sqrt{1780}$
4	$(19, 23)$	58	3380	$\sqrt{3380}$
5	$(25, 29)$	74	5492	$\sqrt{5492}$
6	$(31, 35)$	90	8116	$\sqrt{8116}$
7	$(37, 41)$	106	11252	$\sqrt{11252}$

4.0.3 Complete Distance Sequence in a Running Index

When both zones are merged in the natural order of the indices and labelled with the running index $x = 1, 2, 3, \ldots$, the following unified representation in x is obtained:

Theorem 4.3 (Distance sequence in overall index x)**.** *With the overall index* $x = 1, 2, 3, \ldots$ *(gap at odd, twin at even* x*):*

$$d^2(x) = \begin{cases} 64x^2 + 32x + 20 & x \text{ odd (gap zone)}, \\ 16x^2 + 8x + 5 & x \text{ even (twin zone)}. \end{cases} \tag{4.3}$$

Proof. Let $m = \lceil x/2 \rceil$. For odd $x = 2m - 1$:

$$64x^2 + 32x + 20 = 64(2m - 1)^2 + 32(2m - 1) + 20 = 256m^2 - 192m + 52 = d^2_{\text{Gap}}(m).\ \checkmark$$

For even $x = 2m$:

$$16x^2 + 8x + 5 = 16(2m)^2 + 8(2m) + 5 = 64m^2 + 16m + 5 = d^2_{\text{Twin}}(m).\ \checkmark$$

□

Table 4.3: Complete distance sequence, alternating gap and twin. The precursor value $\sqrt{5}$ (index $x = 0$) corresponds to the distance from the zero point $(-1, 0)$ to the first integer point $(1, 1)$, since $k(-1) = 0$.

x	Pair (n_1, n_2)	Type	d^2	d
0	$(-1,\ 1)$	Precursor	5	$\sqrt{5}$
1	$(1, 5)$	Gap	116	$\sqrt{116}$
2	$(5, 7)$	Twin	85	$\sqrt{85}$
3	$(7, 11)$	Gap	692	$\sqrt{692}$
4	$(11, 13)$	Twin	293	$\sqrt{293}$
5	$(13, 17)$	Gap	1780	$\sqrt{1780}$
6	$(17, 19)$	Twin	629	$\sqrt{629}$
7	$(19, 23)$	Gap	3380	$\sqrt{3380}$
8	$(23, 25)$	Twin	1093	$\sqrt{1093}$
9	$(25, 29)$	Gap	5492	$\sqrt{5492}$
10	$(29, 31)$	Twin	1685	$\sqrt{1685}$
11	$(31, 35)$	Gap	8116	$\sqrt{8116}$
12	$(35, 37)$	Twin	2405	$\sqrt{2405}$
13	$(37, 41)$	Gap	11252	$\sqrt{11252}$
14	$(41, 43)$	Twin	3253	$\sqrt{3253}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Remark 4.4 (Precursor point and zero). The curve $k(n)$ has a zero at $n = -1$:

$$k(-1) = \frac{2(-1)^2 + 3(-1) + 1}{6} = \frac{2 - 3 + 1}{6} = 0.$$

Since $-1 \equiv 5 \pmod 6$, this value satisfies the congruence condition $n \equiv 1, 5 \pmod 6$. However, it lies outside the domain of the sequence ($n \geq 1$) and therefore does not appear as a term of the sequence — not because of the residue class, but because of the restriction to positive indices.

As a geometric precursor point, $(-1, 0)$ nevertheless admits a direct interpretation: the distance to the first genuine sequence term $(1, 1)$ is

$$d = \sqrt{(1 - (-1))^2 + (1 - 0)^2} = \sqrt{5},$$

a value that appears as an initial value of the distance sequence before $k(1) = 1$.

Remark 4.5 (Growth rate). The gap distances grow asymptotically as $16m \sim 16 \cdot \frac{x}{2} = 8x$, the twin distances as $8m \sim 4x$. The ratio

$$\frac{d_{\text{Gap}}(m)}{d_{\text{Twin}}(m)} \xrightarrow{m \to \infty} 2,$$

which directly reflects the ratio of step sizes $\Delta n_{\text{Gap}} / \Delta n_{\text{Twin}} = 4/2 = 2$.

Part II

The Archimedean Spiral — From Pattern to Curve

Chapter 5

From Steps to the Archimedean Spiral

5.1 Geometric Motivation: From the Discrete Step Pattern to the Continuous Spiral

5.1.1 Starting Point and Guiding Question

In Chapter 2 it was shown that the quadratic sequence

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

takes integer values precisely for the index values $n \equiv 1, 5 \pmod 6$. The resulting discrete point set

$$\mathcal{P} = \big\{(n,\, k(n)) \mid n \equiv 1, 5 \pmod 6,\ n \geq 1\big\}$$

has a parabolic shape in Cartesian coordinates, since $k(n)$ is a polynomial of degree two in n. For the objective of this work — the embedding of this discrete structure into a geometrically regular, three-dimensional space — the Cartesian representation is, however, not suitable. It admits no natural periodisation, no angular structure, and it does not make the underlying modular regularity of the index n geometrically visible.

Guiding question of this chapter

Does there exist a partition of the difference sequence of the sequence $k(n)$ into equally-sized blocks such that the resulting block sums can be interpreted as circumferences of concentric circles and the resulting radius sequence defines an Archimedean spiral with a closed-form slope constant? And if so: which block sizes admit it?

The answer, proved below: the block sums form an arithmetic progression precisely for the even block sizes, and $\ell = 2$ is the minimal admissible choice; combined with the circumference interpretation this determines the spiral completely.

5.1.2 The Difference Sequence of the Integer Values

We denote by $(n_k)_{k \geq 1}$ the ascending sequence of all indices with $n_k \equiv 1, 5 \pmod 6$:

$$n_1 = 1, \quad n_2 = 5, \quad n_3 = 7, \quad n_4 = 11, \quad n_5 = 13, \quad n_6 = 17, \quad n_7 = 19, \quad n_8 = 23, \quad \ldots$$

The corresponding function values (integer values of the sequence) are:

$$p_k = k(n_k): \quad 1,\ 11,\ 20,\ 46,\ 63,\ 105,\ 130,\ 188,\ 221, \ldots$$

We define the *first difference sequence* of this value sequence:

$$\delta_k = p_{k+1} - p_k = k(n_{k+1}) - k(n_k), \qquad k \geq 1.$$

The first terms of this difference sequence are:

$$\delta_1 = 10, \quad \delta_2 = 9, \quad \delta_3 = 26, \quad \delta_4 = 17, \quad \delta_5 = 42, \quad \delta_6 = 25, \quad \delta_7 = 58, \quad \delta_8 = 33, \quad \ldots$$

Remark 5.1 (Alternating growth of the differences)*.* The difference sequence (δ_k) exhibits a clear alternating pattern:

- The terms at *odd* positions $(\delta_1, \delta_3, \delta_5, \ldots)$ take the values $10, 26, 42, 58, \ldots$ and form an arithmetic subsequence with common difference 16.
- The terms at *even* positions $(\delta_2, \delta_4, \delta_6, \ldots)$ take the values $9, 17, 25, 33, \ldots$ and form an arithmetic subsequence with common difference 8.

This pattern directly reflects the alternation of the index differences $n_{k+1} - n_k$, which alternate between $\Delta n = 4$ (for $n_k \equiv 1 \pmod 6$, *major steps*) and $\Delta n = 2$ (for $n_k \equiv 5 \pmod 6$, *minor steps*).

5.1.3 The Block-Formation Problem and Its Solution

We pose the following algebraic question: which partition of the difference sequence (δ_k) into consecutive, equally-sized blocks of length $\ell \in \{1, 2, 3, \ldots\}$ produces a sequence of block sums that forms an *arithmetic progression*?

Formally, for a block size $\ell \geq 1$ we define the sequence of block sums

$$S_m^{(\ell)} = \sum_{j=1}^{\ell} \delta_{\ell(m-1)+j}, \qquad m \geq 1,$$

and ask: for which values of ℓ is $(S_m^{(\ell)})_{m \geq 1}$ an arithmetic sequence?

Theorem 5.2 (Even block sizes and minimality of the pairing)**.** *The sequence of block sums $(S_m^{(\ell)})_{m\geq 1}$ is an arithmetic progression if and only if ℓ is even. For $\ell = 2m$ the common difference is*

$$S_{i+1}^{(\ell)} - S_i^{(\ell)} = 24m^2 = 6\ell^2.$$

In particular, $\ell = 2$ is the minimal admissible block size, with common difference 24.

Proof. By Theorem 2.5 the difference sequence is $\delta_{2t-1} = 16t - 6$ (major) and $\delta_{2t} = 8t + 1$ (minor).

Even $\ell = 2m$: shifting an index by ℓ preserves its parity and advances the period counter t by m, so

$$\delta_{j+\ell} - \delta_j = \begin{cases} 16m, & j \text{ odd}, \\ 8m, & j \text{ even}. \end{cases}$$

Each block of length $2m$ contains exactly m odd and m even indices; hence

$$S_{i+1}^{(\ell)} - S_i^{(\ell)} = \sum_{j \in \mathrm{Block}_i} \left(\delta_{j+\ell} - \delta_j\right) = m \cdot 16m + m \cdot 8m = 24m^2 = 6\ell^2,$$

independent of i: the block sums form an arithmetic progression.

Odd ℓ: now j and $j+\ell$ have opposite parity. Writing $\ell = 2m+1$, a direct computation from the two closed formulae gives

$$\delta_{j+\ell} - \delta_j = \begin{cases} -8t + 4\ell + 3, & j = 2t-1 \text{ odd}, \\ 8t + 8\ell + 1, & j = 2t \text{ even}. \end{cases}$$

Since ℓ is odd, the parity of the first index of Block_i alternates with i, so consecutive blocks contain the two term types in the ratios $(m+1) : m$ and $m : (m+1)$ alternately. Summing over a block yields

$$S_{i+1}^{(\ell)} - S_i^{(\ell)} = 6\ell^2 + (-1)^i\,(4\ell i + 3),$$

an expression whose deviation from $6\ell^2$ alternates in sign and grows without bound; $(S_m^{(\ell)})$ is therefore not an arithmetic progression for any odd ℓ. The minimality of $\ell = 2$ among the even block sizes is immediate. $\square$

Remark 5.3 (Numerical illustration). For $\ell = 2$: sums $19, 43, 67, 91$ with constant difference 24. For $\ell = 4$: sums $62, 158, 254, 350$ with constant difference $96 = 6 \cdot 4^2$. For $\ell = 3$: sums $45, 84, 165$ with differences $39, 81$ — not arithmetic, in accordance with the closed form $6\ell^2 + (-1)^i(4\ell i + 3) = 54 + (-1)^i(12i + 3)$.

Remark 5.4 (Connection to the OEIS). The sequence of circumferences $U_m = 24m - 5$ is listed in the OEIS under A350051 [SO24] as part of the trisection of $A017101 = \{3 + 8k\}_{k \geq 0}$. It is congruent to 1 (mod 6) — in agreement with the congruence structure of the sequence $k(n)$.

Remark 5.5 (Compatibility of the minimal pairing). Every even block size yields an arithmetic progression (Theorem 5.2); the pairing $\ell = 2$ is distinguished twice over. It is minimal, and it is the only even block size compatible with the angular assignment of the construction: one block then comprises exactly one major and one minor step, i.e. the six index units $\Delta n = 6$ of a full period, which Chapter 13 maps to one full revolution $\Delta\theta = 2\pi$.

5.1.4 Geometric Interpretation of the Block Sums as Circumferences

Having established that the pairwise block sums $S_m^{(2)} = 24m - 5$ form an arithmetic progression, we now present the decisive geometric argument.

An Archimedean spiral $r(\varphi) = a\varphi$ has the defining property of *constant radial difference*: for each complete revolution $\Delta\varphi = 2\pi$ one has

$$\Delta r = 2\pi a = \text{const.}$$

Equivalently: the circumferences $U_m = 2\pi R_m$ of successive spiral windings themselves form an arithmetic progression with constant difference $\Delta U = 2\pi \Delta R$.

Justification of the circumference identification

The integer points of the sequence $k(n)$ admit a polar arrangement in which the two residue classes occupy fixed angular directions: points with $n_k \equiv 1 \pmod 6$ along one ray, points with $n_k \equiv 5 \pmod 6$ along another (made precise by the twin zones of Chapter 13). This suggests interpreting each pair of consecutive steps — one major and one minor — as a complete *period* of the arrangement, corresponding to one winding.

Under this interpretation the sum of the increments within a period corresponds to the circumference increment of a circle. The identification itself is a definition (Definition 5.6); Theorem 5.2 then shows that the even pairings — minimally $\ell = 2$ — are exactly those for which the sums grow arithmetically. This is the algebraic statement that the identification is *consistent*: with the minimal pairing a spiral with *constant* winding spacing arises, and its consequences are forced, as the following sections show.

Definition 5.6 (Circumferences of the sequence)**.** We identify the pairwise block sums of the difference sequence with the circumferences of a family of concentric circles:

$$U_m := S_m^{(2)} = \delta_{2m-1} + \delta_{2m} = 24m - 5.$$

Remark 5.7 (Classification by placement rule). The circumference reading is one of several possible geometric interpretations of the arithmetic data, and the choice classifies the resulting picture — not by curve type, but by the *placement map* $n \mapsto (r, \theta)$. Reading the linear block sums as circumferences gives $r \propto m$: a constant occupation of six integer points per winding, the construction of this work. Reading cumulative *counts* as areas gives $r \propto \sqrt{N}$: the placement of the Sacks spiral [Sac03], which carries all integers with $2n + 1$ points on the n-th winding — its underlying curve is likewise Archimedean; only the occupation density grows. A root radius combined with a *linear* angle ($r \propto \sqrt{n}$, $\theta \propto n$) yields the Fermat spiral of Vogel's phyllotaxis model. Within the linear branch, the circumference is singled out among the quantities proportional to R by a dimensional property: it is the one choice for which the block sum has the same meaning as the quantity it is identified with — the length traversed per revolution, exactly for the circle and to leading order for the spiral (Chapter 8).

5.2 Derivation of the Major and Minor Step Formulae

5.2.1 Explicit Computation of the Step Sizes

We derive the closed-form formulae for the major and minor step sizes directly from the definition of $k(n)$.

The closed-form step formulae were established arithmetically in Chapter 2 (Theorem 2.5, Definition 2.6):

$$\Delta_{\text{Major}}(m) = 16m - 6, \qquad \Delta_{\text{Minor}}(m) = 8m + 1.$$

The first periods:

Table 5.1: Step sizes of the first periods

Period m	Starting value n	Δ_{Major}	Intermediate value n	Δ_{Minor}	U_m
1	1	10	5	9	19
2	7	26	11	17	43
3	13	42	17	25	67
4	19	58	23	33	91
5	25	74	29	41	115

Remark 5.8 (Verification of the formulae). For $m = 1$: $\Delta_{\text{Major}}(1) = 16 \cdot 1 - 6 = 10$, $\Delta_{\text{Minor}}(1) = 8 \cdot 1 + 1 = 9$. For $m = 2$: $\Delta_{\text{Major}}(2) = 32 - 6 = 26$, $\Delta_{\text{Minor}}(2) = 17$. The formulae match the computed differences.

5.2.2 The Unified Generator Function

The two subsequences can be combined into a single formula that generates all steps of the interleaved sequence (δ_k).

Theorem 5.9 (Closed-form generator of the difference sequence)**.** *The complete difference sequence $(\delta_k)_{k\geq 1}$ in its natural order starting from $n = 1$:*

$$\delta_1 = 10,\ \delta_2 = 9,\ \delta_3 = 26,\ \delta_4 = 17,\ \delta_5 = 42,\ \delta_6 = 25,\ \delta_7 = 58,\ \delta_8 = 33,\ \ldots$$

*(**major, minor, major, minor, . . .**) is generated by the closed-form formula*

$$a_k = \frac{1}{2}\big[(12k + 3) + (-1)^{k+1}(4k + 1)\big], \qquad k \geq 1, \tag{5.1}$$

where $a_1 = 10$ corresponds to the first major step (from $n_1 = 1$ to $n_2 = 5$).

Proof. For $k = 2m - 1$ (odd positions, major steps):

$$\begin{aligned} a_{2m-1} &= \frac{1}{2}\big[(12(2m-1)+3) + (-1)^{2m}(4(2m-1)+1)\big] \\ &= \frac{1}{2}\big[(24m-9) + (8m-3)\big] = \frac{32m-12}{2} = 16m - 6. \end{aligned}$$

This is exactly $\Delta_{\text{Major}}(m) = 16m - 6$. Check: $a_1 = 16 \cdot 1 - 6 = 10$. ✓

For $k = 2m$ (even positions, minor steps):

$$\begin{aligned} a_{2m} &= \frac{1}{2}\big[(12(2m)+3) + (-1)^{2m+1}(4(2m)+1)\big] \\ &= \frac{1}{2}\big[(24m+3) - (8m+1)\big] = \frac{16m+2}{2} = 8m + 1. \end{aligned}$$

This is exactly $\Delta_{\text{Minor}}(m) = 8m + 1$. Check: $a_2 = 8 \cdot 1 + 1 = 9$. ✓

The period sums remain identical:

$$U_m = a_{2m-1} + a_{2m} = (16m - 6) + (8m + 1) = 24m - 5. \qquad ✓$$

□

5.3 From Steps to Circumferences

The step notation $\Delta_{\text{Minor},m}$, $\Delta_{\text{Major},m}$ was introduced in Definition 2.6.

By Definition 5.6, one major and one minor step together form the m-th circumference; by Theorem 5.2 (case $\ell = 2$),

$$U_m = \Delta_{\text{Major},m} + \Delta_{\text{Minor},m} = (16m-6)+(8m+1) = 24m-5, \qquad U_{m+1} - U_m = 24. \tag{5.2}$$

5.4 Radius Sequence

From the circumferences the radius sequence follows via $U_m = 2\pi R_m$:

$$R_m = \frac{U_m}{2\pi} = \frac{24m-5}{2\pi}. \tag{5.3}$$

Theorem 5.10 (Constant radius difference — key result)**.**

$$\boxed{\Delta R = R_{m+1} - R_m = \frac{24}{2\pi} = \frac{12}{\pi} \approx 3.819718634205} \tag{5.4}$$

Proof. Direct computation:

$$R_{m+1} - R_m = \frac{24(m+1)-5}{2\pi} - \frac{24m-5}{2\pi} = \frac{24}{2\pi} = \frac{12}{\pi}.$$

□

5.5 Archimedean Spiral

Definition 5.11. An *Archimedean spiral* is a plane curve in polar coordinates of the form

$$r(\theta) = a\theta,$$

with slope constant $a > 0$. The characterising property of this class of curves is the constant radial difference between successive windings: for each complete revolution $\Delta\theta = 2\pi$,

$$\Delta r = r(\theta + 2\pi) - r(\theta) = 2\pi a.$$

The minimality of the pairing proved in Theorem 5.2 and the resulting constant $\Delta R = 12/\pi$ determine the slope constant a:

$$2\pi a = \Delta R = \frac{12}{\pi}, \tag{5.5}$$

$$a = \frac{12}{2\pi^2} = \frac{6}{\pi^2}. \tag{5.6}$$

Theorem 5.12 (Spiral equation)**.**

$$\boxed{r(\theta) = \frac{6}{\pi^2}\,\theta \approx 0.607927101854 \cdot \theta} \tag{5.7}$$

Remark 5.13 (Connection to the Basel problem). The slope constant

$$a = \frac{6}{\pi^2} = \frac{1}{\zeta(2)}, \qquad \zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

is the reciprocal of the Riemann zeta function at $s = 2$, i.e. the solution of the classical Basel problem. The value $\zeta(2) = \pi^2/6$ was proved by Leonhard Euler (published 1740) [Eul40; HW08]. The fact that the spiral constant, obtained from purely combinatorial-arithmetic considerations on the difference structure of the sequence $k(n)$, coincides with $1/\zeta(2)$ is a structural identity.

Remark 5.14 (The spiral constant as the mean coprimality density). The coincidence of the spiral constant $a = 6/\pi^2$ with the reciprocal of $\zeta(2)$ carries a deeper number-theoretic meaning that extends well beyond the Basel problem.

A classical result of analytic number theory states that the average behaviour of Euler's φ-function satisfies

$$\frac{1}{x} \sum_{n \leqslant x} \frac{\varphi(n)}{n} \longrightarrow \frac{6}{\pi^2} \qquad (x \to \infty) \tag{5.8}$$

(see, e.g., [HW08]). The ratio $\varphi(n)/n = \prod_{p|n}(1 - 1/p)$ measures the *proportion of integers up to n that are coprime to n*. The limit $6/\pi^2$ is therefore the *mean coprimality density* of the natural numbers: if one draws a random pair (a, n) with $1 \leqslant a \leqslant n$, the probability that $\gcd(a, n) = 1$ is asymptotically equal to $6/\pi^2$.

The spiral constant of the present work thus admits three simultaneous interpretations:

1. $1/\zeta(2)$ — the reciprocal of the series $\sum 1/n^2$ *(Basel problem, Euler 1734)*;
2. the mean coprimality density $\overline{\varphi(n)/n}$ *(summatory number theory)*;
3. the slope a of the Archimedean spiral, which follows *uniquely* from the difference structure of $k(n)$ *(present work)*.

This threefold connection is not coincidental: the sequence $k(n)$ takes integer values precisely when $n \equiv 1, 5 \pmod 6$, that is, for those n that are coprime to 6. The proportion of these congruence classes is $\varphi(6)/6 = 2/6 = 1/3$, and the spiral constant $a = 6/\pi^2$ encodes, in a sense, the *universal extension* of this local coprimality ratio to all primes simultaneously.

Remark 5.15 (Distribution function of the error term and the Erdős–Shapiro theorem). Erdős and Shapiro [ES55] studied the error function

$$H(x) = \sum_{n \leqslant x} \frac{\varphi(n)}{n} - \frac{6}{\pi^2} x, \tag{5.9}$$

which measures the deviation of the cumulative values of $\varphi(n)/n$ from the linear main term $6x/\pi^2$. Their principal result is that $H(x)$ possesses a *continuous distribution function*: writing $N(n, u) = \#\{m \leqslant n : H(m) \geqslant u\}$, the limit

$$F(u) = \lim_{n \to \infty} \frac{N(n, u)}{n}$$

exists for every u, and F is continuous.

This result is relevant to the present work for two reasons:

1. The constant $6/\pi^2$ about which $H(x)$ oscillates is *exactly* the spiral constant a. The Erdős–Shapiro theorem shows that the fluctuations around this main term are statistically regular — the coprimality density is thus robust not only on average, but also in a probabilistic sense.

2. The existence and continuity of the distribution function of $H(x)$ relies crucially on the continuity of the distribution function of $\varphi(n)/n$ itself (Erdős–Wintner [EW39]). The primes $p > 3$ satisfy $\varphi(p)/p = 1 - 1/p \to 1$ and therefore cluster at the upper end of this distribution. On the conical helix they appear as a geometrically distinguished subset — the statistical distribution of $\varphi(n)/n$ thereby acquires a spatial interpretation.

The present construction thus *geometrises* an arithmetic quantity whose statistical fine structure has been the subject of deep analytic investigations [ES55; Erd74; EH73].

Remark 5.16 (Visible lattice points and squarefree numbers). Two further classical interpretations of the constant $6/\pi^2$ deserve mention, since both are geometric in nature and thus close to the spirit of the present construction (see [HW08]).

First, a lattice point $(x, y) \in \mathbb{Z}^2$ is called *visible* from the origin if no other lattice point lies on the open segment between the origin and (x, y); this holds precisely when $\gcd(x, y) = 1$. The density of visible points among all lattice points is

$$\lim_{R\to\infty} \frac{\#\{(x,y) : \|(x,y)\| \leqslant R,\ \gcd(x,y) = 1\}}{\#\{(x,y) : \|(x,y)\| \leqslant R\}} = \frac{6}{\pi^2}. \tag{5.10}$$

Second, the density of the squarefree integers is likewise

$$\lim_{x\to\infty} \frac{\#\{n \leqslant x : n \text{ squarefree}\}}{x} = \prod_p \Big(1 - \frac{1}{p^2}\Big) = \frac{1}{\zeta(2)} = \frac{6}{\pi^2}, \tag{5.11}$$

where the Euler product on the left is term by term the same product that yields the spiral constant a in the present work.

The list of interpretations of the slope a thus extends to five: the reciprocal of $\zeta(2)$, the mean coprimality density (Remark 5.14), the density of visible lattice points, the density of squarefree integers, and the slope of the Archimedean spiral derived here. The visible-point interpretation is noteworthy in the present context because it is itself a statement about straight lines through integer points of the plane — the same raw material from which the spiral is built. Whether this analogy can be made structural is an open question of this work, not an established result.

Remark 5.17 (Farey fractions and the Franel–Landau criterion). The Farey sequence F_N consists of all reduced fractions in $[0, 1]$ with denominator at most N. Its cardinality satisfies

$$|F_N| = 1 + \sum_{n\leqslant N} \varphi(n) \ \sim\ \frac{3}{\pi^2} N^2 \ =\ \frac{a}{2} N^2, \tag{5.12}$$

with the same coefficient $a/2$ that appears in the arc-length asymptotics $s(\theta) \approx \frac{a}{2}\theta^2$ of the spiral (Chapter 8). Both statements are counting results governed by the mean of φ; the identical quadratic form is recorded here as a formal correspondence, without a claim that the two counts measure the same quantity.

The connection to the Riemann zeta function is provided by the Franel–Landau criterion [Fra24; Lan24]: writing $f_1 < f_2 < \cdots < f_{|F_N|}$ for the elements of F_N and $\delta_j = f_j - j/|F_N|$ for their deviations from perfect equidistribution, the Riemann hypothesis is *equivalent* to

$$\sum_{j=1}^{|F_N|} |\delta_j| = O\big(N^{1/2+\varepsilon}\big) \qquad \text{for every } \varepsilon > 0. \tag{5.13}$$

The quality of the equidistribution of the reduced fractions — the same arithmetic objects whose count carries the coefficient $a/2$ — therefore encodes the position of the nontrivial zeros of $\zeta(s)$.

Remark 5.18 (Mertens' theorem and the Euler–Mascheroni constant). The spiral constant is the convergent second-power Euler product $a = \prod_p (1-p^{-2})$. Its first-power counterpart does not converge to a positive limit; instead, Mertens' theorem [Mer74] states

$$\prod_{p \leqslant x} \Big(1 - \frac{1}{p}\Big) \sim \frac{e^{-\gamma}}{\ln x} \qquad (x \to \infty), \tag{5.14}$$

where $\gamma = 0.5772156649\ldots$ is the Euler–Mascheroni constant. The partial product on the left is the density of integers free of prime factors $\leqslant x$; for $x = 3$ it equals $(1-\frac{1}{2})(1-\frac{1}{3}) = \frac{1}{3} = \varphi(6)/6$, the local coprimality ratio of the congruence classes $n \equiv 1, 5 \pmod 6$ on which the sequence $k(n)$ is integer-valued.

The two products thus continue the same local datum in two different ways: with squared factors the product converges and yields the spiral constant a; with first powers it decays logarithmically, and γ is the constant that calibrates this decay. The constant γ is tied to ζ at its pole through the Laurent expansion $\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$, while a evaluates $1/\zeta$ at $s = 2$; the pair (γ, a) therefore probes the same function at its two arithmetically distinguished points.

Chapter 6

Simplified Parametrisation of the Spiral

6.1 Motivation for the New Parametrisation

The constants $\frac{6}{\pi\sqrt{3}}$ and $\frac{\pi}{\sqrt{3}}$ arise directly from the asymptotic structure of the arc length of the Archimedean spiral $r(\theta) = a\theta$. The exact arc-length function reads

$$L(\theta) = \frac{a}{2}\big[\theta\sqrt{\theta^2+1} + \operatorname{arcsinh}(\theta)\big].$$

For large θ the term $\theta\sqrt{\theta^2+1}$ dominates, so that

$$L(\theta) \sim \frac{a}{2}\theta^2.$$

With $a = \frac{6}{\pi^2}$ the asymptotic relation

$$L(\theta) \sim \frac{3}{\pi^2}\theta^2$$

follows.

Introducing a parameter p that represents the arc length, the inversion of $p = L(\theta)$ yields

$$\theta(p) \approx \sqrt{\frac{2p}{a}} = \frac{\pi}{\sqrt{3}}\sqrt{p}.$$

The constant $\sqrt{3}$ thus arises directly from the quadratic growth rate of the arc length and represents the scaling factor that ensures the asymptotic identification of the parameter with the arc length.

Since $r = a\theta$, the radius becomes

$$r(p) = a\,\theta(p) = \frac{6}{\pi^2} \cdot \frac{\pi}{\sqrt{3}}\sqrt{p} = \frac{6}{\pi\sqrt{3}}\sqrt{p}.$$

The constant $\sqrt{3}$ thus possesses an intrinsic geometric meaning: it characterises the scaling required to reparametrise the spiral so that the new parameter p produces equidistant arc lengths. The same value appears in the analysis of the quadratic function $f(x) = \frac{1}{6}(2x^2 + 3x + 1)$ as the horizontal stretching factor $\sqrt{3}$ associated with its leading coefficient $\frac{1}{3}$; both occurrences of $\sqrt{3}$ thus trace back to the same fraction 2/6 in the definition of $k(n)$.

Why a new parametrisation is necessary

The standard parametrisation of the Archimedean spiral via the angle θ describes the curve geometrically correctly but does not produce arc lengths that correspond to the arithmetic structure of the sequence $k(n)$. For the present application a parameter is needed whose increment corresponds directly to the arc length measured along the spiral. By introducing the parameter p, which runs through the values of $k(n)$, an equidistant arc-length scale arises. Precisely: p is exactly the value parameter of the sequence and asymptotically its arc length; the deviation is quantified in Chapter 8. The spiral remains geometrically unchanged but acquires a parametrisation that precisely reflects the arithmetic structure of the sequence.

6.2 Polar Form with Parameter p

We introduce a new parameter p that directly corresponds to the integer function values of the sequence $k(n)$.

Theorem 6.1 (Simplified polar representation)**.** *The spiral can be expressed in the form*

$$\boxed{\begin{aligned} r(p) &= \frac{6}{\pi} \cdot \sqrt{\frac{p}{3}} \\ &= \frac{6}{\pi\sqrt{3}} \cdot \sqrt{p}, \\ \theta(p) &= \pi \cdot \sqrt{\frac{p}{3}} \\ &= \frac{\pi}{\sqrt{3}} \cdot \sqrt{p} \end{aligned}} \tag{6.1}$$

where p runs through the values of the sequence $k(n)$.

With numerical values:

$$r(p) = \frac{6}{\pi\sqrt{3}} \cdot \sqrt{p} \approx 1.102657790844 \cdot \sqrt{p}, \tag{6.2}$$

$$\theta(p) = \frac{\pi}{\sqrt{3}} \cdot \sqrt{p} \approx 1.813799364234 \cdot \sqrt{p}. \tag{6.3}$$

Note 6.2. A common source of error is the confusion of the two equivalent forms of the radius function. One has

$$r(p) = \frac{6}{\pi\sqrt{3}}\sqrt{p} = \frac{6}{\pi}\sqrt{\frac{p}{3}}.$$

The factor $\frac{6}{\pi} \approx 1.9099$ is the coefficient of $\sqrt{p/3}$, *not* of $\sqrt{p}$. If it is multiplied directly by $\sqrt{p}$, the resulting coordinates are too large by a factor of $\sqrt{3} \approx 1.732$. For numerical evaluation it is therefore advisable to use the form $r(p) = 1.102657790844 \cdot \sqrt{p}$.

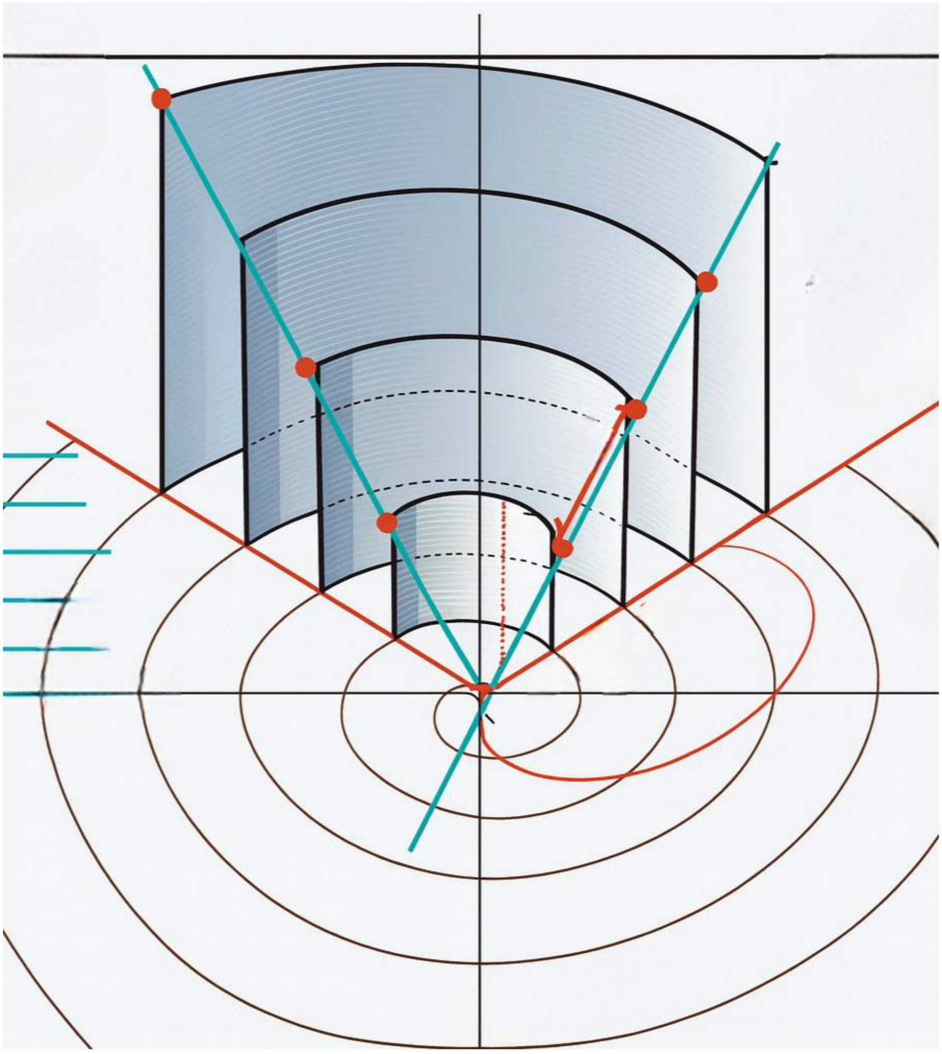


Figure 6.1: Transition from the two-dimensional Archimedean spiral (bottom) to the three-dimensional cylindrical structure (top). The cyan rays mark the radius vector $r(\theta)$ at the integer points of the sequence; the orange lines visualise the corresponding tangential directions. The concentrically arranged circular arcs correspond to the discrete radial increments $\Delta r = 12/\pi$, which, upon unrolling, become the vertical spacings of the 3D surface.

6.3 Parametric Representation in Cartesian Coordinates

$$x(p) = r(p)\cos(\theta(p)) = \frac{6}{\pi\sqrt{3}}\sqrt{p}\,\cos\left(\frac{\pi}{\sqrt{3}}\sqrt{p}\right),$$
$$y(p) = r(p)\sin(\theta(p)) = \frac{6}{\pi\sqrt{3}}\sqrt{p}\,\sin\left(\frac{\pi}{\sqrt{3}}\sqrt{p}\right). \tag{6.4}$$

Remark 6.3 (The angular factor as a Dirichlet L-value: Eisenstein integers). The angular factor $\pi/\sqrt{3}$ of the p-parametrisation admits an exact expression through a Dirichlet L-function. Let χ_{-3} be the nontrivial character modulo 3, i.e. $\chi_{-3}(n) = +1$ for $n \equiv 1 \pmod 3$, -1 for $n \equiv 2 \pmod 3$, and 0 for $3 \mid n$. Its L-series at $s = 1$,

$$L(1,\chi_{-3}) = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots = \frac{\pi}{3\sqrt{3}}, \tag{6.5}$$

is a classical evaluation; by Dirichlet's class number formula it reads

$$L(1,\chi_{-3}) = \frac{2\pi h}{w\sqrt{|d|}}$$

with class number $h = 1$, discriminant $d = -3$ and $w = 6$ units for the field $\mathbb{Q}(\sqrt{-3})$ [IR90]. Hence

$$\frac{\pi}{\sqrt{3}} = 3\,L(1,\chi_{-3}) = \frac{w}{2}\,L(1,\chi_{-3}), \qquad \theta(p) = \frac{w}{2}\,L(1,\chi_{-3})\,\sqrt{p}. \tag{6.6}$$

The ring of integers of $\mathbb{Q}(\sqrt{-3})$ is the ring of Eisenstein integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$ — the hexagonal lattice, whose unit group has order exactly 6. Its arithmetic is organised by the same congruence classes as the present work: a rational prime $p > 3$ splits in $\mathbb{Z}[\omega]$ if $p \equiv 1 \pmod 3$ and remains inert if $p \equiv 2 \pmod 3$, while $p = 3$ ramifies. For odd n, the classes $n \equiv 1, 5 \pmod 6$ on which $k(n)$ is integer-valued are precisely the classes $n \equiv 1, 2 \pmod 3$, so the two strands of the twin structure correspond to the two splitting types. The associated Dedekind zeta function factorises as $\zeta_{\mathbb{Q}(\sqrt{-3})}(s) = \zeta(s)\,L(s,\chi_{-3})$.

It should be stated clearly what is established and what is not: the identities (6.5)–(6.6) are exact, and the correspondence between the residue classes and the splitting law in $\mathbb{Z}[\omega]$ is classical algebraic number theory. That the angular factor of the spiral is forced to be this particular L-value — rather than merely equal to it through the shared origin of both quantities in the modulus 6 — is a conjecture of the present work and remains open.

Remark 6.4 (The angular factor and the reflection formula of the Gamma function). Euler's reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ yields at $z = \frac{1}{3}$

$$\Gamma\left(\tfrac{1}{3}\right)\Gamma\left(\tfrac{2}{3}\right) = \frac{\pi}{\sin(\pi/3)} = \frac{2\pi}{\sqrt{3}}, \qquad \text{hence} \qquad \frac{\pi}{\sqrt{3}} = \frac{1}{2}\,\Gamma\left(\tfrac{1}{3}\right)\Gamma\left(\tfrac{2}{3}\right). \tag{6.7}$$

The arguments $\frac{1}{3}$ and $\frac{2}{3}$ are the two residue classes modulo 3 — normalised to the unit interval — that carry the twin structure of $k(n)$; the angular factor is one half of the product of the Gamma values at exactly these two points.

The appearance of $\sqrt{3}$ in (6.7) and in the L-value of Remark 6.3 is not independent: the Chowla–Selberg formula [CS67] expresses the periods attached to the imaginary quadratic field $\mathbb{Q}(\sqrt{-3})$ in terms of $\Gamma(\frac{1}{3})$, so that the L-value and the Gamma product are two evaluations rooted in the same field. Within the present work these identities are recorded as structure shared with the parametrisation constants; no derivation of the spiral from either side is claimed.

Remark 6.5 (The two roles of $L(s, \chi_{-3})$: value and spectrum). The function $L(s, \chi_{-3})$ of Remark 6.3 enters the arithmetic of the classes $6k \pm 1$ a second time, through its zeros. The nontrivial character modulo 6 is induced by the primitive character χ_{-3} modulo 3, so that

$$L(s, \chi_6) = L(s, \chi_{-3}) \left(1 + 2^{-s}\right), \tag{6.8}$$

and since the elementary factor $1 + 2^{-s}$ has no zeros in the critical strip, both functions possess the same nontrivial zeros (numerically, the lowest ones lie at $\gamma = 8.0397$, 11.2492, 15.7046, 18.2620, $20.4558, \dots$). By the explicit formulae of prime number theory [Dav00], these zeros govern the fluctuation of the prime counts *between* the two residue classes: the combination $\pi(x; 6, 1) - \pi(x; 6, 5)$ is a superposition of oscillations $x^\beta \cos(\gamma \ln x)$ over the zeros $\beta + i\gamma$ of $L(s, \chi_{-3})$, and the generalised Riemann hypothesis is the statement that all these oscillations carry the minimal weight $\beta = \frac{1}{2}$.

The same L-function therefore acts on the present construction in two distinct roles: its *value* at $s = 1$ fixes the angular factor of the parametrisation (Remark 6.3), while its *zeros* form the frequency spectrum that modulates how the primes distribute themselves between the two strands parametrised by it. The first role is a proven identity; that the coincidence of the two roles carries structural content is an open question of this work.

6.4 Derivatives

Theorem 6.6 (Derivatives of the polar coordinates)**.**

$$\begin{aligned} \frac{dr}{dp} &= \frac{d}{dp}\left[\frac{6}{\pi\sqrt{3}}\sqrt{p}\right] = \frac{6}{\pi\sqrt{3}} \cdot \frac{1}{2\sqrt{p}} = \frac{\sqrt{3}}{\pi\sqrt{p}}, \\ \frac{d\theta}{dp} &= \frac{d}{dp}\left[\frac{\pi}{\sqrt{3}}\sqrt{p}\right] = \frac{\pi}{\sqrt{3}} \cdot \frac{1}{2\sqrt{p}} = \frac{\pi}{2\sqrt{3}\sqrt{p}}. \end{aligned} \tag{6.9}$$

Numerically:

$$\frac{dr}{dp} = \frac{\sqrt{3}}{\pi\sqrt{p}} = \frac{0.551328895422}{\sqrt{p}}, \tag{6.10}$$

$$\frac{d\theta}{dp} = \frac{\pi}{2\sqrt{3}\sqrt{p}} = \frac{0.906899682117}{\sqrt{p}}. \tag{6.11}$$

Chapter 7

Arc-Length Integral of the Spiral

7.1 Arc-Length Element

For a parametrised curve $(x(p), y(p))$ the arc-length element is

$$ds = \sqrt{\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2}\, dp. \tag{7.1}$$

Applying the product rule yields:

$$\begin{aligned} \frac{dx}{dp} &= \frac{dr}{dp}\cos(\theta) - r\sin(\theta)\frac{d\theta}{dp}, \\ \frac{dy}{dp} &= \frac{dr}{dp}\sin(\theta) + r\cos(\theta)\frac{d\theta}{dp}. \end{aligned} \tag{7.2}$$

After expanding, the cross terms cancel:

$$\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2 = \left(\frac{dr}{dp}\right)^2 + r^2\left(\frac{d\theta}{dp}\right)^2. \tag{7.3}$$

7.2 Computation of the Terms

Radial Term

$$\left(\frac{dr}{dp}\right)^2 = \left(\frac{\sqrt{3}}{\pi\sqrt{p}}\right)^2 = \frac{3}{\pi^2 p}. \tag{7.4}$$

Angular Term

$$\begin{aligned} r^2\left(\frac{d\theta}{dp}\right)^2 &= \left[\frac{6}{\pi\sqrt{3}}\sqrt{p}\right]^2 \cdot \left[\frac{\pi}{2\sqrt{3}\sqrt{p}}\right]^2 && (7.5) \\ &= \frac{36p}{3\pi^2} \cdot \frac{\pi^2}{12p} && (7.6) \\ &= 1. && (7.7) \end{aligned}$$

Total Speed

This gives:

$$\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2 = \frac{3}{\pi^2 p} + 1 = \frac{\frac{3}{p} + \pi^2}{\pi^2}. \tag{7.8}$$

7.3 Final Form of the Arc-Length Integral

Theorem 7.1 (Arc-length integral)**.**

$$\boxed{L = \int_{p_1}^{p_2} \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\, dp} \tag{7.9}$$

In particular $ds/dp = \sqrt{1 + 3/(\pi^2 p)} \to 1$: the parameter p is asymptotically the arc length itself, with the correction terms quantified in the following two chapters.

Chapter 8

Arc Length of the Plane Archimedean Spiral: Two Methods

In this chapter the arc length of the Archimedean spiral $r(\theta) = \frac{6}{\pi^2}\,\theta$ is computed by two entirely different approaches and the results are compared numerically. This serves both for mutual verification and for the conceptual clarification of how the θ-parametrisation and the p-parametrisation are related. Throughout the numerical tables the decimal point is used consistently.

The intervals under consideration are the minor steps between the integer indices $n \equiv 1, 5 \pmod 6$:

$$(n_1, n_2) \in \{(5,7),\ (11,13),\ (17,19),\ (23,25),\ (29,31),\ (35,37),\ (41,43)\}.$$

In each case $n_1 \equiv 5 \pmod 6$ and $n_2 \equiv 1 \pmod 6$, so these are always minor steps with $\Delta n = 2$.

8.1 Method A: Direct Computation via the Closed-Form Arc-Length Formula

8.1.1 Derivation of the Formula

For a curve in polar coordinates $r = r(\theta)$ the arc-length element reads

$$ds = \sqrt{r(\theta)^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

With $r(\theta) = a\theta$ and $\frac{dr}{d\theta} = a$ (where $a = \frac{6}{\pi^2}$):

$$ds = \sqrt{a^2\theta^2 + a^2}\, d\theta = a\sqrt{\theta^2 + 1}\, d\theta.$$

The arc-length integral between two angles θ_1 and θ_2 is therefore:

$$L_A(\theta_1, \theta_2) = a \int_{\theta_1}^{\theta_2} \sqrt{\theta^2 + 1}\, d\theta.$$

This integral admits a closed form. Using the standard formula

$$\int \sqrt{u^2+1}\,du = \frac{1}{2}\Big[u\sqrt{u^2+1} + \operatorname{arcsinh}(u)\Big] + C$$

we obtain the antiderivative $F(\theta) = \frac{a}{2}\big[\theta\sqrt{\theta^2+1} + \operatorname{arcsinh}(\theta)\big]$ and hence the exact, closed-form arc length:

Theorem 8.1 (Arc length of the Archimedean spiral, closed form)**.** *For $r(\theta) = a\theta$ with $a = \frac{6}{\pi^2}$:*

$$\boxed{L_A(\theta_1,\theta_2) = \frac{a}{2}\Big[\theta\sqrt{\theta^2+1} + \operatorname{arcsinh}(\theta)\Big]_{\theta_1}^{\theta_2}.} \tag{8.1}$$

8.1.2 Assignment of Angle Values to the Indices

The angle values θ_n at the integer points of the helix are

$$\theta_n = \frac{\pi}{3}n + \frac{\pi}{4} = \frac{(4n+3)\pi}{12}.$$

As exact fractions of π:

n_1	$\theta_{n_1} = \frac{(4n_1+3)\pi}{12}$	n_2	$\theta_{n_2} = \frac{(4n_2+3)\pi}{12}$	$\Delta\theta = \frac{2\pi}{3}$
5	$\frac{23\pi}{12}$	7	$\frac{31\pi}{12}$	$\frac{8\pi}{12} = \frac{2\pi}{3}$
11	$\frac{47\pi}{12}$	13	$\frac{55\pi}{12}$	$\frac{2\pi}{3}$
17	$\frac{71\pi}{12}$	19	$\frac{79\pi}{12}$	$\frac{2\pi}{3}$
23	$\frac{95\pi}{12}$	25	$\frac{103\pi}{12}$	$\frac{2\pi}{3}$
29	$\frac{119\pi}{12}$	31	$\frac{127\pi}{12}$	$\frac{2\pi}{3}$
35	$\frac{143\pi}{12}$	37	$\frac{151\pi}{12}$	$\frac{2\pi}{3}$
41	$\frac{167\pi}{12}$	43	$\frac{175\pi}{12}$	$\frac{2\pi}{3}$

Remark 8.2 (Constant angular spacing). All minor steps (i.e. $\Delta n = 2$) produce the same angular spacing $\Delta\theta = \frac{2\pi}{3}$, since

$$\theta_{n+2} - \theta_n = \frac{\pi}{3}(n+2) + \frac{\pi}{4} - \frac{\pi}{3}n - \frac{\pi}{4} = \frac{2\pi}{3}.$$

The spiral traverses exactly $\frac{1}{3}$ of a full revolution per minor step. The angular spacing is thus identical for all intervals under consideration; nevertheless, the arc length grows monotonically, since for the same $\Delta\theta$ the arc on a larger radius is longer.

8.1.3 Numerical Results for Method A

Table 8.1: Arc length by Method A: closed-form formula $L_A = \frac{a}{2}[\theta\sqrt{\theta^2+1} + \operatorname{arcsinh}(\theta)]_{\theta_1}^{\theta_2}$ with $a = 6/\pi^2$.

(n_1, n_2)	θ_1 (num.)	θ_2 (num.)	L_A
$(5, 7)$	6.02138592	8.11578102	9.09026487
$(11, 13)$	12.30457123	14.39896633	17.04771118
$(17, 19)$	18.58775653	20.68215164	25.03243246
$(23, 25)$	24.87094184	26.96533694	33.02456693
$(29, 31)$	31.15412715	33.24852225	41.01977219
$(35, 37)$	37.43731245	39.53170755	49.01654352
$(41, 43)$	43.72049776	45.81489286	57.01422134

Remark 8.3 (Growth rate of the arc length). The arc lengths grow approximately proportionally to the mean θ-value. For large θ the linear term in $\sqrt{\theta^2+1} \approx \theta$ dominates, so that

$$L_A \approx a \int_{\theta_1}^{\theta_2} \theta \, d\theta = \frac{a}{2}(\theta_2^2 - \theta_1^2) = \frac{a}{2}(\theta_2 + \theta_1)\Delta\theta.$$

With $\Delta\theta = \frac{2\pi}{3}$ and increasing $\bar{\theta} = \frac{\theta_1+\theta_2}{2}$, a linear growth in $\bar{\theta}$ results. The difference between two consecutive arc lengths is therefore approximately $a \cdot \frac{2\pi}{3} \cdot 2\pi = a \cdot \frac{4\pi^2}{3} = 8$ per period of the indices (from n to $n+6$). Numerically: $25.032 - 17.048 = 7.985$, $33.025 - 25.032 = 7.993$ etc. — this convergence confirms the asymptotics.

8.2 Method B: Arc Length via the p-Parametrisation

8.2.1 Derivation of the Integral

In Chapter 7 the arc-length integral in the p-parametrisation was derived. Since $r(p) = \frac{6}{\pi\sqrt{3}}\sqrt{p}$ and $\theta(p) = \frac{\pi}{\sqrt{3}}\sqrt{p}$, the computation of the arc-length element yields:

$$\left(\frac{dr}{dp}\right)^2 = \frac{3}{\pi^2 p}, \qquad r^2\left(\frac{d\theta}{dp}\right)^2 = 1,$$

and hence:

$$L_B(p_1, p_2) = \int_{p_1}^{p_2} \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\, dp. \tag{8.2}$$

The points in the p-parametrisation correspond to the function values $p = k(n)$:

n_1	$p_1 = k(n_1)$	n_2	$p_2 = k(n_2)$	$\Delta p = p_2 - p_1$
5	11	7	20	9
11	46	13	63	17
17	105	19	130	25
23	188	25	221	33
29	295	31	336	41
35	426	37	475	49
41	581	43	638	57

Remark 8.4 (Growth of the Δp values). The values $\Delta p = p_2 - p_1$ are precisely the minor steps of the difference sequence:

$$\Delta p_m = \Delta_{\text{Minor}}(m) = 8m + 1, \qquad m = 1, 2, 3, \ldots$$

For $m = 1$: $\Delta p = 9$, for $m = 2$: $\Delta p = 17$, and so on. The function values p grow quadratically in n, while Δp grows only linearly. This implies that for large p the quotient $\Delta p/p$ tends to zero — the integrand term $\frac{3}{p}$ then becomes negligible compared with π^2, and the integral approaches the Euclidean value Δp.

8.2.2 Numerical Results for Method B

Table 8.2: Arc length by Method B: numerical integral $L_B = \int_{p_1}^{p_2} \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\,dp$.

(n_1, n_2)	$p_1 = k(n_1)$	$p_2 = k(n_2)$	Δp	L_B
$(5, 7)$	11	20	9	9.09039285
$(11, 13)$	46	63	17	17.04772970
$(17, 19)$	105	130	25	25.03243825
$(23, 25)$	188	221	33	33.02456944
$(29, 31)$	295	336	41	41.01977350
$(35, 37)$	426	475	49	49.01654429
$(41, 43)$	581	638	57	57.01422182

8.3 Direct Comparison: Method A vs. Method B

Table 8.3: Comparison of arc lengths: Method A (closed-form θ-formula) vs. Method B (p-integral). The deviation $\delta = (L_B - L_A)/L_A$ is positive and decreases monotonically through the computed range towards zero.

(n_1, n_2)	L_A	L_B	$L_B - L_A$	δ (%)
$(5, 7)$	9.09026487	9.09039285	+0.00012798	+0.00141
$(11, 13)$	17.04771118	17.04772970	+0.00001852	+0.00011
$(17, 19)$	25.03243246	25.03243825	+0.00000579	+0.00002
$(23, 25)$	33.02456693	33.02456944	+0.00000251	+0.00001
$(29, 31)$	41.01977219	41.01977350	+0.00000131	+0.00000
$(35, 37)$	49.01654352	49.01654429	+0.00000077	+0.00000
$(41, 43)$	57.01422134	57.01422182	+0.00000048	+0.00000

Proposition 8.5 (Agreement of the methods). *The two computational methods yield numerically almost identical results. The relative deviation $\delta = (L_B - L_A)/L_A$ is positive in all computed intervals and converges to zero:*

$$\delta(5 \to 7) = 0.00141\,\%, \quad \delta(11 \to 13) = 0.00011\,\%, \quad \delta(41 \to 43) < 0.000005\,\%.$$

Remark 8.6 (Explanation of the small positive deviation). The deviation stems from the fact that the p-parametrisation is only *asymptotically* arc-length preserving: the inversion $p = L(\theta)$ holds asymptotically for large θ, not for small ones. For the first interval ($p_1 = 11$, $p_2 = 20$) the correction is still measurable; for $p \gtrsim 100$ it is negligible for all practical purposes.

Analytically, this can be understood as follows: the arc-length element in the p-parametrisation is (as derived in Chapter 7)

$$ds = \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\,dp = \sqrt{1 + \frac{3}{\pi^2 p}}\,dp,$$

which shows that the speed converges to 1 as $p \to \infty$. The term $\frac{3}{\pi^2 p}$ produces the positive residual deviation; it is approximately 0.028 for $p_1 = 11$, but only about 0.0005 for $p_1 = 581$.

8.3.1 Why the Equidistant Parametrisation Works

The central conceptual statement of this comparison is that the integer values $p = k(n)$ of the quadratic sequence already constitute a very good approximation to the arc-length values of the Archimedean spiral. Specifically: the step $\Delta p = p_2 - p_1 = \Delta_{\mathrm{Minor}}(m)$ is numerically almost equal to the arc length L_A of the corresponding spiral arc:

Table 8.4: Comparison of Δp (arithmetic step) vs. L_A (arc length). The ratio $L_A/\Delta p$ converges to 1.

(n_1, n_2)	Δp	L_A	$L_A/\Delta p$	Deviation
$(5,7)$	9	9.09026	1.01003	+1.00 %
$(11,13)$	17	17.04771	1.00280	+0.28 %
$(17,19)$	25	25.03243	1.00130	+0.13 %
$(23,25)$	33	33.02457	1.00074	+0.07 %
$(29,31)$	41	41.01977	1.00048	+0.05 %
$(35,37)$	49	49.01654	1.00034	+0.03 %
$(41,43)$	57	57.01422	1.00025	+0.02 %

Remark 8.7 (Geometric significance). This table is the numerical heart of the entire construction: it shows that the arithmetic steps $\Delta p = 8m+1$ of the minor difference sequence are not only integers but also increasingly accurate approximations of the arc lengths of the corresponding spiral arcs. For $m \to \infty$ one has $L_A/\Delta p \to 1$ — the discrete arithmetic structure of the sequence $k(n)$ and the continuous geometry of the Archimedean spiral merge asymptotically into a single description.

Put differently: the construction succeeds because the function values $k(n)$ are not only integers that enforce the correct spiral constant, but simultaneously approximate arc-length parameters of the resulting spiral. This is a direct consequence of the asymptotic arc-length formula $L(\theta) \sim \frac{a}{2}\theta^2$ and the choice $p = k(n) \sim \frac{a}{2}\theta_n^2$.

Chapter 9

Arc Length of the Quadratic Curve as a Periodic Sequence

In the preceding chapters the arc length of the Archimedean spiral was investigated. Here we return to the original quadratic curve

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

and systematically compute its arc length along the integer indices (i.e. $n \equiv 1, 5 \pmod 6$, cf. Chapter 2). The result is a closed-form formula that reproduces the same periodic structure of twin and gap zones that already appeared in Chapter 2 as a difference sequence — now, however, as an *arc-length sequence* of the parabola.

9.1 Derivation of the General Arc-Length Formula

9.1.1 Arc-Length Element of the Parabola

For a real-valued function $k(n)$ the arc-length element is

$$ds = \sqrt{1 + \left[k'(n)\right]^2}\, dn.$$

The derivative of the quadratic sequence is:

$$k'(n) = \frac{4n + 3}{6}.$$

This yields:

$$ds = \sqrt{1 + \frac{(4n+3)^2}{36}}\, dn = \frac{1}{6}\sqrt{36 + (4n+3)^2}\, dn.$$

The arc length between two indices n_1 and n_2 is therefore:

$$L(n_1, n_2) = \frac{1}{6} \int_{n_1}^{n_2} \sqrt{36 + (4n+3)^2}\, dn. \tag{9.1}$$

9.1.2 Substitution and Closed Form

The integral (9.1) is reduced to a standard integral by the linear substitution

$$u = 4n + 3, \qquad du = 4\,dn$$

yielding:

$$L = \frac{1}{6} \cdot \frac{1}{4} \int_{u_1}^{u_2} \sqrt{u^2 + 36}\, du = \frac{1}{24} \int_{u_1}^{u_2} \sqrt{u^2 + 36}\, du,$$

where $u_i = 4n_i + 3$.

The indefinite integral of $\sqrt{u^2 + a^2}$ is well known:

$$\int \sqrt{u^2 + a^2}\, du = \frac{1}{2}\Big[u\sqrt{u^2 + a^2} + a^2 \ln\!\big(u + \sqrt{u^2 + a^2}\big)\Big] + C.$$

With $a = 6$ and the identity $\ln(u + \sqrt{u^2 + 36}) = \operatorname{arcsinh}(u/6) + \ln 6$ (the constant cancels upon evaluation), we obtain the antiderivative:

$$G(u) = \frac{1}{2}\Big[u\sqrt{u^2 + 36} + 36 \operatorname{arcsinh}\Big(\frac{u}{6}\Big)\Big].$$

Theorem 9.1 (Closed-form arc-length formula for the parabola)**.** *Let $u_i = 4n_i + 3$. The arc length of the curve $k(n)$ between n_1 and n_2 is:*

$$L(n_1, n_2) = \frac{1}{48}\Big[u\sqrt{u^2 + 36} + 36 \operatorname{arcsinh}\Big(\frac{u}{6}\Big)\Big]_{u_1}^{u_2}. \tag{9.2}$$

Remark 9.2 (Alternative notation with csch^{-1}). Since $\operatorname{arcsinh}(x) = \operatorname{csch}^{-1}(1/x)$, the formula can also be written as

$$L = \frac{1}{48}\Big[u\sqrt{u^2 + 36} + 36 \operatorname{csch}^{-1}\Big(\frac{6}{u}\Big)\Big]_{u_1}^{u_2}.$$

This form arises naturally from hyperbolic substitutions and is the preferred output in some computer algebra systems.

9.2 Twin-Zone Sequence (Minor Steps, $\Delta n = 2$)

9.2.1 Derivation of the Periodic Formula

The minor steps connect $n_1 = 6m - 1$ (with $n_1 \equiv 5 \pmod 6$) and $n_2 = 6m + 1$ (with $n_2 \equiv 1 \pmod 6$), for $m = 1, 2, 3, \ldots$ The corresponding u-values are:

$$u_1 = 4(6m - 1) + 3 = 24m - 1, \qquad u_2 = 4(6m + 1) + 3 = 24m + 7.$$

Theorem 9.3 (Twin-zone arc length)**.** *The arc length of the parabola $k(n)$ over the m-th minor step ($n : 6m - 1 \to 6m + 1$) is:*

$$\begin{aligned} L_{\text{Twin}}(m) = \frac{1}{48}\Big[&(24m + 7)\sqrt{(24m + 7)^2 + 36} + 36 \operatorname{arcsinh}\Big(\frac{24m + 7}{6}\Big) \\ &- (24m - 1)\sqrt{(24m - 1)^2 + 36} - 36 \operatorname{arcsinh}\Big(\frac{24m - 1}{6}\Big)\Big]. \end{aligned} \tag{9.3}$$

Proof. Direct application of Theorem 8.1 with the bounds $u_1 = 24m - 1$ and $u_2 = 24m + 7$. ∎

9.2.2 Numerical Values

Table 9.1: Twin-zone arc lengths $L_{\text{Twin}}(m)$ for the first seven periods. The arc length grows approximately linearly with m, with the ratio $L_{\text{Twin}}(m)/\Delta_{\text{Minor}}(m)$ converging monotonically to 1.

m	$n_1 \to n_2$	u_1	u_2	$\Delta_{\text{Minor}}(m)$	$L_{\text{Twin}}(m)$	L/Δ_{Minor}
1	$5 \to 7$	23	31	9	9.2210749306	1.024564
2	$11 \to 13$	47	55	17	17.1174799293	1.006911
3	$17 \to 19$	71	79	25	25.0799476641	1.003198
4	$23 \to 25$	95	103	33	33.0605833402	1.001836
5	$29 \to 31$	119	127	41	41.0487686495	1.001189
6	$35 \to 37$	143	151	49	49.0408093942	1.000833
7	$41 \to 43$	167	175	57	57.0350833164	1.000615

Remark 9.4 (Convergence to the minor step)*.* The last column shows that $L_{\text{Twin}}(m) \to \Delta_{\text{Minor}}(m)$ as $m \to \infty$. This is geometrically plausible: for large m the curve $k(n)$ in the short interval $[6m-1, 6m+1]$ is nearly rectilinear (the curvature decreases), so the arc length converges to the chord length. Since for very steep curves ($k'(n) \gg 1$) the arc length is asymptotically dominated by the k'-term:

$$L_{\text{Twin}}(m) \approx \int_{6m-1}^{6m+1} \frac{4n+3}{6}\, dn = \frac{1}{6}\left[\frac{(4n+3)^2}{8}\right]_{6m-1}^{6m+1} \approx 8m+1 = \Delta_{\text{Minor}}(m).$$

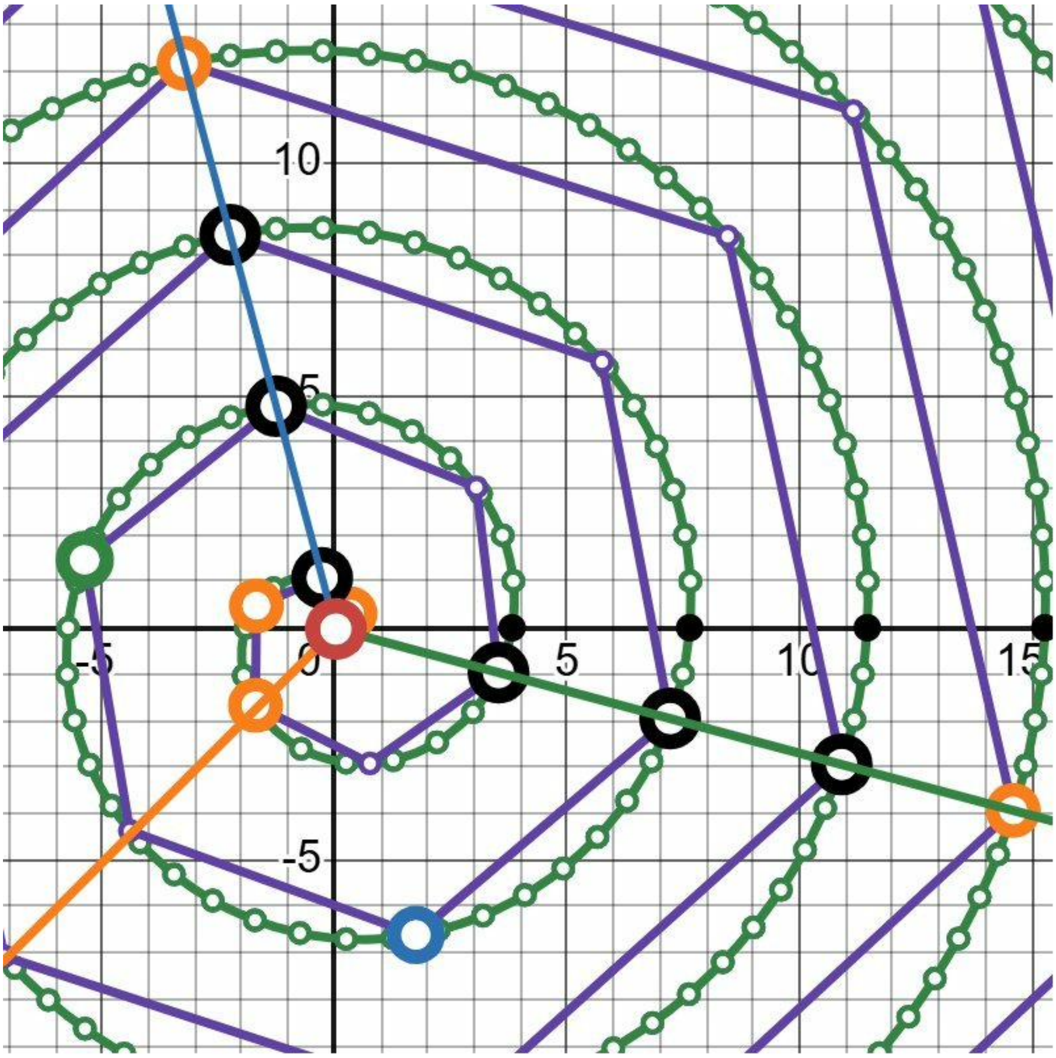


Figure 9.1: Enlarged plan view of the Archimedean spiral $r = \frac{6}{\pi^2}\,\theta$ with equidistant arc-length scaling (1 unit = 1). The small green circular markers are placed at constant arc spacing 1 and are thus directly countable. The large hollow black dots mark the integer sequence values $k(n)$ on the spiral; the coloured hollow dots identify the same values on their respective diagonal ray. The counted arc segments confirm $\Delta_{\text{Minor}}(m) = 8m + 1$ directly: 1st twin zone ($n = 5 \to 7$): **9 segments** $= 8{\cdot}1{+}1$; 2nd twin zone ($n = 11 \to 13$): **17 segments** $= 8 \cdot 2 + 1$; 3rd twin zone ($n = 17 \to 19$): **25 segments** $= 8 \cdot 3 + 1$. The coloured diagonals form the hexagonal ray lattice of the congruence classes $n \equiv 1, 5 \pmod 6$ with regular 60 angular spacing — a direct consequence of the 6-periodic structure of the sequence $k(n)$.

9.3 Gap-Zone Sequence (Major Steps, $\Delta n = 4$)

9.3.1 Derivation of the Periodic Formula

The major steps connect $n_1 = 6m - 5$ (with $n_1 \equiv 1 \pmod 6$) and $n_2 = 6m - 1$ (with $n_2 \equiv 5 \pmod 6$), for $m = 1, 2, 3, \ldots$ The u-values are:

$$u_1 = 4(6m-5) + 3 = 24m - 17, \qquad u_2 = 4(6m-1) + 3 = 24m - 1.$$

Theorem 9.5 (Gap-zone arc length)**.** *The arc length of the parabola $k(n)$ over the m-th major step ($n : 6m - 5 \to 6m - 1$) is:*

$$\begin{aligned} L_{\text{Gap}}(m) = \frac{1}{48}\Bigg[&(24m-1)\sqrt{(24m-1)^2+36} + 36\,\operatorname{arcsinh}\!\left(\frac{24m-1}{6}\right) \\ &- (24m-17)\sqrt{(24m-17)^2+36} - 36\,\operatorname{arcsinh}\!\left(\frac{24m-17}{6}\right)\Bigg]. \end{aligned} \tag{9.4}$$

Proof. Direct application of Theorem 8.1 with $u_1 = 24m - 17$ and $u_2 = 24m - 1$. □

9.3.2 Numerical Values

Table 9.2: Gap-zone arc lengths $L_{\text{Gap}}(m)$ for the first seven periods.

m	$n_1 \to n_2$	u_1	u_2	$\Delta_{\text{Major}}(m)$	$L_{\text{Gap}}(m)$
1	$1 \to 5$	7	23	10	10.8394091489
2	$7 \to 11$	31	47	26	26.3101623759
3	$13 \to 17$	55	71	42	42.1910659377
4	$19 \to 23$	79	95	58	58.1381553675
5	$25 \to 29$	103	119	74	74.1082162014
6	$31 \to 35$	127	143	90	90.0889489962
7	$37 \to 41$	151	167	106	106.0755084967

Remark 9.6 (Relation to the Euclidean distances). The value $\Delta_{\text{Major}}(m) = 16m - 6$ in the table is the Euclidean distance $d = \sqrt{(\Delta n)^2 + (\Delta k)^2}$, *not* the arc length. The arc length $L_{\text{Gap}}(m)$ always exceeds the Euclidean distance, but likewise converges asymptotically to $\Delta_{\text{Major}}(m)$ as $m \to \infty$.

9.4 Complete Sequence and Connection to the Difference Structure

The following table juxtaposes both zones and displays the complete arc-length sequence in the order in which the indices $n \equiv 1, 5 \pmod 6$ succeed one another.

Table 9.3: Complete arc-length sequence of the parabola $k(n)$, alternating between gap zone (major, $\Delta n = 4$) and twin zone (minor, $\Delta n = 2$). For comparison, the difference values Δ_{Major} and Δ_{Minor} from Chapter 2 are given.

$n_1 \to n_2$	Type	Arc length L	Difference Δk	$L/\Delta k$
$1 \to 5$	Gap (m=1)	10.839409	10	1.0839
$5 \to 7$	Twin (m=1)	9.221075	9	1.0246
$7 \to 11$	Gap (m=2)	26.310162	26	1.0119
$11 \to 13$	Twin (m=2)	17.117480	17	1.0069
$13 \to 17$	Gap (m=3)	42.191066	42	1.0045
$17 \to 19$	Twin (m=3)	25.079948	25	1.0032
$19 \to 23$	Gap (m=4)	58.138155	58	1.0024
$23 \to 25$	Twin (m=4)	33.060583	33	1.0018
$25 \to 29$	Gap (m=5)	74.108216	74	1.0015
$29 \to 31$	Twin (m=5)	41.048769	41	1.0012
$31 \to 35$	Gap (m=6)	90.088949	90	1.0010
$35 \to 37$	Twin (m=6)	49.040809	49	1.0008
$37 \to 41$	Gap (m=7)	106.075508	106	1.0007
$41 \to 43$	Twin (m=7)	57.035083	57	1.0006

Main observation: arc length $\approx$ difference value

The rightmost column shows that the ratio $L/\Delta k$ is close to 1 for all periods and converges monotonically to 1. This means: the differences $\Delta_{\text{Minor}}(m) = 8m + 1$ and $\Delta_{\text{Major}}(m) = 16m - 6$ of the sequence $k(n)$ are simultaneously very good approximations of the arc lengths of the parabola on the respective sub-intervals. This arises from the fact that the slope $k'(n) = (4n + 3)/6$ is considerably larger than 1 for the relevant n-values, so that the contribution of the horizontal component ($\Delta n \leq 4$) to the arc-length element $\sqrt{1 + (k')^2}$ becomes ever smaller:

$$\frac{ds}{dn} = \sqrt{1 + \frac{(4n+3)^2}{36}} \xrightarrow{n\to\infty} \frac{4n+3}{6} = k'(n).$$

The arc length thus approaches asymptotically the integral of $k'(n)$, that is, the function-value difference Δk itself.

Proposition 9.7 (Asymptotic arc-length–difference equivalence). *As $m \to \infty$:*

$$\frac{L_{\text{Twin}}(m)}{\Delta_{\text{Minor}}(m)} \to 1, \qquad \frac{L_{\text{Gap}}(m)}{\Delta_{\text{Major}}(m)} \to 1.$$

Proof. For large m, $u = 24m + O(1)$ is large, and the expansion $\sqrt{u^2 + 36} = u\sqrt{1 + 36/u^2} =$

$u + 18/u + O(u^{-3})$ holds. Thus:

$$G(u_2) - G(u_1) = \frac{1}{2}\Big[(u_2^2 - u_1^2) + 36 \ln\frac{u_2}{u_1} + O(u^{-1})\Big].$$

For the twin zone: $u_2^2 - u_1^2 = (u_2 + u_1)(u_2 - u_1) = (48m + 6) \cdot 8$, so that $L_{\text{Twin}} \approx \frac{1}{24}(48m + 6) \cdot 8 \cdot \frac{1}{2} = 8m + 1 = \Delta_{\text{Minor}}(m)$. The logarithmic and correction terms are $O(\ln m/m)$ and hence relatively negligible. The analogous argument applies to the gap zone. □

Remark 9.8 (Connection to the spiral construction). In Chapter 5 the differences Δk were interpreted as circle circumferences, which led to the Archimedean spiral. The asymptotic equivalence $L \approx \Delta k$ proved in this chapter provides a further justification for this identification: the differences are not only combinatorial quantities of the sequence, but simultaneously approximate the geometric length of the curve on the respective segments. The construction is thus geometrically consistent in a twofold sense — arithmetically through the congruence structure and analytically through the arc length.

Chapter 10

Taylor Expansion of the Arc Length

10.1 Taylor Expansion of $\sqrt{1+x}$

The Taylor series about $x = 0$ reads:

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{5x^4}{128} + \mathcal{O}(x^5). \tag{10.1}$$

Setting

$$x = \frac{3}{\pi^2 p}$$

we obtain:

$$\sqrt{1 + \frac{3}{\pi^2 p}} = 1 + \frac{3}{2\pi^2 p} - \frac{9}{8\pi^4 p^2} + \frac{27}{16\pi^6 p^3} + \mathcal{O}(p^{-4}). \tag{10.2}$$

10.2 Term-by-Term Integration

We integrate the individual terms:

1. $\displaystyle\int 1\, dp = p$

2. $\displaystyle\int \frac{3}{2\pi^2 p}\, dp = \frac{3}{2\pi^2} \ln(p)$

3. $\displaystyle\int -\frac{9}{8\pi^4 p^2}\, dp = \frac{9}{8\pi^4 p}$

4. $\displaystyle\int \frac{27}{16\pi^6 p^3}\, dp = -\frac{27}{32\pi^6 p^2}$

10.3 Approximation Formula

Theorem 10.1 (Taylor Approximation of the Arc Length)**.**

$$\boxed{L \approx (p_2 - p_1) + \frac{3}{2\pi^2} \ln\left(\frac{p_2}{p_1}\right) + \frac{9}{8\pi^4}\left(\frac{1}{p_1} - \frac{1}{p_2}\right)} \tag{10.3}$$

10.4 Coefficients — Numerical Values

$$\frac{3}{2\pi^2} = \frac{3}{2 \cdot 9.869604401089} = 0.151981775464, \tag{10.4}$$

$$\frac{9}{8\pi^4} = \frac{9}{8 \cdot 97.409091034002} = 0.011549230037. \tag{10.5}$$

The neglected remainder in the boxed expansion is of order $O(p_1^{-2})$.

10.5 Convergence Behaviour

For large p we have:

$$\frac{L}{p_2 - p_1} = 1 + \frac{3}{2\pi^2} \cdot \frac{\ln(p_2/p_1)}{p_2 - p_1} + \mathcal{O}\left(\frac{1}{p}\right).$$

Hence the arc length converges to the Euclidean distance:

$$\lim_{p\to\infty} [L(p, p + \Delta p) - \Delta p] = 0.$$

Part III

The Cone — The Three-Dimensional Envelope

Chapter 11

The Conical Form as the 3D Envelope of the Spiral

11.1 From the 2D Model to the Conical Form

The spiral investigated previously possesses a pronounced arithmetic structure: the points P_n lie on radii whose distances are determined by the quadratic sequence

$$k(n) = \frac{2n^2 + 3n + 1}{6}.$$

Particularly distinguished are the values

$$n \equiv 1, 5 \pmod 6,$$

for which the mean values $k(n)$ are integers. These two classes form an alternating ray system that repeats every six steps.

The observation that these rays exhibit a regular structure in the plane naturally leads to the question of whether there exists a three-dimensional form that geometrically bundles this structure. The answer is a cone: it forms an envelope in which the alternating rays appear as lines on the surface and the spiral itself can be interpreted as a space curve on this surface.

11.2 The 3D Form as a Cone of Revolution

The three-dimensional structure arises from a specific geometric process. The starting point is the two-dimensional term curve

$$r = r(p), \qquad \varphi = \varphi(p),$$

whose path we cut along the vertical axis. If this curve segment is erected perpendicular to the plane and then wound spirally about the z-axis along its length, the resulting three-dimensional surface has a straight generatrix — a *cone of revolution.*

This can be justified as follows:

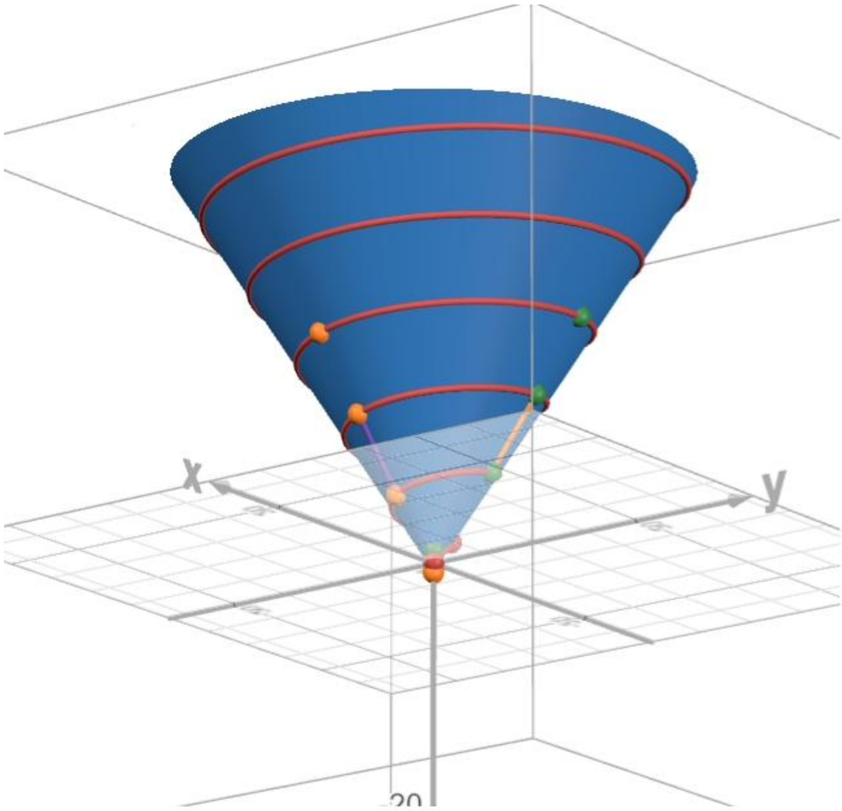


Figure 11.1: The conical helix: the red spiral curve winds upward along the blue cone surface $z = -\frac{3}{4} + \frac{\pi}{2}\,r$. The orange points mark the integer values $k(n)$ at $n \equiv 1 \pmod 6$, and the green points the values at $n \equiv 5 \pmod 6$. The apex of the cone is located at $P_0 = (0,\,0,\,-\frac{3}{4})$; the light triangle in the foreground shows the radial cross-section and illustrates the constant slope $\Delta z/\Delta r = \pi/2$ of the generatrix.

- The radial increments of the sequence are constant:

$$\Delta r = \frac{12}{\pi}.$$

  A surface on which equal radial increments lie along a fixed direction is necessarily a conical surface.

- The sequence possesses a periodic structure with

$$\Delta n = 6.$$

  This periodicity is translated into a constant slope along the height when the curve is wound up.

- When a planar curve with constant radial increment is wound up, the resulting surface has a straight generatrix. This is precisely the defining property of a cone.

Thus the three-dimensional form is a direct geometric consequence of winding up the 2D curve. The spiral that arises on this surface is therefore a *conical helix*, and the cone itself is the envelope of the original two-dimensional structure.

11.3 Radial Structure and 6-Periodicity

Between two rays of the same class (i.e. two values of n with $n \equiv 1 \pmod 6$ or two values with $n \equiv 5 \pmod 6$) there is always

$$\Delta n = 6.$$

The corresponding radial increment of the spiral is

$$\Delta r = \frac{12}{\pi}.$$

These two values form the basis of the cone geometry: they determine the slope of the cone and hence its opening angle.

11.4 Derivation of the Cone Slope

We consider the cross-section of the cone with a plane through the axis of revolution. This yields a right triangle with

$$\text{horizontal leg: } \Delta r = \frac{12}{\pi}, \qquad \text{vertical leg: } \Delta z = 6.$$

The slope of the cone is given by

$$\frac{\Delta z}{\Delta r} = \frac{6}{12/\pi} = \frac{\pi}{2}.$$

Thus the generatrix of the cone has the equation

$$z = \frac{\pi}{2} r + b$$

in the (r, z)-cross-section.

11.5 The Vertex of the Quadratic Sequence as the Cone Apex

The constant b follows directly from the structure of the quadratic sequence. The vertex of the parabola

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

is located at

$$n_0 = -\frac{3}{4}.$$

This value marks the origin of the construction: it is the point of minimal slope of the sequence and hence the geometric starting point of the cone projection.

We identify

$$z_0 = n_0 = -\frac{3}{4}, \qquad r_0 = 0.$$

The point $(0, -\frac{3}{4})$ must therefore lie on the cone. This identification is the single normalisation of the construction; the slope of the height function carries no such freedom (Proposition 13.5 in Chapter 13). Substituting into the cone equation yields

$$-\frac{3}{4} = \frac{\pi}{2} \cdot 0 + b,$$

hence

$$b = -\frac{3}{4}.$$

The final cone equation is therefore:

$$\boxed{z = -\frac{3}{4} + \frac{\pi}{2}\, r}$$

11.6 Cone Surface and Generatrix

The cone surface is the set of all points (r, φ, z) satisfying the linear relationship between radius and height:

$$z(r) = -\frac{3}{4} + \frac{\pi}{2}\, r.$$

Each line on the surface corresponds to a fixed angular direction φ. The alternating rays of the sequence lie exactly on such generatrices. Their 6-periodicity is encoded by the constant slope of the cone.

11.7 Verification: The Conical Helix Lies Entirely on the Cone Surface

Theorem 11.1 (Helix Lies on the Cone)**.** *The conical helix defined in Chapter 13,*

$$\gamma(\theta) = \left(\tfrac{6}{\pi^2}\theta\cos\theta,\ \tfrac{6}{\pi^2}\theta\sin\theta,\ -\tfrac{3}{4} + \tfrac{3}{\pi}\theta\right),$$

lies entirely on the cone surface $z = -\frac{3}{4} + \frac{\pi}{2}\, r$ for all $\theta \geq 0$.

Proof. We must show that the z-coordinate of the helix equals $-\frac{3}{4} + \frac{\pi}{2}\, r(\theta)$. We have:

$$r(\theta) = \frac{6}{\pi^2}\,\theta,$$

and therefore

$$-\frac{3}{4} + \frac{\pi}{2}\, r(\theta) = -\frac{3}{4} + \frac{\pi}{2} \cdot \frac{6}{\pi^2}\,\theta = -\frac{3}{4} + \frac{3}{\pi}\,\theta = z(\theta). \qquad \square$$

Remark 11.2. The identity $\frac{\pi}{2} \cdot \frac{6}{\pi^2} = \frac{3}{\pi}$ is the algebraic core of the entire 3D model: the cone slope $\frac{\pi}{2}$ and the spiral parameter $a = \frac{6}{\pi^2}$ are exactly matched so as to produce the height function $z = \frac{3}{\pi}\theta$. This relationship follows from the choice $\Delta z = 6$ and $\Delta r = \frac{12}{\pi}$, since

$$\frac{\Delta z}{\Delta r} = \frac{6}{12/\pi} = \frac{\pi}{2} \qquad \text{and} \qquad \frac{\pi}{2} \cdot a = \frac{\pi}{2} \cdot \frac{6}{\pi^2} = \frac{3}{\pi}.$$

In Chapter 16 this cone conformity is verified purely algebraically via the universal quantity $Q(k) = \sqrt{48k + 1}$ (Theorem 16.8).

Chapter 12

Geometry of the Cone Generators: Planes, Angles, and Growth

12.1 Overview

The integer points of the sequence $k(n)$ lie on discrete generators of the cone. Every two adjacent generators span a triangular face whose position in space is completely determined by the cone parameters. In this chapter we derive the plane equation, the tilt angle, the angles between the generators, and the growth rate of the radii. As a concrete example, the triangular area at $n = 19$ is computed.

12.2 Fundamentals: Generators of the Cone

The cone is defined by the equation

$$z = -\frac{3}{4} + \frac{\pi}{2} r$$

with apex at the point $P_0 = (0, 0, -\frac{3}{4})$. A generator at azimuthal angle φ runs from P_0 in the direction

$$\vec{v}(\varphi) = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ \pi/2 \end{pmatrix}.$$

The factor $\pi/2$ in the z-component follows directly from the cone slope $dz/dr = \pi/2$.

The length of this direction vector (per unit $\Delta r = 1$) is:

$$|\vec{v}| = \sqrt{1^2 + \left(\frac{\pi}{2}\right)^2} = \sqrt{1 + \frac{\pi^2}{4}} \approx 1.86209589.$$

Integer Points and Their Radii

At height $z = n$ the point lies on the generator at radius

$$r_n = \frac{z + \frac{3}{4}}{\pi/2} = \frac{2}{\pi}\left(n + \frac{3}{4}\right) = \frac{2n}{\pi} + \frac{3}{2\pi}.$$

The radius grows *linearly* in n at the rate

$$\frac{dr_n}{dn} = \frac{2}{\pi} \approx 0.63661977. \tag{12.1}$$

The following table shows the radii for the first integer indices $n \equiv 1, 5 \pmod 6$:

Table 12.1: Radii $r_n = \frac{2}{\pi}n + \frac{3}{2\pi}$ of the integer sequence values on the cone.

n	$r_n = \frac{2n}{\pi} + \frac{3}{2\pi}$	r_n (numerical)
1	$\frac{2}{\pi} + \frac{3}{2\pi} = \frac{7}{2\pi}$	1.11408460
5	$\frac{10}{\pi} + \frac{3}{2\pi} = \frac{23}{2\pi}$	3.66056369
7	$\frac{14}{\pi} + \frac{3}{2\pi} = \frac{31}{2\pi}$	4.93380324
11	$\frac{22}{\pi} + \frac{3}{2\pi} = \frac{47}{2\pi}$	7.48028233
13	$\frac{26}{\pi} + \frac{3}{2\pi} = \frac{55}{2\pi}$	8.75352187
17	$\frac{34}{\pi} + \frac{3}{2\pi} = \frac{71}{2\pi}$	11.30000096
19	$\frac{38}{\pi} + \frac{3}{2\pi} = \frac{79}{2\pi}$	12.57324050

Remark 12.1 (Arithmetic of the radius column). The table encodes three layers of arithmetic beyond the linear growth.

(i) *Integer circumferences.* Combining the two fractions gives $r_n = (4n+3)/(2\pi) = Q/(2\pi)$, hence

$$2\pi r_n = 4n + 3 = Q = \sqrt{48\,k(n) + 1}\,: \tag{12.2}$$

the circumference of the cone at each integer point is an integer, namely the universal quantity Q (Section 16.5.5). The numerators $7, 23, 31, 47, \ldots$ in the middle column are these circumferences. Modulo 12 they separate the two residue classes: $Q \equiv 7$ for $n \equiv 1 \pmod 6$ and $Q \equiv 11$ for $n \equiv 5 \pmod 6$ — the circumference reads off the class membership.

(ii) *OEIS anchors.* The index sequence $n = 1, 5, 7, 11, \ldots$ is A007310 in [SO24]; the numerators $2n = 2, 10, 14, 22, \ldots$ of the linear term $2n/\pi$ run through A091999, the numbers congruent to 2 or 10 modulo 12 — equivalently, the even numbers divisible by neither 3 nor 4. Doubling maps the classes $\{1, 5\}$ mod 6 bijectively onto $\{2, 10\}$ mod 12; the twin pattern of gaps $(4, 2)$ becomes $(8, 4)$. Each element $2n$ with $n > 1$ is the sum of exactly four consecutive integers, $2n = m + (m+1) + (m+2) + (m+3)$ with $m = (n-3)/2$, and the starting values again track the classes: $m \equiv 1 \pmod 3$ for $n \equiv 5 \pmod 6$ and $m \equiv 2 \pmod 3$ for $n \equiv 1 \pmod 6$.

(iii) *An infinite product evaluates the angular quantum.* With $a(j) = 6j - 3 + (-1)^j$ enumerating A091999, one has the identity [SO24]

$$\prod_{j=1}^{\infty}\left(1 + \frac{(-1)^j}{a(j)}\right) = 2\sin\frac{\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{2} = 0.51763809\ldots, \tag{12.3}$$

confirmed here numerically to seven digits with $2 \cdot 10^6$ factors. The angle $\pi/12$ is precisely the angle per Q-unit of the parametrisation, $d\theta = (\pi/12)\,dQ$ (Theorem 15.2), and by the reflection formula $\Gamma(\frac{1}{12})\,\Gamma(\frac{11}{12}) = \pi/\sin(\pi/12)$ the identity is the twelfth-argument sibling of Remark 6.4. The product evaluation itself is classical; the connection to the constants of the present construction is an observation of this work.

Remark 12.2 (Linear Growth Rate). The radial difference between two consecutive generators is constant:

$$\Delta r_{\text{Minor}} = \frac{2}{\pi} \cdot 2 = \frac{4}{\pi} \approx 1.27323954, \qquad \Delta r_{\text{Major}} = \frac{2}{\pi} \cdot 4 = \frac{8}{\pi} \approx 2.54647909.$$

This is a direct consequence of the linearity (12.1).

12.3 Angular Separations of the Generators in the xy-Plane

The azimuthal angles of the integer points follow from $\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}$. Consecutive points differ by:

$$\begin{aligned} \Delta\varphi_{\text{Minor}} &= \frac{\pi}{3} \cdot 2 = \frac{2\pi}{3} = 120^\circ, \\ \Delta\varphi_{\text{Major}} &= \frac{\pi}{3} \cdot 4 = \frac{4\pi}{3} = 240^\circ. \end{aligned}$$

Together they yield a complete revolution: $\Delta\varphi_{\text{Minor}} + \Delta\varphi_{\text{Major}} = 2\pi$.

12.4 Angle Between Two Generators in Space

Two generators with azimuthal difference $\Delta\varphi$ have the direction vectors

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ \pi/2 \end{pmatrix}, \qquad \vec{v}_2 = \begin{pmatrix} \cos\Delta\varphi \\ \sin\Delta\varphi \\ \pi/2 \end{pmatrix}.$$

The dot product reads:

$$\vec{v}_1 \cdot \vec{v}_2 = \cos\Delta\varphi + \left(\frac{\pi}{2}\right)^2 = \cos\Delta\varphi + \frac{\pi^2}{4}.$$

Since $|\vec{v}_1| = |\vec{v}_2| = \sqrt{1 + \pi^2/4}$, we obtain:

$$\cos\theta = \frac{\cos\Delta\varphi + \pi^2/4}{1 + \pi^2/4}.$$

For minor and major steps, $\cos(120^\circ) = \cos(240^\circ) = -\frac{1}{2}$, hence:

$$\cos\theta = \frac{-\frac{1}{2} + \frac{\pi^2}{4}}{1 + \frac{\pi^2}{4}} = \frac{\pi^2 - 2}{\pi^2 + 4} \approx 0.56739934.$$

Theorem 12.3 (Angle Between the Generators)**.** *The spatial angle between two consecutive generators (both minor and major steps) is:*

$$\theta = \arccos\left(\frac{\pi^2 - 2}{\pi^2 + 4}\right) \approx 55.43092771^\circ. \tag{12.4}$$

Remark 12.4. Although the azimuthal angles 120° and 240° are different, the spatial angle between the generators is the same in both cases. The reason: $\cos(120^\circ) = \cos(240^\circ) = -\frac{1}{2}$, so that the z-component $\pi/2$ has the same effect in both cases, and the different opening angles in the plane lead to the same spatial angle.

12.5 Plane Equation and Tilt Angle of the Triangular Face

Two generators span a triangular face. Its normal vector is obtained via the cross product:

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 1 & 0 & \pi/2 \\ \cos\Delta\varphi & \sin\Delta\varphi & \pi/2 \end{vmatrix}.$$

Expanding yields:

$$\begin{aligned} n_x &= 0 \cdot \tfrac{\pi}{2} - \tfrac{\pi}{2} \cdot \sin\Delta\varphi = -\frac{\pi}{2}\sin\Delta\varphi, \\ n_y &= \tfrac{\pi}{2} \cdot \cos\Delta\varphi - 1 \cdot \tfrac{\pi}{2} = \frac{\pi}{2}(\cos\Delta\varphi - 1), \\ n_z &= 1 \cdot \sin\Delta\varphi - 0 \cdot \cos\Delta\varphi = \sin\Delta\varphi. \end{aligned}$$

For $\Delta\varphi = 120^\circ$ ($\sin 120^\circ = \frac{\sqrt{3}}{2}$, $\cos 120^\circ = -\frac{1}{2}$):

$$\vec{n} = \begin{pmatrix} -\dfrac{\pi\sqrt{3}}{4} \\ -\dfrac{3\pi}{4} \\ \dfrac{\sqrt{3}}{2} \end{pmatrix} \approx \begin{pmatrix} -1.36034952 \\ -2.35619449 \\ 0.86602540 \end{pmatrix}, \qquad |\vec{n}| \approx 2.85520635.$$

The tilt angle of the triangular face relative to the xy-plane is the angle between $\vec{n}$ and the z-axis:

$$\gamma = \arccos\left(\frac{|n_z|}{|\vec{n}|}\right) = \arccos\left(\frac{\sqrt{3}/2}{|\vec{n}|}\right) \approx \arccos(0.30331447).$$

Theorem 12.5 (Tilt Angle of the Triangular Face)**.**

$$\gamma = \arccos\left(\frac{|n_z|}{|\vec{n}|}\right) \approx 72.34321285^\circ. \tag{12.5}$$

The triangular face is therefore nearly perpendicular to the xy-plane, consistent with the image of a steeply rising cone.

12.6 Area of the Triangle at $n = 19$

As a concrete example we compute the triangular area spanned by the two generators of the twin zone at $n = 19$.

Step 1 — Determine the Vertices

The radius at $z = 19$:

$$r_{19} = \frac{2 \cdot 19}{\pi} + \frac{3}{2\pi} = \frac{79}{2\pi} \approx 12.57324050.$$

The three vertices of the triangle (apex A, and two points B, C on the cone surface with $\Delta\varphi = 120°$):

$$\begin{aligned}
A &= \left(0,\ 0,\ -\tfrac{3}{4}\right), \\
B &= (r_{19},\ 0,\ 19) \approx (12.573241,\ 0,\ 19), \\
C &= (r_{19}\cos 120°,\ r_{19}\sin 120°,\ 19) \approx (-6.286621,\ 10.888746,\ 19).
\end{aligned}$$

Step 2 — Determine the Side Lengths

Sides $|AB|$ and $|AC|$ (generators):

Since both generators lead to the same height $z = 19$, $|AB| = |AC|$. The generator length is computed as:

$$|AB| = r_{19} \cdot |\vec{v}| = \frac{79}{2\pi} \cdot \sqrt{1 + \frac{\pi^2}{4}} \approx 12.573241 \cdot 1.862096 \approx 23.41257946.$$

Base $|BC|$ (chord at radius r_{19} and angle $120°$):

$$|BC| = 2\,r_{19} \sin\!\left(\frac{120°}{2}\right) = 2\,r_{19} \sin 60° = 2\,r_{19} \cdot \frac{\sqrt{3}}{2} = r_{19}\sqrt{3} = \frac{79\sqrt{3}}{2\pi} \approx 21.77749137.$$

Step 3 — Area via Cross Product

The edge vectors from point A:

$$\overrightarrow{AB} = B - A = \begin{pmatrix} r_{19} \\ 0 \\ 19 + \frac{3}{4} \end{pmatrix} = \begin{pmatrix} 79/(2\pi) \\ 0 \\ 79/4 \end{pmatrix}, \qquad \overrightarrow{AC} = C - A = \begin{pmatrix} -r_{19}/2 \\ r_{19}\sqrt{3}/2 \\ 79/4 \end{pmatrix}.$$

The cross product:

$$\overrightarrow{AB} \times \overrightarrow{AC} \approx (-215.052727,\ -372.482250,\ 136.906818), \qquad \left|\overrightarrow{AB} \times \overrightarrow{AC}\right| \approx 451.36922682.$$

The area of the triangle:

$$F = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| \approx \frac{451.369}{2}.$$

Theorem 12.6 (Triangular Area at $n = 19$)**.**

$$\boxed{F_{19} = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| \approx 225.68461341} \tag{12.6}$$

Verification via Heron's Formula

With side lengths $a = |BC| \approx 21.77749137$, $b = |AC| \approx 23.41257946$, $c = |AB| \approx 23.41257946$ and $s = \frac{a+b+c}{2} \approx 34.30132515$:

$$F = \sqrt{s(s-a)(s-b)(s-c)} \approx \sqrt{34.301 \cdot 12.524 \cdot 10.889 \cdot 10.889} \approx 225.68461341 \quad \checkmark$$

12.7 Summary of All Angles and Dimensions

Table 12.2: Geometric characteristics of the cone generators.[1]

Quantity	Exact Form	Numerical
Cone slope	$dz/dr = \pi/2$	1.57079633
Generator length per $\Delta r = 1$	$\sqrt{1+\pi^2/4}$	1.86209589
Growth rate dr/dn	$2/\pi$	0.63661977
Δr minor step	$4/\pi$	1.27323954
Δr major step	$8/\pi$	2.54647909
Azimuthal angle minor	$2\pi/3$	120°
Azimuthal angle major	$4\pi/3$	240°
Angle between generators (both)	$\arccos\left(\frac{\pi^2-2}{\pi^2+4}\right)$	55.43092771°
Tilt angle of triangular face	$\arccos\left(\frac{\lvert n_z\rvert}{\lvert\vec{n}\rvert}\right)$	72.34321285°
Radius at $n = 19$	$79/(2\pi)$	12.57324050
Generator length at $n = 19$	$\frac{79}{2\pi}\sqrt{1+\pi^2/4}$	23.41257946
Base side $\lvert BC\rvert$ at $n = 19$	$\frac{79\sqrt{3}}{2\pi}$	21.77749137
Triangular area at $n = 19$	$\frac{1}{2}\lvert\vec{AB} \times \vec{AC}\rvert$	225.68461341
Half-opening angle α (to the axis)	$\arctan(2/\pi)$	32.48163659°
Inclination angle β (to the base)	$\arctan(\pi/2)$	57.51836341°
Development angle φ (unrolled mantle)	$4\pi/\sqrt{\pi^2+4}$	193.33053797°

[1] Two angles complete the description of the cone. The inclination angle $\beta = 90° - \alpha = \arctan(\pi/2)$ is the angular form of the cone slope $dz/dr = \pi/2$. Unrolling the lateral surface along a generatrix produces a plane sector whose central angle — the development angle — is $\varphi = 2\pi \sin\alpha = 4\pi/\sqrt{\pi^2+4} = 3.37425443$ rad $= 193.33053797°$, independent of the height, as required for a cone. Two exact consequences of $\tan\beta = \pi/2$: at every lattice point the height above the apex is $h = z + \frac{3}{4} = Q/4$, while the circumference is $2\pi r = Q$ (Remark 12.1). The apex height is therefore exactly one quarter of the circumference — for $n = 19$: circumference 79, apex height 79/4 — and the unrolled net of the cone is a sector of 193.33° whose arc through the lattice point n has the integer length Q.

Part IV

The Conical Helix — Complete Curve Analysis

Chapter 13

The Conical Helix

13.1 The Helix as a Trace on the Cone Surface

When the original spiral is coupled with the linear height function $z = n$, a space curve arises that winds along the cone surface. This curve is a *conical helix*: it combines the radial structure of the spiral with the vertical structure of the sequence and constitutes the three-dimensional shape of the construction.

13.2 Complete Helix Parametrisation

Theorem 13.1 (Conical Helix)**.**

$$\boxed{\gamma(\theta) = \begin{pmatrix} \frac{6}{\pi^2}\theta\cos(\theta) \\ \frac{6}{\pi^2}\theta\sin(\theta) \\ -\frac{3}{4} + \frac{3}{\pi}\theta \end{pmatrix}} \tag{13.1}$$

13.3 Derivation of the Velocity and the Arc-Length Integral

We consider the conical helix

$$\gamma(t) = \left(\frac{6}{\pi^2}\,t\cos t,\ \frac{6}{\pi^2}\,t\sin t,\ -\frac{3}{4} + \frac{3}{\pi}\,t\right),$$

which consists of a linear height function and a radially growing circular component. To determine the arc length of this curve, we first require the speed $\|\gamma'(t)\|$.

1. Differentiation of the Components

We differentiate each component separately. For the first two components we apply the product rule, since each is a product of t and a trigonometric function.

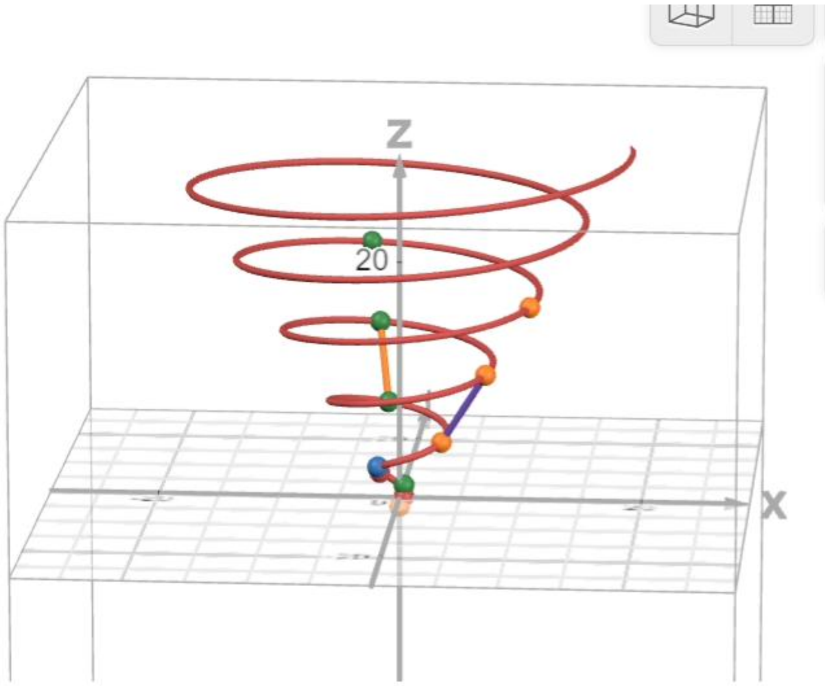


Figure 13.1: The conical helix $\gamma(\theta)$ in a perspective view along the z-axis. The red curve winds upward with increasing radius. The green points mark the integer height values $z = n$ with $n \equiv 5 \pmod 6$, and the orange points the values with $n \equiv 1 \pmod 6$. The orange and blue vectors in the lower region illustrate the alternating angular spacings $\Delta\theta_{\text{major}} = 4\pi/3$ and $\Delta\theta_{\text{minor}} = 2\pi/3$ between consecutive integer points.

First component.

$$x(t) = \frac{6}{\pi^2}\, t\cos t \quad \Rightarrow \quad x'(t) = \frac{6}{\pi^2}(\cos t - t\sin t).$$

Second component.

$$y(t) = \frac{6}{\pi^2}\, t\sin t \quad \Rightarrow \quad y'(t) = \frac{6}{\pi^2}(\sin t + t\cos t).$$

Third component. The height function is linear, so:

$$z(t) = -\frac{3}{4} + \frac{3}{\pi}t \quad \Rightarrow \quad z'(t) = \frac{3}{\pi}.$$

2. Speed of the Curve

The speed is defined as

$$\|\gamma'(t)\| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}.$$

Substituting the derivatives computed above:

$$\|\gamma'(t)\|^2 = \left(\frac{6}{\pi^2}\right)^2 (\cos t - t\sin t)^2 + \left(\frac{6}{\pi^2}\right)^2 (\sin t + t\cos t)^2 + \left(\frac{3}{\pi}\right)^2.$$

3. Simplification of the Trigonometric Expression

The two squared terms can be simplified by expanding:

$$(\cos t - t\sin t)^2 = \cos^2 t - 2t\cos t\sin t + t^2\sin^2 t,$$

$$(\sin t + t\cos t)^2 = \sin^2 t + 2t\sin t\cos t + t^2\cos^2 t.$$

Adding both expressions, the mixed terms cancel:

$$-2t\cos t\sin t + 2t\sin t\cos t = 0.$$

There remains:

$$(\cos t - t\sin t)^2 + (\sin t + t\cos t)^2 = \cos^2 t + \sin^2 t + t^2(\sin^2 t + \cos^2 t).$$

Since $\cos^2 t + \sin^2 t = 1$ always holds, we obtain:

$$(\cos t - t\sin t)^2 + (\sin t + t\cos t)^2 = 1 + t^2.$$

4. Final Form of the Speed

It follows that:

$$\|\gamma'(t)\|^2 = \frac{36}{\pi^4}(1 + t^2) + \frac{9}{\pi^2}.$$

Factoring out $\frac{9}{\pi^4}$ and taking the square root yields the compact form:

$$\boxed{\|\gamma'(t)\| = \frac{3}{\pi^2}\sqrt{4(1 + t^2) + \pi^2}}.$$

This representation is particularly well suited for computing the arc length.

5. Arc-Length Integral

The arc length between two parameter values t_1 and t_2 is therefore:

$$\boxed{L(t_1, t_2) = \int_{t_1}^{t_2} \frac{3}{\pi^2} \sqrt{4(1+t^2)+\pi^2}\, dt.}$$

This integral will be evaluated numerically for specific intervals in what follows.

13.4 Arc Length of the Helix in the p-Parametrisation

It is important to emphasise that the formula derived in Chapter 7

$$L_{\text{Spiral}} = \int_{p_1}^{p_2} \frac{1}{\pi} \sqrt{\frac{3}{p} + \pi^2}\, dp \tag{13.2}$$

computes the arc length of the *two-dimensional spiral* in the p-parametrisation. The arc length of the three-dimensional helix (with the θ-parametrisation above) is a *different quantity*, since it includes the additional vertical component $z'(\theta) = 3/\pi$. The numerical differences between the two arc lengths observed in Chapter 15 arise because they are arc lengths of *different curves* (the planar spiral on the one hand, the space curve on the other), not from an error in the parametrisation.

13.5 Arc Length Between Integer n-Values

For integer n we have:

$$\theta(n) = \frac{\pi n}{3} + \frac{\pi}{4}. \tag{13.3}$$

The arc length along the helix is:

$$L = \int_{\theta(n_1)}^{\theta(n_2)} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 + \left(\frac{dz}{d\theta}\right)^2}\, d\theta. \tag{13.4}$$

13.6 Example Table

Table 13.1: Arc lengths between consecutive n-values

$n_1 \to n_2$	Δn	Type	$\Delta\theta$	Arc length
$1 \to 5$	4	Major	$\frac{4\pi}{3}$	11.15519925
$5 \to 7$	2	Minor	$\frac{2\pi}{3}$	9.30913741
$7 \to 11$	4	Major	$\frac{4\pi}{3}$	26.43477180
$11 \to 13$	**2**	**Minor**	$\frac{2\pi}{3}$	**17.16486194**
$13 \to 17$	4	Major	$\frac{4\pi}{3}$	42.26825081
$17 \to 19$	2	Minor	$\frac{2\pi}{3}$	25.11227648

Table 13.2: Parameter values θ_n, reduced angles, and radii of the conical helix. All radii were computed exactly as $r_n = (6/\pi^2)\,\theta_n$ and rounded to eight decimal places.

n	θ_n exact	θ_n numerical (rad)	Angle mod 360°	Radius $r_n = \frac{6}{\pi^2}\,\theta_n$
5	$\frac{23\pi}{12}$	6.02138592	345°	3.66056369
7	$\frac{31\pi}{12}$	8.11578102	105°	4.93380324
11	$\frac{47\pi}{12}$	12.30457123	345°	7.48028233
13	$\frac{55\pi}{12}$	14.39896633	105°	8.75352187
17	$\frac{71\pi}{12}$	18.58775653	345°	11.30000096
19	$\frac{79\pi}{12}$	20.68215164	105°	12.57324050
23	$\frac{95\pi}{12}$	24.87094184	345°	15.11971959
25	$\frac{103\pi}{12}$	26.96533694	105°	16.39295914
29	$\frac{119\pi}{12}$	31.15412715	345°	18.93943823
31	$\frac{127\pi}{12}$	33.24852225	105°	20.21267777

13.7 Full Revolution of the Helix

A full revolution about the z-axis corresponds to $\Delta\theta = 2\pi$.

$$L_{360^\circ} = \int_0^{2\pi} \sqrt{\left(\frac{6}{\pi^2}\right)^2 (1+\theta^2) + \left(\frac{3}{\pi}\right)^2}\, d\theta = \boxed{14.55066317457281468 63}. \tag{13.5}$$

13.7.1 Height Gain per Revolution

$$\Delta z = \frac{3}{\pi} \cdot 2\pi = 6. \tag{13.6}$$

The helix thus rises by exactly 6 units per revolution.

13.8 Geometric Properties of the Helix

13.8.1 Tangent Vector

$$\gamma'(\theta) = \begin{pmatrix} \frac{6}{\pi^2}(\cos\theta - \theta\sin\theta) \\ \frac{6}{\pi^2}(\sin\theta + \theta\cos\theta) \\ \frac{3}{\pi} \end{pmatrix} \tag{13.7}$$

13.8.2 Curvature

Remark 13.2 (Planar Spiral Curvature as an Approximation). The curvature of a space curve $\gamma(t)$ is defined as $\kappa = |\gamma' \times \gamma''| \,/\, |\gamma'|^3$ [Car76; Str88] and depends, in the case of the conical helix, on the height component $z'(\theta) = 3/\pi$. As a tractable approximation one may use the formula for the planar Archimedean spiral, which for large θ converges rapidly to the exact 3D curvature (deviation below 2 % for $\theta \gtrsim 11$; at $\theta = 2$ it is still 47 %):

$$\gamma(\theta) = \big(a\theta\cos\theta,\ a\theta\sin\theta,\ c_0 + c_1\theta\big), \qquad a = 6/\pi^2, \quad c_1 = 3/\pi,$$

$$\kappa(\theta) = \frac{\theta^2 + 2}{a\,(\theta^2+1)^{3/2}}, \qquad a = \frac{6}{\pi^2}. \tag{13.8}$$

Sample values at the first integer points (each at $\theta = \theta_n$):

$$\kappa(\theta_{11}) = \kappa\big(\tfrac{47\pi}{12}\big) \approx 0.134120, \tag{13.9}$$

$$\kappa(\theta_{13}) = \kappa\big(\tfrac{55\pi}{12}\big) \approx 0.114512. \tag{13.10}$$

The curvature decreases monotonically with increasing θ — the helix becomes progressively straighter along its course.

Theorem 13.3 (Exact 3D Curvature and Torsion)**.** *For the conical helix the exact formulae*

$$\kappa_{3\mathrm{D}}(\theta) = \frac{a\sqrt{c_1^2\,(\theta^2+4) + a^2\,(\theta^2+2)^2}}{\big(a^2(1+\theta^2)+c_1^2\big)^{3/2}}, \tag{13.11}$$

$$\tau(\theta) = \frac{c_1\,(\theta^2+6)}{c_1^2\,(\theta^2+4) + a^2\,(\theta^2+2)^2} \tag{13.12}$$

hold.

Proof. The derivatives of the parametrisation are

$$\begin{aligned}
\gamma' &= \big(a(\cos\theta - \theta\sin\theta),\ a(\sin\theta + \theta\cos\theta),\ c_1\big),\\
\gamma'' &= \big(-a(2\sin\theta + \theta\cos\theta),\ a(2\cos\theta - \theta\sin\theta),\ 0\big),\\
\gamma''' &= \big(a(-3\cos\theta + \theta\sin\theta),\ a(-3\sin\theta - \theta\cos\theta),\ 0\big).
\end{aligned}$$

By direct computation of the cross product we obtain

$$\gamma' \times \gamma'' = \big(-ac_1(2\cos\theta - \theta\sin\theta),\ -ac_1(2\sin\theta + \theta\cos\theta),\ a^2(\theta^2+2)\big)$$

with $|\gamma'\times\gamma''|^2 = a^2\big[c_1^2(\theta^2+4) + a^2(\theta^2+2)^2\big]$ and $|\gamma'|^2 = a^2(1+\theta^2)+c_1^2$. The curvature follows from $\kappa = |\gamma'\times\gamma''|/|\gamma'|^3$.

For the torsion, the scalar triple product gives $(\gamma'\times\gamma'')\cdot\gamma''' = a^2c_1(\theta^2+6)$, whence $\tau = (\gamma'\times\gamma'')\cdot\gamma'''\,/\,|\gamma'\times\gamma''|^2$. ☐

Remark 13.4. For $\theta\to\infty$ we have asymptotically

$$\kappa_{3\mathrm{D}}(\theta) \sim \frac{1}{a\theta} = \frac{1}{r(\theta)}, \qquad \tau(\theta) \sim \frac{c_1}{a^2\theta^2} = \frac{\pi^3}{12\,\theta^2},$$

$$\text{i.e.}\quad \lim_{\theta\to\infty}\theta^2\,\tau(\theta) = \frac{c_1}{a^2} = \frac{\pi^3}{12} = 2.583856\ldots$$

The torsion is positive for every θ — in contrast to the planar spiral, whose torsion vanishes identically — but decays quadratically to zero, while the curvature approaches that of a circle of the current radius $r(\theta)$: locally the helix straightens its screw component faster than its bending. Numerically, $\theta^2\tau(\theta) = 2.5692\ldots$ at $\theta = 10$ and $2.583856\ldots$ at $\theta = 10^4$.

At the integer points $\theta_n = \pi n/3 + \pi/4 = \pi Q/12$ with $Q = 4n+3$ (Section 16.5.5), the identity $Q^2 = 48k(n) + 1$ turns the asymptotics into an arithmetic statement:

$$\tau(\theta_n) \;\approx\; \frac{12\pi}{Q^2} \;=\; \frac{12\pi}{48\,k(n)+1}, \tag{13.13}$$

with relative error $1.9 \cdot 10^{-4}$ at $n = 47$ and $4.5 \cdot 10^{-5}$ at $n = 97$. The torsion at a lattice point is thus asymptotically the reciprocal of the same quantity $48k+1$ whose square root Q parametrises the construction (Section 16.5.5); the differential-geometric screw rate and the arithmetic of the sequence are read off from one number. Note that τ is a smooth, monotone function of θ: it does not distinguish the two residue classes, nor primes from composites — that information lies in the occupation of the points, not in the geometry of the curve.

13.9 Integer Points on the Helix

Solving $z(\theta_n) = n$:

$$-\frac{3}{4} + \frac{3}{\pi}\theta_n = n, \tag{13.14}$$

$$\theta_n = \frac{\pi}{3}\left(n + \frac{3}{4}\right), \tag{13.15}$$

$$\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}. \tag{13.16}$$

Thus all $n \equiv 1, 5 \pmod 6$ occur as discrete points on the helix.

Proposition 13.5 (Determination of the height function)**.** *Let $z(\theta) = b\theta + c$ be an affine height function. The requirement of one period per revolution, $z(\theta + 2\pi) = z(\theta) + 6$, forces $b = 6/(2\pi) = 3/\pi$. The remaining constant is fixed by the vertex–apex identification $z(0) = n_0 = -\frac{3}{4}$ of Chapter 11, giving $c = -\frac{3}{4}$. No further normalisation enters: slope and offset of the height function are determined by one structural requirement and one explicit identification, and with these values the integer heights are attained exactly at $\theta_n = \pi n/3 + \pi/4$.*

The complete analysis of these integer points — including the prime number corollary — follows in Chapter 14.

Chapter 14

Prime Numbers on the Helix

This chapter shows how the conical helix provides a geometric embedding of the congruence classes $n \equiv 1, 5 \pmod 6$. Since all primes $p > 3$ lie in precisely these classes (Theorem 2.4), they appear as a distinguished subset of the integer helix points.

14.1 Integer Points on the Helix

We begin by determining at which angles θ the helix intersects integer heights $z = n$.

Theorem 14.1 (Integer Points)**.** *The helix*

$$\gamma(\theta) = \begin{pmatrix} \frac{6}{\pi^2}\theta\cos\theta \\ \frac{6}{\pi^2}\theta\sin\theta \\ -\frac{3}{4} + \frac{3}{\pi}\theta \end{pmatrix}$$

intersects the plane $z = n$ at the angle

$$\boxed{\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}} \tag{14.1}$$

for every $n \in \mathbb{Z}$; the intersection point carries an integer sequence value $k(n)$ precisely when $n \equiv 1, 5 \pmod 6$.

Proof. We solve the equation $z(\theta_n) = n$:

$$-\frac{3}{4} + \frac{3}{\pi}\theta_n = n \tag{14.2}$$

$$\frac{3}{\pi}\theta_n = n + \frac{3}{4} \tag{14.3}$$

$$\theta_n = \frac{\pi}{3}\left(n + \frac{3}{4}\right) \tag{14.4}$$

$$\theta_n = \frac{\pi}{3}n + \frac{\pi}{4} \tag{14.5}$$

This formula holds for all $n \in \mathbb{R}$. However, the points on the helix correspond to integer values of the original sequence $k(n)$ only when $n \equiv 1, 5 \pmod 6$. $\square$

Example 14.2 (First Integer Points)**.** The first integer points are:

$$
\begin{aligned}
n = 1 : \theta_1 &= \frac{\pi}{3} + \frac{\pi}{4} = \frac{7\pi}{12} \approx 1.833 \\
n = 5 : \theta_5 &= \frac{5\pi}{3} + \frac{\pi}{4} = \frac{23\pi}{12} \approx 6.021 \\
n = 7 : \theta_7 &= \frac{7\pi}{3} + \frac{\pi}{4} = \frac{31\pi}{12} \approx 8.117 \\
n = 11 : \theta_{11} &= \frac{11\pi}{3} + \frac{\pi}{4} = \frac{47\pi}{12} \approx 12.305 \\
n = 13 : \theta_{13} &= \frac{13\pi}{3} + \frac{\pi}{4} = \frac{55\pi}{12} \approx 14.398
\end{aligned}
$$

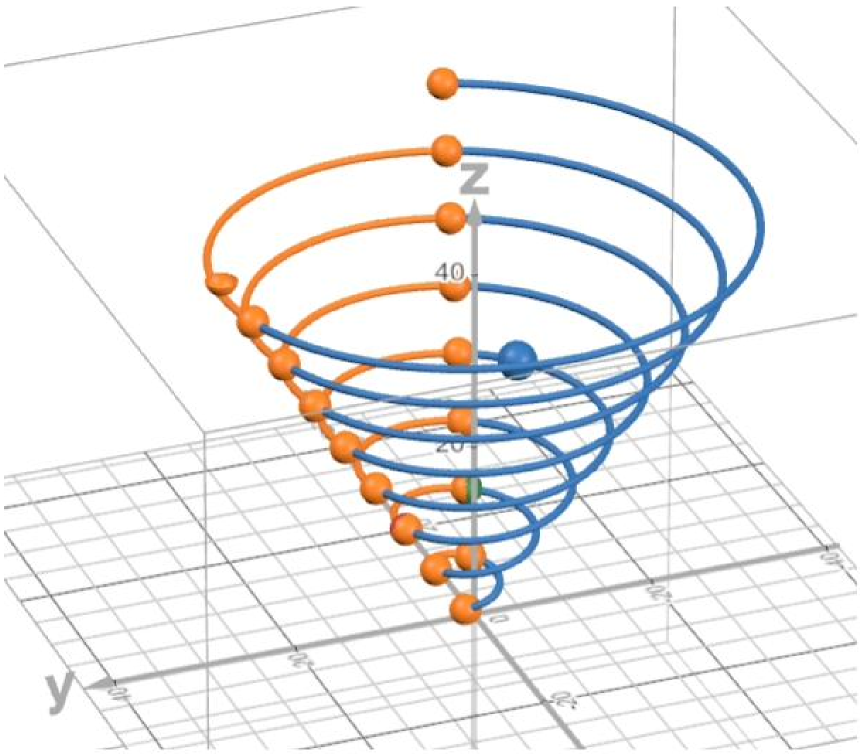


Figure 14.1: Two superimposed conical helices: the blue curve connects the integer points with $n \equiv 1 \pmod 6$ (major steps, $\Delta\theta = 4\pi/3$), and the orange curve connects the points with $n \equiv 5 \pmod 6$ (minor steps, $\Delta\theta = 2\pi/3$). The orange spheres mark all integer values $k(n)$ on the helix; both primes and composite numbers of the sequence are represented. The differing winding densities of the two curves reflect the alternating gap–twin pattern of the sequence.

14.2 The Angular Spacing Pattern

Proposition 14.3 (Alternating Pattern). *The angular spacings between consecutive integer points follow an alternating pattern:*

$$\Delta\theta_{major} = \theta_{n+4} - \theta_n = \frac{4\pi}{3} \approx 4.189 \quad \textit{(for } n \equiv 1 \pmod 6\textit{)} \tag{14.6}$$

$$\Delta\theta_{minor} = \theta_{n+2} - \theta_n = \frac{2\pi}{3} \approx 2.094 \quad \textit{(for } n \equiv 5 \pmod 6\textit{)} \tag{14.7}$$

Proof. Direct computation:

$$\theta_{n+4} - \theta_n = \left(\frac{\pi}{3}(n+4) + \frac{\pi}{4}\right) - \left(\frac{\pi}{3}n + \frac{\pi}{4}\right) = \frac{4\pi}{3} \tag{14.8}$$

$$\theta_{n+2} - \theta_n = \left(\frac{\pi}{3}(n+2) + \frac{\pi}{4}\right) - \left(\frac{\pi}{3}n + \frac{\pi}{4}\right) = \frac{2\pi}{3} \tag{14.9}$$

□

Remark 14.4. This alternating pattern of $\frac{4\pi}{3}$ and $\frac{2\pi}{3}$ corresponds geometrically to the alternating "major steps" and "minor steps" in the original difference sequence.

14.3 Geometric Corollary: All Primes on the Helix

We now formulate the central geometric corollary of the preceding construction.

Theorem 14.5 (Primes as a Subset of the Helix). *All primes $p > 3$ lie as discrete points on the conical helix at angles*

$$\boxed{\theta_p = \frac{\pi}{3}p + \frac{\pi}{4}} \tag{14.10}$$

Proof. The proof is a direct corollary of two known results:

Step 1 (Classical number theory, Theorem 2.4). All primes $p > 3$ satisfy $p \equiv 1$ or $5 \pmod 6$.

Step 2 (Geometric construction, Theorem 14.1). The helix intersects the plane $z = n$ at the angle $\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}$ for all $n \equiv 1, 5 \pmod 6$.

Conclusion. Since every prime $p > 3$ satisfies the condition from Step 1, and the helix by Step 2 attains integer heights precisely at these congruence classes, every prime $p > 3$ lies on the helix at the angle $\theta_p = \frac{\pi}{3}p + \frac{\pi}{4}$. □

Important caveat. The converse does not hold: not every integer point on the helix corresponds to a prime (Section 14.5); the helix is a geometric carrier of the primes, not a generator.

Why is this of interest? The statement that all primes exceeding 3 fall into the classes $6k \pm 1$ holds, of course, in any representation of the residue classes — a number line with two colours suffices. What distinguishes the present carrier is that it organises radius, height, angle and circumference *simultaneously* through the single integer $Q = 4n + 3$: at every lattice point, class membership ($Q \equiv 7$ or $11 \bmod 12$), the circumference (Q itself), the height above the apex ($Q/4$) and the angular position ($\theta = \pi Q/12$) are read off from one number (Remark 12.1, Section 16.5.5).

Example 14.6 (First Primes on the Helix)**.** The first primes greater than 3 and their positions:

$$\begin{aligned}
p = 5 : \theta_5 &= \frac{5\pi}{3} + \frac{\pi}{4} = \frac{23\pi}{12} \approx 6.021 \\
p = 7 : \theta_7 &= \frac{7\pi}{3} + \frac{\pi}{4} = \frac{31\pi}{12} \approx 8.117 \\
p = 11 : \theta_{11} &= \frac{11\pi}{3} + \frac{\pi}{4} = \frac{47\pi}{12} \approx 12.305 \\
p = 13 : \theta_{13} &= \frac{13\pi}{3} + \frac{\pi}{4} = \frac{55\pi}{12} \approx 14.398 \\
p = 17 : \theta_{17} &= \frac{17\pi}{3} + \frac{\pi}{4} = \frac{71\pi}{12} \approx 18.587 \\
p = 19 : \theta_{19} &= \frac{19\pi}{3} + \frac{\pi}{4} = \frac{79\pi}{12} \approx 20.682 \\
p = 23 : \theta_{23} &= \frac{23\pi}{3} + \frac{\pi}{4} = \frac{95\pi}{12} \approx 24.870
\end{aligned}$$

14.4 Geometric Interpretation

Corollary 14.7 (Prime Distribution on the Helix)**.** *The primes greater than 3 form a discrete subset of the helix whose spacings are determined by the prime gaps in the congruence classes* $1, 5 \pmod 6$.

Remark 14.8 (Relation to Prime Gaps). If p and q are consecutive primes with $p, q > 3$, the arc length along the helix between these two points depends on the prime gap $q - p$.

For twin primes ($q = p + 2$):

$$\Delta\theta = \theta_q - \theta_p = \frac{2\pi}{3}$$

For cousin primes ($q = p + 4$):

$$\Delta\theta = \theta_q - \theta_p = \frac{4\pi}{3}$$

For a general prime gap $q - p = g$ the angular distance is $\Delta\theta = g\,\pi/3$; equal angular distance does not mean equal arc length, since the arc element grows with the radius.

14.5 Composite Numbers on the Helix

The helix does not contain only primes. Since it geometrically embeds all indices of the congruence classes 1 and 5 modulo 6, all composite numbers belonging to these classes also lie on the curve.

Example 14.9 (First Composite Number on the Helix)**.** The first composite number on the helix is $n = 25 = 5^2$:

$$\theta_{25} = \frac{25\pi}{3} + \frac{\pi}{4} = \frac{103\pi}{12} \approx 26.964$$

Remark 14.10 (Complete Characterisation). The helix contains exactly the set $\{n \in \mathbb{Z}_{>0} : n \equiv 1, 5 \pmod 6\}$. The primes $p > 3$ form a distinguished subset therein — infinite, of relative density zero (Proposition 14.11), and not separated from the composites by the geometry. The numbers 2, 3, and all n with $n \equiv 0, 2, 3, 4 \pmod 6$ do not lie on the helix.

14.6 Asymptotic Density

Proposition 14.11 (Density of Primes on the Helix)**.** *By the prime number theorem, the density of primes among the integer points of the helix tends asymptotically to zero:*

$$\lim_{N\to\infty} \frac{\#\{p \leq N : p \text{ prime}, p > 3\}}{\#\{n \leq N : n \equiv 1, 5 \pmod 6\}} = 0$$

Proof. The numerator grows as $\sim \frac{N}{\ln N}$ (prime number theorem), while the denominator grows as $\sim \frac{N}{3}$ (since exactly $\frac{1}{3}$ of all integers satisfy $n \equiv 1, 5 \pmod 6$). Hence:

$$\frac{N/\ln N}{N/3} = \frac{3}{\ln N} \to 0 \quad \text{as } N \to \infty$$

□

14.7 Visualisation

Remark 14.12 (Comparison with Other Prime Spirals). In contrast to the Ulam spiral or the Sacks spiral, which represent empirically discovered patterns, our helix is *derived* from the arithmetic properties of a specific quadratic sequence. The appearance of all primes > 3 is a *theorem*, not an observation.

14.8 Open Questions

1. Can the helix structure yield insights into prime gaps or the distribution of twin primes?
2. Is there a natural extension to other congruence-based sequences?

Chapter 15

Arc-Length Comparison: Parabola vs. Helix

15.1 Principle of the Comparison

The quadratic curve $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ was geometrically wound onto a cone, where it forms the helix $\gamma(\theta)$. The central question is: how closely do the arc lengths agree?

Arc length of the parabola between $n = n_1$ and $n = n_2$:

$$L_{\text{Parabola}} = \int_{n_1}^{n_2} \sqrt{1 + \left(\frac{dk}{dn}\right)^2}\, dn = \int_{n_1}^{n_2} \frac{1}{6}\sqrt{36 + (4n+3)^2}\, dn.$$

Arc length of the helix with the direct θ-parametrisation:

$$\gamma(\theta) = \begin{pmatrix} \frac{6}{\pi^2}\theta\cos\theta \\ \frac{6}{\pi^2}\theta\sin\theta \\ -\frac{3}{4} + \frac{3}{\pi}\theta \end{pmatrix}, \qquad \theta_n = \frac{\pi}{3}n + \frac{\pi}{4}.$$

Differentiation yields:

$$\|\gamma'(\theta)\| = \sqrt{\left(\frac{6}{\pi^2}\right)^2 (1 + \theta^2) + \left(\frac{3}{\pi}\right)^2}.$$

The arc length is therefore:

$$L_{\text{Helix}} = \int_{\theta_{n_1}}^{\theta_{n_2}} \|\gamma'(\theta)\|\, d\theta.$$

15.2 Numerical Examples

All arc lengths were computed numerically to high precision. As a detailed example, the first step ($n = 1 \to n = 5$, major step / gap zone) is worked out step by step; the remaining results are summarised in Table 15.1.

15.2.1 Detailed Example: $n = 1 \to n = 5$ (Major Step)

Step 1 — Determine parameters:

$$n_1 = 1, \quad n_2 = 5, \quad \Delta n = 4, \quad \theta_1 = \frac{\pi}{3} \cdot 1 + \frac{\pi}{4} = 1.83259571, \quad \theta_2 = \frac{\pi}{3} \cdot 5 + \frac{\pi}{4} = 6.02138592.$$

Step 2 — Arc length of the parabola:

$$L_{\text{Parabola}} = \int_1^5 \frac{1}{6}\sqrt{36 + (4n+3)^2}\, dn = \boxed{10.83940915}.$$

Step 3 — Arc length of the helix:

$$L_{\text{Helix}} = \int_{1.83259571}^{6.02138592} \sqrt{\left(\frac{6}{\pi^2}\right)^2 (1 + \theta^2) + \left(\frac{3}{\pi}\right)^2}\, d\theta = \boxed{11.15519925}.$$

Step 4 — Relative deviation:

$$\varepsilon = \frac{L_{\text{Helix}} - L_{\text{Parabola}}}{L_{\text{Parabola}}} = \frac{0.31579010}{10.83940915} = \boxed{+2.913\,\%}.$$

Remark 15.1. The helix is *longer* than the parabola, since the spiral motion in the (x, y)-plane (especially at small θ) generates additional path length.

15.2.2 Comparison Table

Table 15.1: Arc-length comparison: parabola vs. helix (true θ-parametrisation).

No.	$n_1 \to n_2$	Type	L_{Parabola}	L_{Helix}	Diff.	Dev.
1	$1 \to 5$	Major	10.83940915	11.15519925	+0.31579010	+2.913 %
2	$5 \to 7$	Minor	9.22107493	9.30913741	+0.08806248	+0.955 %
3	$11 \to 13$	Minor	17.11747993	17.16486194	+0.04738201	+0.277 %
4	$17 \to 19$	Minor	25.07994766	25.11227648	+0.03232881	+0.129 %
5	$23 \to 25$	Minor	33.06058334	33.08510518	+0.02452184	+0.074 %
6	$35 \to 37$	Minor	49.04080939	49.05733914	+0.01652975	+0.034 %

15.3 Unified Representation via the Universal Quantity Q

The convergence of the arc lengths can be explained analytically by expressing both integrals in terms of the universal quantity $Q(k) = \sqrt{48k+1} = 4n + 3$ (introduced formally in Section 16.5.5). With $dQ = 4\,dn$ and $\theta = \pi Q/12$, hence $d\theta = (\pi/12)\,dQ$, one obtains:

Theorem 15.2 (Q-Parametrisation of the Arc Lengths)**.** *The arc-length elements of the parabola $k(n)$ and of the conical helix can be expressed uniformly as*

$$ds = \frac{1}{24}\sqrt{Q^2 + C}\, dQ \tag{15.1}$$

where

$$C_{\text{Parabola}} = 36, \qquad C_{\text{Helix}} = 36 + \frac{144}{\pi^2} \approx 50.59.$$

Proof. **Parabola.** The arc-length element of the curve $(n, k(n))$ is $ds = \sqrt{1 + k'(n)^2}\, dn$ with $k'(n) = (4n + 3)/6$. Substituting $Q = 4n + 3$, $dn = dQ/4$:

$$ds = \sqrt{1 + \frac{Q^2}{36}}\, \frac{dQ}{4} = \frac{1}{4} \cdot \frac{\sqrt{36 + Q^2}}{6}\, dQ = \frac{1}{24}\sqrt{Q^2 + 36}\, dQ.$$

Helix. With $a = 6/\pi^2$, $c_1 = 3/\pi$, and $\theta = \pi Q/12$ we have $d\theta = (\pi/12)\, dQ$. The arc-length element is $ds = \sqrt{a^2(1 + \theta^2) + c_1^2}\, d\theta$. Substituting $\theta = \pi Q/12$:

$$\begin{aligned} a^2(1 + \theta^2) + c_1^2 &= \frac{36}{\pi^4}\Big(1 + \frac{\pi^2 Q^2}{144}\Big) + \frac{9}{\pi^2} \\ &= \frac{36}{\pi^4} + \frac{Q^2}{4\pi^2} + \frac{9}{\pi^2} = \frac{1}{4\pi^2}\Big(Q^2 + 36 + \frac{144}{\pi^2}\Big). \end{aligned}$$

With $d\theta = (\pi/12)\, dQ$ it follows that:

$$ds = \frac{1}{2\pi}\sqrt{Q^2 + 36 + \frac{144}{\pi^2}} \cdot \frac{\pi}{12}\, dQ = \frac{1}{24}\sqrt{Q^2 + 36 + \frac{144}{\pi^2}}\, dQ. \qquad \square$$

Remark 15.3 (Geometric Meaning). The difference $C_{\text{Helix}} - C_{\text{Parabola}} = 144/\pi^2 \approx 14.59$ is the exact contribution of the spiral motion in the (x, y)-plane. For $Q \to \infty$ the term Q^2 dominates in both cases, and the ratio of the arc-length elements converges:

$$\frac{ds_{\text{Helix}}}{ds_{\text{Parabola}}} = \sqrt{\frac{Q^2 + 36 + 144/\pi^2}{Q^2 + 36}} \xrightarrow{Q \to \infty} 1.$$

This explains the convergence $\varepsilon \to 0$ observed in Table 15.1 analytically, not merely numerically.

15.4 Interpretation of the Results

Proposition 15.4 (Asymptotic Arc-Length Convergence)**.** *The helix arc length is consistently* longer *than the arc length of the corresponding parabola segments, with the relative difference converging monotonically to zero:*

$$\frac{L_{\text{Helix}}(n, n + 2) - L_{\text{Parabola}}(n, n + 2)}{L_{\text{Parabola}}(n, n + 2)} \xrightarrow{n \to \infty} 0.$$

This follows directly from the uniform Q-representation (Theorem 15.2), since $C_{\text{Helix}}/Q^2 \to 0$ and $C_{\text{Parabola}}/Q^2 \to 0$ as $Q \to \infty$.

Corollary 15.5 (Practical Accuracy)**.** *For all interval pairs from $n = 11$ onwards — in particular for all primes $p \geq 11$ — the deviation is below* 0.3 %*. For applications in the visualisation of prime distributions this accuracy is fully sufficient.*

Part V

The Direct Solution — From Parabola to Helix

Chapter 16

Fundamental Identity, Ray Structure, and Universal Parametrisation

16.1 Equidistant Parametrisation of the Spiral

On the Archimedean spiral $r(\theta) = \frac{6}{\pi^2}\,\theta$, points are to be placed that follow one another at *constant arc-length spacing* $\Delta s = 1$. This equidistant parametrisation leads to a characteristic geometric scaling factor.

16.1.1 Derivation of the Scaling Factor $\sqrt{3}$

The arc length of the spiral from 0 to θ is:

$$s(\theta) = \frac{a}{2}\left[\theta\sqrt{\theta^2+1} + \operatorname{arcsinh}(\theta)\right], \qquad a = \frac{6}{\pi^2}.$$

For large θ the quadratic term dominates:

$$s(\theta) \;\approx\; \frac{a}{2}\,\theta^2 \;=\; \frac{3}{\pi^2}\,\theta^2.$$

For the k-th point to lie at $s_k = k$ (unit spacing), we require:

$$\frac{3}{\pi^2}\,\theta_k^2 = k \quad\Longrightarrow\quad \theta_k = \frac{\pi}{\sqrt{3}}\,\sqrt{k}.$$

The angle thus grows with the *square root* of the index k, and the **scaling factor** $\sqrt{3}$ arises directly from the geometry of the spiral as a consequence of the condition $\Delta s = 1$.

The corresponding radial coordinate:

$$r_k = \frac{6}{\pi^2}\cdot\theta_k = \frac{6}{\pi^2}\cdot\frac{\pi\sqrt{k}}{\sqrt{3}} = \frac{6\sqrt{k}}{\pi\sqrt{3}} = \frac{2\sqrt{3k}}{\pi}.$$

Table 16.1: Actual arc-length spacing Δs between consecutive equidistant points—convergence towards 1.

k	$\theta_k = \pi\sqrt{k}/\sqrt{3}$	$r_k = 2\sqrt{3k}/\pi$	$\Delta s_k \to 1$
10	5.7357	3.4869	1.01589
100	18.1380	11.0266	1.00153
1000	57.3574	34.8691	1.00015
10000	181.3799	110.2658	1.00002

16.2 $\sqrt{3}$ as the Common Scaling Factor of $k(n)$ and the Spiral

The scaling factor $\sqrt{3}$ of the equidistant construction simultaneously appears as the *stretching factor of the quadratic sequence* $k(n)$. This dual nature of $\sqrt{3}$ reveals a shared structural character of both structures.

16.2.1 Step 1 — Leading Coefficient of $k(n)$

The sequence $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ has the leading term:

$$k(n) = \frac{1}{3}n^2 + \frac{1}{2}n + \frac{1}{6}.$$

The stretching factor of the parabola is $\frac{1}{3} = \frac{1}{(\sqrt{3})^2}$: the factor $\sqrt{3}$ appears in the denominator of the quadratic coefficient.

16.2.2 Step 2 — Stretching Factor of the Arc Length

The asymptotic arc length of the spiral reads:

$$s(\theta) \approx \frac{3}{\pi^2}\theta^2.$$

Here the numerator carries the factor $3 = (\sqrt{3})^2$—precisely the reciprocal of the parabola coefficient.

Reciprocal Relationship of the Stretching Factors

$$\underbrace{\frac{1}{3}}_{\text{Parabola coeff. in } k(n)} \cdot \underbrace{3}_{\text{Arc-length coeff. of the spiral}} = 1.$$

Both structures carry the same factor $\sqrt{3}$, but in *reciprocal roles*: the parabola stretches by $1/(\sqrt{3})^2$, the spiral by $(\sqrt{3})^2$.

16.2.3 Step 3 — $\theta_{k(n)} = \theta_n$ (asymptotically)

Substituting $k = k(n)$ into the equidistant angular formula and expanding $\sqrt{k(n)}$ via Taylor:

$$\sqrt{k(n)} = \sqrt{\frac{n^2}{3} + \frac{n}{2} + \frac{1}{6}} = \frac{n}{\sqrt{3}}\sqrt{1 + \frac{3}{2n} + \frac{1}{2n^2}}$$
$$\approx \frac{n}{\sqrt{3}}\left(1 + \frac{3}{4n}\right) = \frac{n}{\sqrt{3}} + \frac{3}{4\sqrt{3}}.$$

This yields the equidistant angle at the position $k = k(n)$:

$$\theta_{k(n)} = \frac{\pi}{\sqrt{3}} \cdot \sqrt{k(n)} \approx \frac{\pi}{\sqrt{3}}\left(\frac{n}{\sqrt{3}} + \frac{3}{4\sqrt{3}}\right) = \frac{\pi n}{3} + \frac{3\pi}{4 \cdot 3} = \frac{\pi n}{3} + \frac{\pi}{4}.$$

This is the actual helix angle $\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}$:

Theorem 16.1 ($\sqrt{3}$-Invariance)**.** *The equidistant spiral parametrisation, evaluated at the positions $k = k(n)$, approximately reproduces the helix angles:*

$$\theta_{k(n)} = \frac{\pi}{\sqrt{3}}\sqrt{k(n)} \approx \frac{\pi}{3}n + \frac{\pi}{4} = \theta_n.$$

The error is of order $O(1/n)$ and converges to 0.

Table 16.2: Comparison of $\theta_{k(n)}$ (equidistant, asymptotic) and θ_n (helix)—the difference converges to 0.

n	$k(n)$	$\theta_{k(n)}$	$\theta_n = \frac{\pi n}{3} + \frac{\pi}{4}$	Difference
5	11	6.015692	6.021386	−0.005694
11	46	12.301786	12.304571	−0.002785
19	130	20.680495	20.682152	−0.001657
47	760	50.002998	50.003683	−0.000685
97	3185	102.363226	102.363561	−0.000335

Remark 16.2 (Geometric Interpretation). The statement $\theta_{k(n)} \approx \theta_n$ means: if one places equidistant points with unit spacing on the Archimedean spiral and accesses them via the index $k = k(n)$, one arrives approximately at exactly the same angles as the integer points of the helix. The sequence $k(n)$ is *the indexing induced by the common factor* $\sqrt{3}$ of the equidistant spiral parametrisation through the common scaling factor $\sqrt{3}$.

16.3 Three Equivalent Representations in Desmos

The same equidistant parametrisation can be expressed in Desmos [Des24] through different formulae. Depending on the chosen form, the slider parameter h takes a different numerical value, even though all three versions generate *identical points* on the spiral:

Table 16.3: Three equivalent formula versions for the equidistant parametrisation. All produce exactly the same points at unit spacing $\Delta s = 1$.

Form	Formula θ_k	h for $\Delta s = 1$	Occurrence
A	$\dfrac{\pi\sqrt{k}}{\sqrt{3}}$	$\sqrt{3} \approx 1.732$	Project standard (`Power_equidistant_Spiral_v6`)
B	$\dfrac{2\pi\sqrt{k}}{h}$	$2\sqrt{3} \approx 3.464$	`equidistant_Spiral_v5`
C	$\dfrac{\pi\sqrt{3k}}{h}$	3	Alternative notation

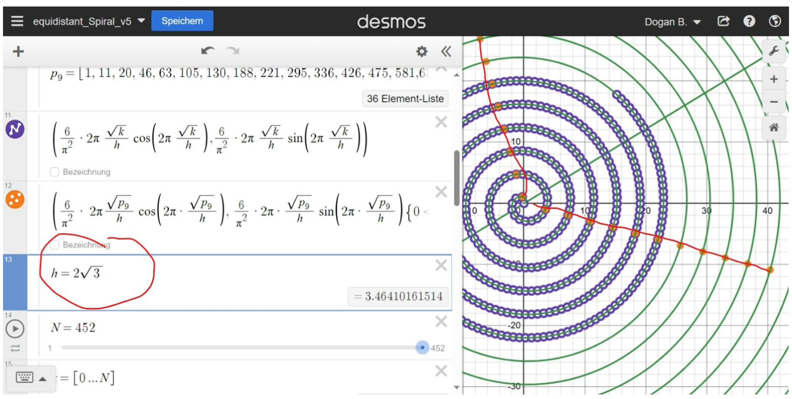
Remark 16.3 (Why Three Different h-Values?). The three forms arise from algebraic rearrangement of the same expression:

$$\underbrace{\frac{\pi\sqrt{k}}{\sqrt{3}}}_{\text{Form A}} = \underbrace{\frac{2\pi\sqrt{k}}{2\sqrt{3}}}_{\text{Form B}} = \underbrace{\frac{\pi\sqrt{3k}}{3}}_{\text{Form C}}.$$

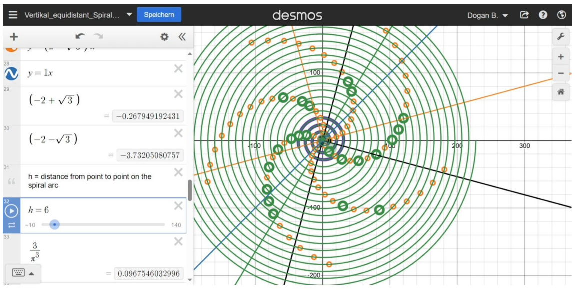
The project uses **Form A** with the derived scaling factor $\sqrt{3}$, since this emerges directly from the derivation. The simplified representation $h = \sqrt{3}$ is thus the geometrically motivated constant of the spiral.

16.4 Ray Structure as a Function of the Formula Term

The following comparison shows how the visual impression of the spiral depends on the choice of the h-parameter in Form B. The slider parameter h changes the angular spacing between the placed points and thus the pattern of visible diagonals (parastichies).



$h = 2\sqrt{3}$ **(Form B)**
Project standard: corresponds to $h_A = \sqrt{3}$ in Form A.
Arc-length spacing $\Delta s \approx 1$, uniform point distribution.



$h = 6$ **(Form B)**
Corresponds to $h_A = 3$ in Form A.
Arc-length spacing $\Delta s \approx 1/3$.
Integer density distributes over 5 parastichy arms.

$h = 11$ **(Form B)**
Arc-length spacing $\Delta s \approx 12/121 \approx 0.1$.
Expansive diagonal structure. Integers distribute over 9 lines.

All four representations show the same Archimedean spiral $r = \frac{6}{\pi^2}\theta$ (Desmos, $N \approx 952$ points). The slider parameter h follows Formula B: $\theta_k = 2\pi\sqrt{k}/h$. The project standard $h_A = \sqrt{3}$ (Form A) corresponds to $h_B = 2\sqrt{3}$ in these images. The circle markers correspond to the sequence values $k(n)$; the coloured diagonals run along the residue classes $n \equiv 1, 5 \pmod 6$.

16.5 Direct Derivation of the Height Function from the Quadratic Sequence

16.5.1 Two Paths — One Statement: Why Both Derivations Are Valuable

In the preceding section, the helix angle θ_n was obtained through a Taylor expansion of $\sqrt{k(n)}$ *asymptotically* (Theorem 16.1). This approach is deliberately preserved as an independent method—for it works for *arbitrary quadratic polynomials*, without presupposing the specific arithmetic structure of $k(n)$. This makes it a general tool for other researchers.

Methodological Note for Other Polynomials

For an arbitrary quadratic sequence $p(n) = an^2 + bn + c$, the asymptotic procedure applies analogously:

$$\sqrt{p(n)} \approx \sqrt{a}\, n + \frac{b}{2\sqrt{a}},$$

from which the spiral angle grows linearly in n; its slope $\sqrt{2a/A}$ is determined jointly by the leading coefficient a and the slope A of the associated spiral (for $k(n)$: $\pi/3$). The approximation method requires only knowledge of the leading coefficient—*no* completing the square and *no* integrality property. It is therefore applicable to a substantially broader class of sequences.

The exact method (following section) requires instead that $\deg(p) \cdot (\text{prefactor}) \cdot p(n) + 1$ be a perfect square—a property that $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ happens to possess, but most polynomials do not.

In the following it is shown that for $k(n)$ itself an *exact* derivation is possible—without approximation and with complete error resolution. The key is the invertibility of $k(n)$ and a hidden perfect-square property.

16.5.2 The Sequence $k(n)$ and Its Exact Inverse

The starting point is the quadratic sequence

$$k(n) = \frac{2n^2 + 3n + 1}{6} = \frac{(n+1)(2n+1)}{6}, \qquad n \equiv 1, 5 \pmod 6. \tag{16.1}$$

This formula assigns to each index n (the helix height) a spiral index k. The question is: What is the *inverse function*—that is, n as a function of k?

The naive answer via the quadratic formula yields:

$$2n^2 + 3n + 1 = 6k \implies n = \frac{-3 + \sqrt{9 - 8(1 - 6k)}}{4} = \frac{-3 + \sqrt{48k + 1}}{4}. \tag{16.2}$$

This is an *exact* formula—but it contains a square root that is irrational for general k. The key identity: at the integer positions $k = k(n)$ this square root is always an integer.

16.5.3 Fundamental Identity: The $(4n + 3)^2$ Relation

Theorem 16.4 (Fundamental Identity)**.** *For all $n \geq 0$ the following holds exactly:*

$$\boxed{(4n + 3)^2 = 48\, k(n) + 1.} \tag{16.3}$$

Proof. Direct substitution of (16.1):

$$48\, k(n) + 1 = 48 \cdot \frac{2n^2 + 3n + 1}{6} + 1 = 8(2n^2 + 3n + 1) + 1 = 16n^2 + 24n + 9 = (4n + 3)^2. \quad \square$$

The identity explains why formula (16.2) yields integer values at the positions $k = k(n)$: the polynomial $k(n)$ is constructed so that $48k(n) + 1$ is always a perfect square, namely $(4n + 3)^2$.

Remark 16.5 (The Derivative as a Bridge). The derivative $k'(n) = (4n+3)/6$ exhibits the same expression $4n+3$. Identity (16.3) can therefore also be written as:

$$\big[6\,k'(n)\big]^2 = 48\,k(n) + 1,$$

i.e., the growth rate of the sequence and its value are linked by an exact algebraic relation—a property that does *not* hold for general quadratic polynomials.

16.5.4 Exact Inverse Formula: Height from Spiral Index

From Theorem 16.4 the exact inverse of $k(n)$ follows immediately.

Theorem 16.6 (Exact Height Function). *The unique non-negative solution of the equation $k(n) = k$ is:*

$$\boxed{n(k) = \frac{\sqrt{48k+1}-3}{4}.} \tag{16.4}$$

This is the exact height function of the helix: $z(k) = n(k)$.

Proof. From $(4n+3)^2 = 48k+1$ it follows that $4n+3 = \sqrt{48k+1}$ (since $4n+3 > 0$), hence $n = (\sqrt{48k+1}-3)/4$. $\square$

Remark 16.7 (Comparison: Asymptotic vs. Exact). The asymptotic approximation from the previous section reads $n \approx \sqrt{3k} - 3/4$. Comparison with the exact formula makes the difference visible and quantifies the error:

$$\begin{aligned}
n_{\text{exact}}(k) &= \frac{\sqrt{48k+1}-3}{4} = \frac{4\sqrt{3k}\sqrt{1+\frac{1}{48k}}-3}{4} = \sqrt{3k}\,\underbrace{\sqrt{1+\frac{1}{48k}}}_{\approx 1+\frac{1}{96k}} - \frac{3}{4} \\
&= \sqrt{3k} + \underbrace{\frac{\sqrt{3}}{96\sqrt{k}}}_{\text{correction term}} - \frac{3}{4}, \\
n_{\text{asympt.}}(k) &= \sqrt{3k} - \frac{3}{4}.
\end{aligned}$$

The absolute error of the approximation is $+\frac{\sqrt{3}}{96\sqrt{k}}$ and is always *positive*: the asymptotic formula systematically underestimates the true value n. For $k = k(n)$ this error is of order $O(n^{-1})$.

16.5.5 The Universal Quantity $Q(k)$: All Helix Quantities from a Single Root

The fundamental identity (16.3) raises the question whether all previously derived geometric quantities can be expressed through a single scalar. The identity suggests treating the expression under the square root as exactly this quantity:

$$Q(k) := \sqrt{48k+1}. \tag{16.5}$$

Universality is meant construction-internally: every geometric quantity of the helix developed in this work — index, angle, radius, circumference, apex height, generator length, asymptotic torsion — is an explicit function of Q alone, as the following table shows; no universality beyond this construction is claimed.

At the lattice points $k = k(n)$, Q always takes the integer value $Q(k(n)) = 4n+3$. Thus Q becomes the universal bridge between the spiral index k and all geometric quantities of the helix.

Table 16.4: All geometric characteristics of the helix as exact functions of $Q(k) = \sqrt{48k+1}$. At the lattice points $k = k(n)$, $Q = 4n + 3$ (integer); the third column gives the value there directly.

Characteristic	Exact formula in Q	Value at $k = k(n)$	Unit
Height (inverse of k)	$z = \dfrac{Q-3}{4}$	n	—
Height above apex	$h = \dfrac{Q}{4}$		
Radius (cone)	$r = \dfrac{Q}{2\pi}$	$\dfrac{4n+3}{2\pi}$	—
Azimuthal angle (spiral)	$\theta = \dfrac{\pi Q}{12}$	$\dfrac{\pi}{3}n + \dfrac{\pi}{4}$	rad
Derivative $k'(n)$	$k' = \dfrac{Q}{6}$	$\dfrac{4n+3}{6}$	—
Generator length	$\ell = \dfrac{Q}{2\pi}\sqrt{1+\frac{\pi^2}{4}}$	$r_n\,\lvert\vec{v}\rvert$	—

The table makes visible what the asymptotic method captures only *approximately*: the helix is completely determined by the *single* function $Q(k) = \sqrt{48k+1}$. The asymptotic formula used in Chapter 16, $\theta \approx (\pi/\sqrt{3})\sqrt{k}$, is nothing other than the leading approximation of $\theta = \pi Q/12 \approx \pi \cdot 4\sqrt{3k}/12 = (\pi/\sqrt{3})\sqrt{k}$ for large k.

16.5.6 Verification of the Cone and Spiral Equations (Exact)

Both core geometric statements—cone conformity and spiral conformity—hold exactly with the Q-representation, not merely asymptotically.

Theorem 16.8 (Cone Conformity). *The helix points $(r(k), z(k))$ lie exactly on the cone $z = -\frac{3}{4} + \frac{\pi}{2} r$:*

$$-\frac{3}{4} + \frac{\pi}{2} \cdot \frac{Q}{2\pi} = -\frac{3}{4} + \frac{Q}{4} = \frac{Q-3}{4} = z(k). \qquad \square$$

Theorem 16.9 (Spiral Conformity). *The helix points $(r(k), \theta(k))$ lie exactly on the Archimedean spiral $r = \frac{6}{\pi^2}\theta$:*

$$\frac{6}{\pi^2} \cdot \frac{\pi Q}{12} = \frac{Q}{2\pi} = r(k). \qquad \square$$

In both cases every approximation vanishes: the identities are algebraic consequences of the structure of $k(n)$ and hold for all k, not only for $k \to \infty$.

16.5.7 ODE Characterisation of the Height Function

The exact formula (16.4) possesses a direct interpretation as the solution of a differential equation.

Theorem 16.10 (ODE of the Height Function)**.** *The height function $z = z(k)$ is the unique solution of the initial value problem*

$$\frac{dz}{dk} = \frac{6}{4z+3}, \qquad z\left(\tfrac{1}{6}\right) = 0. \tag{16.6}$$

The explicit solution reads:

$$2z^2 + 3z + 1 = 6k, \tag{16.7}$$

which upon solving for $z \geq 0$ coincides exactly with (16.4) *and is nothing other than the sequence equation $k(z) = k$.*

Proof. Separation of variables: $(4z+3)\,dz = 6\,dk$, integration: $2z^2 + 3z = 6k + C$. The initial condition $z(1/6) = 0$ yields $C = -1$, hence (16.7). □

Remark 16.11 (Asymptotics of the ODE). The slope $dz/dk = 6/(4z+3) = 1/k'(z)$ is the reciprocal of the growth rate of $k(n)$. For large z it approaches $dz/dk \approx 3/(2z)$, whence $z^2 \approx 3k$, i.e., $z \approx \sqrt{3k}$—exactly the asymptotic approximation of the previous section. The ODE thus makes visible why the approximation is good: it correctly captures the leading term of the growth.

16.5.8 Numerical Table: Asymptotic vs. Exact

Table 16.5 directly compares both methods. The systematically negative error of the approximation (it always underestimates n) and its $O(n^{-1})$ convergence are clearly visible.

Table 16.5: Comparison of asymptotic approximation ($n \approx \sqrt{3k} - \frac{3}{4}$) and exact formula ($n = (\sqrt{48k+1} - 3)/4$) for selected n. The error is always negative and converges as $O(n^{-1})$ to 0.

n	$k(n)$	$Q = 4n+3$	n_{exact}	$n_{\text{asympt.}}$	Error	θ_n (rad)
1	1	7	1.0000	0.9821	−1.79%	1.8326
5	11	23	5.0000	4.9946	−0.11%	6.0214
7	20	31	7.0000	6.9960	−0.06%	8.1158
11	46	47	11.0000	10.9973	−0.02%	12.3046
13	63	55	13.0000	12.9977	−0.02%	14.3990
17	105	71	17.0000	16.9982	−0.01%	18.5878
19	130	79	19.0000	18.9984	−0.01%	20.6822
47	760	191	47.0000	46.9993	<0.01%	50.0037

16.5.9 Complete Exact Helix Parametrisation

The result of all preceding steps can be summarised in a single closed formula.

Theorem 16.12 (Exact Helix Parametrisation). *Let* $Q(k) = \sqrt{48k+1}$. *The helix is described exactly for* $k \in \{k(n) : n \equiv 1, 5 \pmod 6\}$ *by:*

$$\boxed{\vec{p}(k) = \begin{pmatrix} \dfrac{Q(k)}{2\pi}\cos\Bigl(\dfrac{\pi\, Q(k)}{12}\Bigr) \\ \dfrac{Q(k)}{2\pi}\sin\Bigl(\dfrac{\pi\, Q(k)}{12}\Bigr) \\ \dfrac{Q(k)-3}{4} \end{pmatrix}, \qquad Q(k) = \sqrt{48k+1}.} \tag{16.8}$$

All three coordinates are exact functions of the single parameter k.

Remark 16.13 (Three Invariants — One Origin). The parametrisation (16.8) unites three simultaneously valid exact invariants, all of which follow from the same underlying structure $Q(k)$:

1. **Arithmetic invariant**: $(4n+3)^2 = 48\,k(n) + 1$ (Theorem 16.4),
2. **Cone invariant**: $z + \frac{3}{4} = \frac{\pi}{2}\, r$ (Theorem 16.8),
3. **Spiral invariant**: $r = \frac{6}{\pi^2}\,\theta$ (Theorem 16.9).

The helix is thus triply lawful—and all three laws are algebraic consequences of the *quadratic sequence* $k(n)$ alone.

Summary: Both Derivation Paths at a Glance

The angular formula $\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}$ can be derived in two fundamentally different ways that complement each other:

Path 1 (asymptotic, generally applicable):

$$\theta_{k(n)} = \frac{\pi}{\sqrt{3}}\sqrt{k(n)} \xrightarrow{\text{Taylor}} \frac{\pi n}{3} + \frac{\pi}{4} + O\left(\frac{1}{n}\right).$$

Works for arbitrary quadratic $p(n)$. Error is $O(1/n)$.

Path 2 (exact, specific to $k(n)$):

$$(4n+3)^2 = 48\,k(n) + 1 \implies Q(k(n)) = 4n+3 \implies \theta = \frac{\pi Q}{12} = \frac{\pi n}{3} + \frac{\pi}{4}.$$

No error. Requires that $48k(n) + 1$ be a perfect square.

The four equivalent expressions for the height function:

$$\begin{aligned}
\text{(algebraic)} \quad & (4n+3)^2 = 48\,k(n) + 1, \\
\text{(analytic)} \quad & n(k) = \tfrac{1}{4}\Big(\sqrt{48k+1} - 3\Big), \\
\text{(differential)} \quad & (4z+3)\,\tfrac{dz}{dk} = 6, \quad z\big(\tfrac{1}{6}\big) = 0, \\
\text{(geometric)} \quad & z + \tfrac{3}{4} = \tfrac{\pi}{2}\,r = \tfrac{\pi^2}{12}\,\theta.
\end{aligned}$$

All four are exactly equivalent. Together they show: the quadratic sequence $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ carries the complete geometric helix information within itself—encoded through $Q(k) = \sqrt{48k+1}$.

Chapter 17

Direct Derivation of the Height Function from the Quadratic Sequence

17.1 Introduction

The construction of the conical helix is based on an interplay between arithmetic and geometric structures. The starting point is the quadratic sequence

$$k(n) = \frac{1}{6}(2n^2 + 3n + 1),$$

whose analytic continuation is a parabola with a clearly pronounced symmetry and a well-defined vertex. This quadratic structure not only provides an ordering of the integer function values but also determines the scaling and shift parameters that appear in the height component of the helix.

The aim of this chapter is to derive the height function of the conical helix

$$z(\theta) = -\frac{3}{4} + \frac{3}{\pi}\theta$$

directly from the properties of the sequence. The derivation proceeds without recourse to intermediate constructions and shows that both the shift by $-\frac{3}{4}$ and the slope $\frac{3}{\pi}$ are immediate consequences of the underlying quadratic arithmetic. The additive constant reflects the position of the vertex of the parabola, while the slope arises from the residue class structure modulo 6, which enforces a linear coupling between the angular parameter and the arithmetic progression.

This makes visible that the height component of the helix is a direct and structurally necessary consequence of the arithmetic properties of the sequence $k(n)$. This chapter formulates this connection precisely and anchors the helix parametrisation in the internal logic of the quadratic sequence.

17.2 Derivation from the Parabola

The quadratic sequence

$$k(n) = \frac{1}{6}(2n^2 + 3n + 1)$$

has as its underlying continuous extension the parabola

$$f(x) = \frac{1}{6}(2x^2 + 3x + 1).$$

The vertex of this parabola is located at

$$x_{\mathrm{v}} = -\frac{b}{2a} = -\frac{3}{4},$$

so that the constant $-\frac{3}{4}$ constitutes the reference for the arithmetic symmetry of the sequence. This shift forms the starting point for the height component of the subsequent helix.

For the construction of the conical helix a linear relationship between the angular parameter θ and the arithmetic scale of the sequence is assumed. The central requirement is that a full revolution of the helix ($\Delta\theta = 2\pi$) corresponds to an increment of 6 in the underlying integer structure. This reflects the residue class structure modulo 6, which is characteristic of the sequence $k(n)$ and the primes $p > 3$.

For a linear height function

$$z(\theta) = \alpha\,\theta + \beta$$

the condition

$$\alpha \cdot 2\pi = 6$$

immediately yields

$$\alpha = \frac{3}{\pi}.$$

The additive constant β is fixed by the vertex–apex identification (Proposition 13.5); it is the single normalisation of the construction. Its value:

$$\beta = -\frac{3}{4}.$$

The height component of the conical helix is therefore

$$z(\theta) = -\frac{3}{4} + \frac{3}{\pi}\,\theta,$$

which geometrically encodes both the position of the vertex of the quadratic sequence and the arithmetic periodicity modulo 6. The function $z(\theta)$ thus arises directly from the structure of the sequence $k(n)$, without requiring an intermediate construction via the Archimedean spiral.

17.3 Generalisation: The Construction Scheme for Arbitrary Functions

The derivation developed in the preceding section possesses a structural property: at its core it is independent of the specific form of the initial function $f(x)$. The decomposition into a radial component $r(\theta)$, which determines the winding character, and an axial

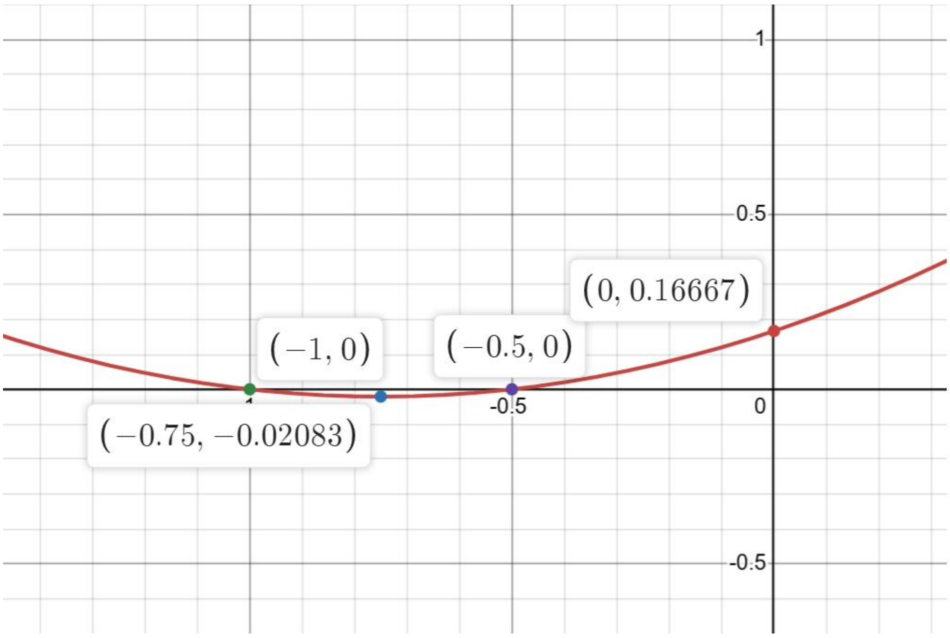


Figure 17.1: The parabola $f(x) = \frac{1}{6}(2x^2 + 3x + 1)$ in the neighbourhood of the vertex. Shown are the four geometrically distinguished points: the vertex $(-\frac{3}{4}, -\frac{1}{48})$, the zero $(-1, 0)$, the second intersection with the x-axis $(-\frac{1}{2}, 0)$, and the y-intercept $(0, \frac{1}{6})$. The vertex provides the shift parameter $\beta = -\frac{3}{4}$ of the height function, while the zero $(-1, 0)$ represents the predecessor point of the Euclidean distance sequence.

component $z(\theta)$, which encodes the height progression, applies equally to every continuous, monotonically increasing function. This section formulates the general construction scheme and shows that the present work can be understood as a model solution for an entire class of helix constructions.

17.3.1 The General Parametrisation Scheme

Definition 17.1 (Function-Helix)**.** Let $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be a continuous function, $T > 0$ a freely chosen revolution period, and $\beta \in \mathbb{R}$ a reference value. The associated *function-helix* is the space curve

$$\gamma(\theta) = \begin{pmatrix} f\left(\dfrac{T}{2\pi}\theta\right)\cos\theta \\ f\left(\dfrac{T}{2\pi}\theta\right)\sin\theta \\ \beta + \dfrac{T}{2\pi}\theta \end{pmatrix}, \qquad \theta \geq 0. \tag{17.1}$$

The parametrisation (17.1) contains three structurally independent constituents, each contributing a distinct geometric effect.

The **radial component** $r(\theta) = f\big(\frac{T}{2\pi}\theta\big)$ completely determines the type of the generating spiral in the xy-plane. The function f specifies how rapidly the distance from the z-axis increases with the rotation angle θ. Each choice of f corresponds to a classifiable spiral type.

The **axial component** $z(\theta) = \beta + \alpha\theta$ with $\alpha = T/(2\pi)$ is, by contrast, structurally identical for all functions f. It is an affine-linear function of θ whose slope α depends solely on the chosen revolution period T. The choice $T = 6$ reproduces exactly the slope $\alpha = 3/\pi$ of the conical helix. This linearity is not a peculiarity of the quadratic initial function but a universal consequence of the requirement of constant height gain per revolution.

The **shift parameter** β anchors the curve at the distinguished reference point of f — the vertex for a parabola, the inflection point for a cubic, the zero for a polynomial with a sign-changing derivative. It ensures that $\theta = 0$ is geometrically meaningful.

17.3.2 The Distinguished Role of the Parabola

Within the class described by Definition 17.1, the linear radial profile $f(x) = (6/\pi^2) \cdot x$ — the profile induced by the parabola $k(n)$ — occupies a distinguished position. The affine radial profiles are precisely those for which the condition

$$\frac{dz}{dr} = \frac{dz/d\theta}{dr/d\theta} = \text{const}$$

is satisfied. Since $r(\theta) = (6/\pi^2)\theta$ is linear in θ — just as $z(\theta)$ is — their quotient is constant:

$$\frac{dz}{dr} = \frac{3/\pi}{6/\pi^2} = \frac{\pi}{2}.$$

This constancy means geometrically that the helix lies on a *circular cone*. For all non-affine functions f, the quantity $dr/d\theta$ varies with θ, so that the quotient dz/dr becomes position-dependent and the curve lies on a curved surface of revolution rather than on a cone.

17.3.3 Classification by Spiral Type

Depending on the choice of the function f, qualitatively different helix types arise, distinguished by their asymptotic growth behaviour and the associated surface of revolution. The following table summarises the principal cases, each with period $T = 6$ and $\alpha = 3/\pi$:

Table 17.1: Classification of function-helices by choice of the radial function $f(x)$ with fixed period $T = 6$. The axial component $z(\theta) = \beta + (3/\pi)\theta$ is structurally identical in all cases; only β and f vary.

$f(x)$	Spiral type	Surface of revolution	β
$\frac{6}{\pi^2}x$ (Archimedean)	Archimedean, $r \sim \theta$	Circular cone $z \sim r$	$-\frac{3}{4}$
$x^{1/3}$	Cube-root spiral, $r \sim \theta^{1/3}$	$z \sim r^3$	0
$\sqrt{x}$	Fermat (root) spiral, $r \sim \theta^{1/2}$	$z \sim r^2$ (paraboloid)	0
$\ln x$	$r \sim \ln\theta$ (log. growth)	$z \sim e^r$	0
x^2	Galilean spiral, $r \sim \theta^2$	$z \sim r^{1/2}$	0

Remark 17.2 (Universality of the Height Function). The table illustrates the central structural finding: the height function $z(\theta) = \beta + (3/\pi)\theta$ is *identical* for all helix types listed — apart from the scalar shift parameter β. It is not specific to the quadratic sequence but a universal affine-linear function that follows solely from the requirement of a constant slope $\Delta z/\Delta\theta = T/(2\pi)$. What distinguishes the present construction from other helices is not the height function itself but the origin of its parameters: α and β, which are derived from the internal arithmetic structure of the sequence $k(n)$. It is precisely this arithmetic anchoring that makes the conical helix a mathematically privileged object within the family described by Definition 17.1.

17.3.4 The Construction Scheme as a Model Solution

The foregoing analysis allows the procedure developed in this chapter to be formulated as a general construction scheme that can be transferred to arbitrary initial functions. The scheme consists of three mutually independent steps.

Step 1 — Choice of the function and the reference point. Let a continuous function $f(x)$ be given. The reference point x_0 is the distinguished point of f — vertex, inflection point, or zero — and yields the shift parameter $\beta = -\alpha\,\varphi$, where φ is the phase offset of the angular coupling $\theta_{x_0} = 0$.

Step 2 — Choice of the revolution period T. The period T specifies which increment in x corresponds to a full revolution. It determines the slope $\alpha = T/(2\pi)$ of the height function and the linear coupling $x(\theta) = (T/2\pi)\,\theta$. For the quadratic sequence $k(n)$, the value $T = 6$ is distinguished by the congruence structure $n \equiv 1, 5 \pmod 6$.

Step 3 — Parametrisation of the helix. The complete curve follows immediately from Equation (17.1). The winding character is determined by f, the height progression by T, and the position in space by β. No further construction assumptions are required.

Remark 17.3 (Geometric Intuition). The scheme can be understood intuitively as follows: the curve $y = f(x)$ in the plane is regarded as a flexible strip that is wound up along the z-axis — similar to a strip of paper rolled into a spiral. The slope during winding is α, the radius at each height follows from the function value $f(x(\theta))$, and the starting point of the winding corresponds to β. The result is always a well-parametrised space curve whose geometric properties are completely determined by f, T, and β.

Chapter 18

Summary and Outlook

18.1 Main Results

This work connects arithmetic properties of a quadratic sequence with classical geometry and shows how a simple Diophantine structure gives rise to a three-dimensional helix. The principal results are:

1. Quadratic sequence

$$k(n) = \frac{1}{6}(2n^2 + 3n + 1)$$

with integer values precisely for

$$n \equiv 1, 5 \pmod 6.$$

2. Alternating step sizes (minor/major steps) with explicit formula

$$a_k = \frac{1}{2}\big[(12k + 3) + (-1)^{k+1}(4k + 1)\big], \qquad k \geq 1.$$

3. Circle circumferences

$$U_m = 24m - 5$$

with constant difference

$$\Delta R = \frac{12}{\pi}.$$

4. Archimedean spiral

$$r(\theta) = \frac{6}{\pi^2}\theta.$$

5. Simplified parametrisation via p:

$$r(p) = \frac{6}{\pi\sqrt{3}}\sqrt{p}, \qquad \theta(p) = \frac{\pi}{\sqrt{3}}\sqrt{p}.$$

6. Exact arc-length integral

$$L = \int_{p_1}^{p_2} \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\, dp.$$

7. Taylor approximation with high accuracy:

$$L \approx (p_2 - p_1) + \frac{3}{2\pi^2}\ln\left(\frac{p_2}{p_1}\right) + \frac{9}{8\pi^4}\left(\frac{1}{p_1} - \frac{1}{p_2}\right).$$

8. Conical helix with half-angle

$$\alpha = \arctan\left(\frac{2}{\pi}\right) \approx 32.48^\circ.$$

9. Integer points on the helix:

$$\theta_n = \frac{\pi}{3}n + \frac{\pi}{4}.$$

10. All primes $p > 3$ appear as discrete points on the helix. This follows as a corollary from the fact that all primes $p > 3$ satisfy the congruence condition $p \equiv 1, 5 \pmod 6$ by a classical result of number theory, and the helix is constructed so as to contain all indices from these congruence classes. The helix additionally contains all composite numbers with $n \equiv 1, 5 \pmod 6$; the primes form a distinguished subset therein.

18.2 Methodological Significance

This work demonstrates that arithmetic structures can not only be visualised but *geometrically forced*. The connection between:

- Number theory (congruences, prime numbers),
- Analysis (integrals, Taylor series),
- Geometry (spirals, cones, helices),
- Classical constants (π, $\zeta(2)$)

leads to a surprisingly coherent three-dimensional structure.
The helix follows necessarily from:

$$\Delta n = 6, \qquad \Delta r = \frac{12}{\pi}, \qquad r \sim \sqrt{p}, \qquad \theta \sim \sqrt{p}.$$

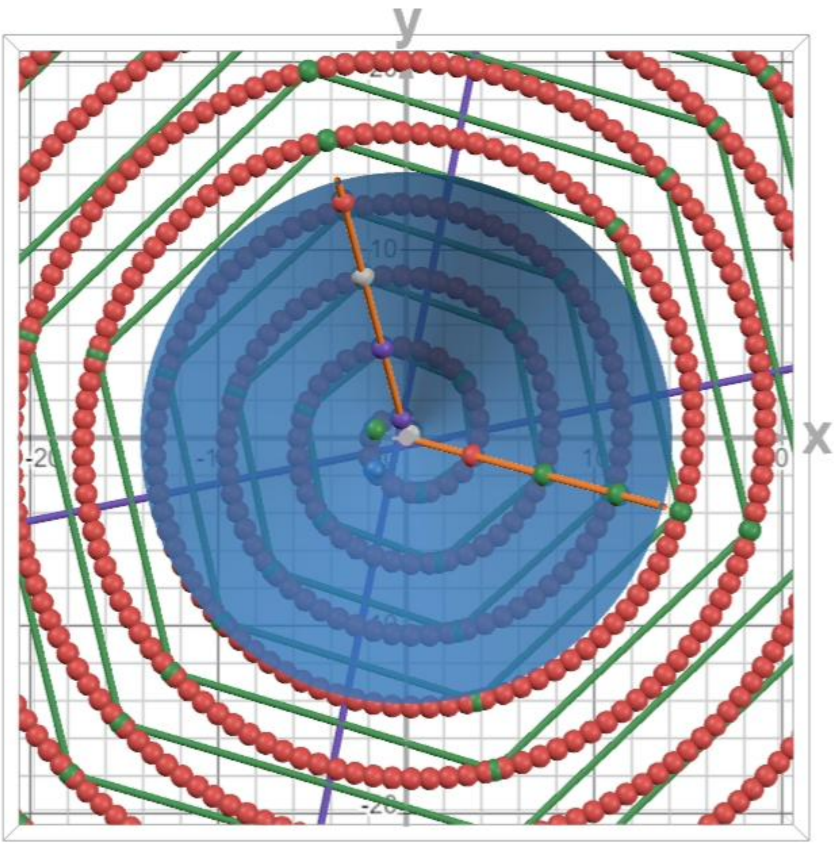


Figure 18.1: Top view of the conical helix (projection onto the xy-plane). The red spheres mark the equidistantly distributed points of the sequence on the spiral path. The green points (arranged hexagonally on six rays) show the function values $k(n)$ for all residue classes modulo 6; only the two rays with $n \equiv 1, 5 \pmod 6$ yield integer values. The two diagonals run along these residue classes and make the 6-periodicity of the arithmetic structure visually apparent. The blue disc shows the cone surface in projection; the spiral grows from inside to outside proportionally to $r \sim \theta$ (Archimedean).

18.3 Outlook

The construction opens up several avenues for further investigation:

- **Prime gaps on the helix:** How do the distances between primes behave in the 3D model?

- **Systematic comparison with the Ulam and Sacks spirals:** While the Ulam spiral [SUW64] and the Sacks spiral [Sac03] are empirically motivated, the present construction possesses a closed-form derivation. A detailed comparison of the prime distributions across all three spiral types remains open.

- **Distribution-theoretic refinements:** The spiral constant $a = 6/\pi^2$ coincides with the asymptotic mean of $\varphi(n)/n$ (Remark 5.14). Erdős and Shapiro [ES55] proved that the error term $H(x) = \sum_{n\leqslant x} \varphi(n)/n - 6x/\pi^2$ possesses a continuous distribution function; the Erdős–Wintner theory [EW39] yields analogous results for $\varphi(n)/n$ itself. A natural follow-up question is whether the fluctuations of $\varphi(n)/n$ around the mean $6/\pi^2$ can be read off geometrically from the helix representation — for instance, as radial deviations of the integer-indexed points from the ideal spiral curve. Erdős himself formulated the differentiability properties of this distribution function as an open problem [Erd74], which remains unsettled to this day.

- **Zeros of ζ as arc lengths:** The arc-length parametrisation of Chapter 6 permits placing the imaginary parts γ_n of the nontrivial zeros of $\zeta(s)$ on the spiral as marked points, with winding number $\theta(\gamma_n)/2\pi = \sqrt{\gamma_n/12}$. A numerical experiment with the first 200 zeros shows their angular positions $\theta(\gamma_n) \bmod 2\pi$ to be equidistributed (Weyl sums at or below the $1/\sqrt{N}$ level; Kolmogorov–Smirnov statistic $D = 0.049$ against a 5% critical value of 0.096): with respect to this ruler the zeros behave angularly generically. Whether deviations from equidistribution appear at larger N, for the three-dimensional helix ruler, or for the zeros of $L(s, \chi_{-3})$ in place of ζ, is open.

- **Generalisation to higher polynomial degrees:** Which geometries arise from cubic or quartic sequences?

- **Modular interpretations:** Is there a connection to modular forms or lattice structures?

- **Dynamical systems:** Can the helix be interpreted as the trajectory of a flow? In the elementary sense the answer is affirmative and explicit: in cylindrical coordinates the curve satisfies the autonomous system

$$\dot{r} = a, \qquad \dot{\varphi} = 1, \qquad \dot{z} = c_1,$$

  with θ as time — it is the trajectory of the constant vector field $(a,\, 1,\, c_1) = (6/\pi^2,\, 1,\, 3/\pi)$, and the position-independence of the winding distance d is the time-translation invariance of this flow. The integer points sample the trajectory at equal time steps $\Delta\theta = \pi/3$, six steps per revolution, so the residue classes modulo 6 appear as the six phases $\varphi \equiv (3 + 4r)\pi/12 \pmod{2\pi}$, $r = 0, \ldots, 5$, of a stroboscopic

(Poincaré) section; the integer-valued classes $n \equiv 1, 5 \pmod 6$ occupy the two fixed phases $7\pi/12$ and $23\pi/12$ — the twin zones of Chapter 13. The mod-6 arithmetic of the sequence is thus the return-map bookkeeping of the flow. What remains open is the substantive part: this flow is linear and integrable, whereas a dynamics carrying arithmetic information in the sense of the spectral programme of Berry–Keating [BK99] — periodic orbits with periods $\ln p$, resonances at the zeros of ζ — would have to be hyperbolic. Whether the cone can serve as the geometric carrier of such a dynamics is unknown.

- **Asymptotic analysis:** The exact 3D curvature (Theorem 13.3) and torsion are now known; their finer asymptotic analysis, particularly at the prime points, remains open.

This work shows that even simple sequences such as

$$k(n) = \frac{2n^2 + 3n + 1}{6}$$

give rise to a coherent geometric structure when consistently transferred into three-dimensional space. The contribution of this work is the synthesis: a single derivation path connecting known arithmetic building blocks — congruences, differences, circumference, cone, and helix — into one model.

Appendix A

Formula Collection

This section provides a compact overview of all central formulae of this work.

A.1 Quadratic Sequence

Main function:

$$k(n) = \frac{1}{6}(2n^2 + 3n + 1) \tag{A.1}$$

Congruence condition:

$$k(n) \in \mathbb{Z} \quad \Longleftrightarrow \quad n \equiv 1, 5 \pmod 6 \tag{A.2}$$

y-**intercept:**

$$k(0) = \frac{1}{6} \tag{A.3}$$

A.2 Difference Sequence

Generator function (Theorem 5.4, Ch. 5):

$$a_k = \frac{1}{2}\big[(12k + 3) + (-1)^{k+1}(4k + 1)\big], \qquad k \geq 1. \tag{A.4}$$

The sequence begins with $a_1 = 10$ (major), $a_2 = 9$ (minor), $a_3 = 26$, ...

Major steps (odd $k = 2m - 1$):

$$a_{2m-1} = \Delta_{\text{Major}}(m) = 16m - 6 \tag{A.5}$$

Minor steps (even $k = 2m$):

$$a_{2m} = \Delta_{\text{Minor}}(m) = 8m + 1 \tag{A.6}$$

A.3 Circles and Radii

Circle circumferences:

$$U_m = a_{2m-1} + a_{2m} = 24m - 5 \tag{A.7}$$

Constant difference:

$$\Delta U = 24 \tag{A.8}$$

Radius sequence:

$$R_m = \frac{U_m}{2\pi} = \frac{24m - 5}{2\pi} \tag{A.9}$$

Radial difference (key result):

$$\boxed{\Delta R = \frac{12}{\pi} = 3.819718634205\ldots} \tag{A.10}$$

A.4 Archimedean Spiral

Standard form:

$$r(\theta) = a\theta, \qquad a = \frac{6}{\pi^2} \tag{A.11}$$

Spiral equation:

$$\boxed{r(\theta) = \frac{6}{\pi^2}\theta} \tag{A.12}$$

Connection to the Basel problem:

$$a = \frac{6}{\pi^2} = \frac{1}{\zeta(2)}, \qquad \zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{A.13}$$

A.5 Simplified Parametrisation

Polar form with parameter p:

$$\boxed{\begin{aligned} r(p) &= \frac{6}{\pi\sqrt{3}}\sqrt{p}, \\ \theta(p) &= \frac{\pi}{\sqrt{3}}\sqrt{p} \end{aligned}} \tag{A.14}$$

Cartesian coordinates:

$$\begin{aligned} x(p) &= \frac{6}{\pi\sqrt{3}}\sqrt{p}\cos\left(\frac{\pi}{\sqrt{3}}\sqrt{p}\right), \\ y(p) &= \frac{6}{\pi\sqrt{3}}\sqrt{p}\sin\left(\frac{\pi}{\sqrt{3}}\sqrt{p}\right) \end{aligned} \tag{A.15}$$

Derivatives:

$$\begin{aligned} \frac{dr}{dp} &= \frac{\sqrt{3}}{\pi\sqrt{p}}, \\ \frac{d\theta}{dp} &= \frac{\pi}{2\sqrt{3}\sqrt{p}} \end{aligned} \tag{A.16}$$

A.6 Arc-Length Integral

Exact form:

$$\boxed{L = \int_{p_1}^{p_2} \frac{1}{\pi}\sqrt{\frac{3}{p} + \pi^2}\, dp} \tag{A.17}$$

Alternative form:

$$L = \int_{p_1}^{p_2} \sqrt{1 + \frac{3}{\pi^2 p}}\, dp \tag{A.18}$$

Taylor approximation:

$$\boxed{L \approx (p_2 - p_1) + \frac{3}{2\pi^2}\ln\left(\frac{p_2}{p_1}\right) + \frac{9}{8\pi^4}\left(\frac{1}{p_1} - \frac{1}{p_2}\right)} \tag{A.19}$$

Coefficients:

$$\begin{aligned} \frac{3}{2\pi^2} &= 0.151981775464\ldots, \\ \frac{9}{8\pi^4} &= 0.011549230037\ldots \end{aligned} \tag{A.20}$$

A.7 Conical Helix

Cone equation:

$$z = -\frac{3}{4} + \frac{\pi}{2}r \tag{A.21}$$

Height function:

$$z(\theta) = -\frac{3}{4} + \frac{3}{\pi}\theta \tag{A.22}$$

Half-angle:

$$\tan(\alpha) = \frac{2}{\pi} \quad \Rightarrow \quad \boxed{\alpha = \arctan\left(\frac{2}{\pi}\right) = 32.481636590530^\circ} \tag{A.23}$$

Helix parametrisation:

$$\boxed{\gamma(\theta) = \begin{pmatrix} \frac{6}{\pi^2}\theta\cos\theta \\ \frac{6}{\pi^2}\theta\sin\theta \\ -\frac{3}{4} + \frac{3}{\pi}\theta \end{pmatrix}} \tag{A.24}$$

Integer points:

$$\theta_n = \frac{\pi}{3}n + \frac{\pi}{4} \qquad (n \equiv 1, 5 \pmod 6) \tag{A.25}$$

Tangent vector:

$$\gamma'(\theta) = \begin{pmatrix} \dfrac{6}{\pi^2}(\cos\theta - \theta\sin\theta) \\ \dfrac{6}{\pi^2}(\sin\theta + \theta\cos\theta) \\ \dfrac{3}{\pi} \end{pmatrix} \tag{A.26}$$

Planar curvature (approximation):

$$\kappa_{2\mathrm{D}}(\theta) = \frac{\theta^2+2}{a\,(\theta^2+1)^{3/2}}, \qquad a = \frac{6}{\pi^2} \tag{A.27}$$

Exact 3D curvature and torsion (Theorem 13.3):

$$\kappa_{3\mathrm{D}}(\theta) = \frac{a\sqrt{c_1^2(\theta^2+4) + a^2(\theta^2+2)^2}}{\left(a^2(1+\theta^2)+c_1^2\right)^{3/2}}, \tag{A.28}$$

$$\tau(\theta) = \frac{c_1\,(\theta^2+6)}{c_1^2\,(\theta^2+4) + a^2\,(\theta^2+2)^2}, \qquad c_1 = \frac{3}{\pi}. \tag{A.29}$$

A.8 Numerical Constants

Table A.1: Important constants (15 decimal places)

Constant	**Value**	**Meaning**
π	3.141592653589793	Circle constant
π^2	9.869604401089358	
$\sqrt{3}$	1.732050807568877	
$\frac{6}{\pi^2}$	0.607927101854027	Spiral parameter
$\frac{12}{\pi}$	3.819718634205488	Radial difference
$\frac{6}{\pi\sqrt{3}}$	1.102657790843584	Radius factor
$\frac{\pi}{\sqrt{3}}$	1.813799364234218	Angular factor
$\frac{3}{2\pi^2}$	0.151981775464066	Taylor coefficient 1
$\frac{9}{8\pi^4}$	0.011549230036520	Taylor coefficient 2
$\arctan(\frac{2}{\pi})$	32.481636590530°	Cone half-angle

Appendix B

Figures and Plots

This chapter collects all relevant figures that illustrate the geometric structures of the spiral, the cone, and the helix.

B.1 2D Spiral

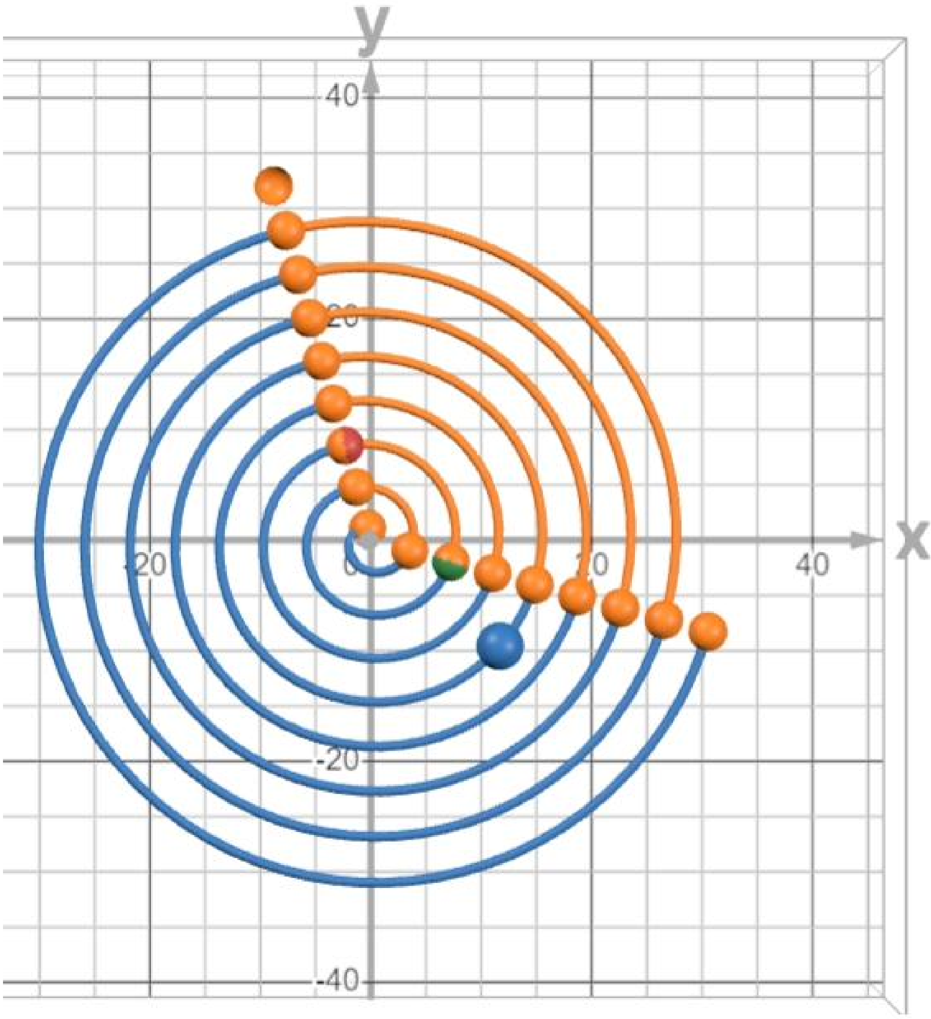


Figure B.1: Archimedean spiral $r(\theta) = \frac{6}{\pi^2}\theta$ with the integer points of the sequence $k(n)$ marked. The green-orange point corresponds to $n = 11$, the red-orange point to $n = 13$. The blue point is a freely placeable reference point on the spiral path.

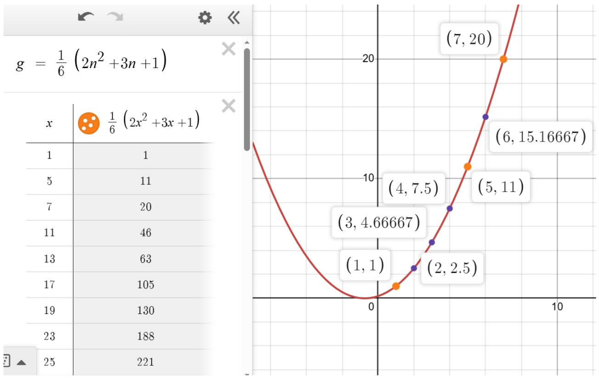


Figure B.3: Parabolic curve $k(n) = \frac{1}{6}(2n^2 + 3n + 1)$ with integer points at $n \equiv 1, 5 \pmod 6$

B.2 3D Conical Helix

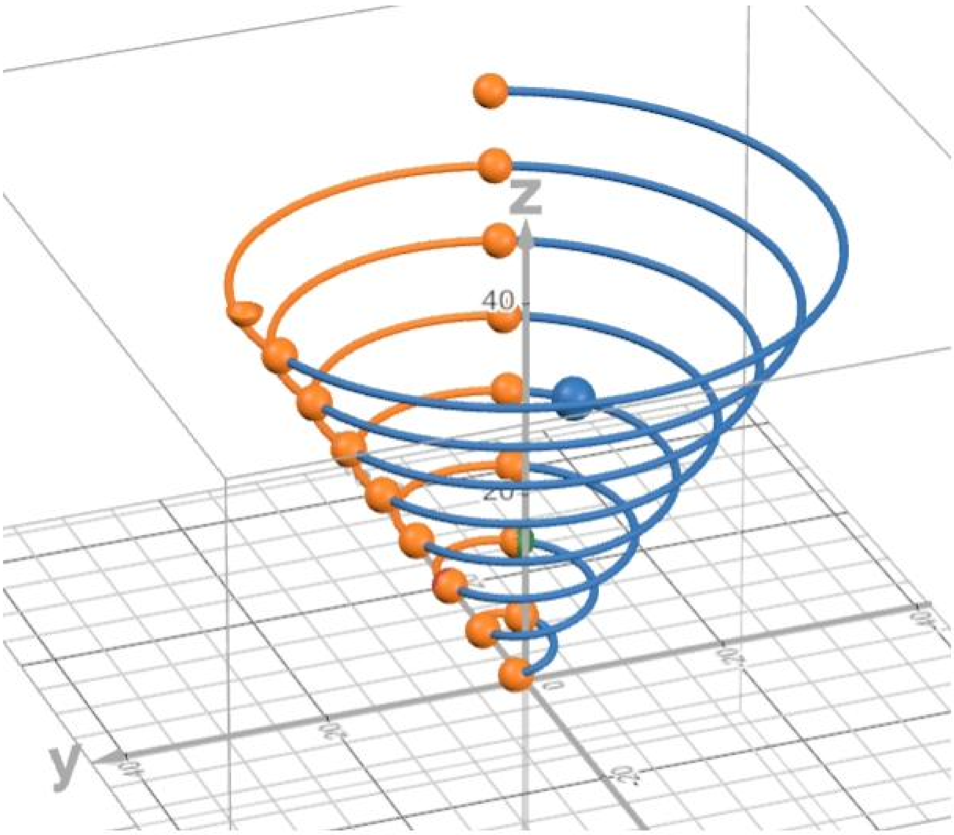


Figure B.2: Conical helix $\gamma(\theta)$ in three-dimensional space; primes $p > 3$ appear as highlighted discrete points. The orange-coloured arc segments represent the arc length over the minor steps ($\Delta n = 2$), while the blue-coloured arc segments represent the arc length over the major steps ($\Delta n = 4$).

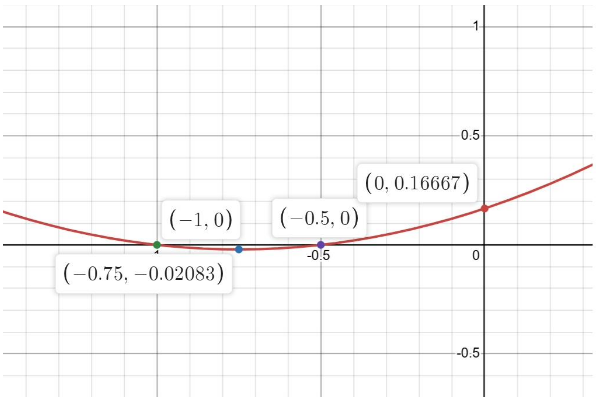


Figure B.4: Zeros of the function $k(n)$: $n_1 = -\frac{1}{2}$ and $n_2 = -1$, together with the y-intercept and the vertex of the curve.

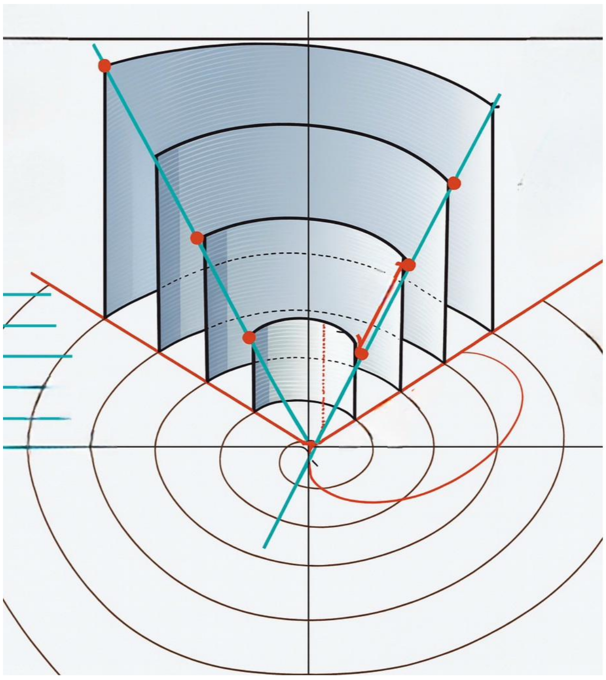


Figure B.5: Curve segments of the minor steps ($\Delta n = 2$) on the parabolic curve (twin zones). The major segments have been omitted for clarity.

Appendix C

Appendix

C.1 Numerical Computations: Python Code

The following Python code reproduces the numerical values of this work (arc lengths, torsion, curvature). It can be executed directly and requires only the standard libraries `numpy` and `scipy`.

```
import numpy as np
from scipy import integrate

# ---------------------------------------------------
# Basic functions
# ---------------------------------------------------
def k(n):
    """Quadratic mean-value sequence."""
    return (2*n**2 + 3*n + 1) / 6

def theta_n(n):
    """Angular parameter for integer points on the helix."""
    return (np.pi / 3) * n + (np.pi / 4)

# Helix parameters
a = 6 / np.pi**2   # Spiral slope
b = 3 / np.pi       # Height slope

# ---------------------------------------------------
# Arc length of the spiral (p-parametrisation)
# ---------------------------------------------------
def arc_length_spiral(p1, p2):
    """Exact arc-length integral of the spiral."""
    integrand = lambda p: (1/np.pi) * np.sqrt(3/p + np.pi**2)
    L, _ = integrate.quad(integrand, p1, p2)
    return L

def arc_length_spiral_taylor(p1, p2):
```

```
    """Taylor approximation of the arc length."""
    return ((p2 - p1)
            + (3 / (2*np.pi**2)) * np.log(p2/p1)
            + (9 / (8*np.pi**4)) * (1/p1 - 1/p2))

# ------------------------------------------------
# Arc length of the helix (theta-parametrisation)
# ------------------------------------------------
def arc_length_helix(t1, t2):
    """Arc-length integral of the conical helix."""
    integrand = lambda t: (3/np.pi**2) * np.sqrt(4*(1+t**2) + np.pi**2)
    L, _ = integrate.quad(integrand, t1, t2)
    return L

def arc_length_parabola(n1, n2):
    """Arc length of the quadratic parabola k(n)."""
    integrand = lambda n: (1/6) * np.sqrt(36 + (4*n+3)**2)
    L, _ = integrate.quad(integrand, n1, n2)
    return L

# ------------------------------------------------
# Curvature and torsion
# ------------------------------------------------
def curvature(t):
    """Planar spiral curvature (2D approx., cf. Ch. 13)."""
    return (np.pi**2 * (t**2 + 2)) / (6 * (t**2 + 1)**1.5)

def curvature_3d(t):
    """Exact 3D curvature of the conical helix (Theorem 13.9)."""
    c1 = 3 / np.pi
    num = a * np.sqrt(c1**2 * (t**2 + 4) + a**2 * (t**2 + 2)**2)
    den = (a**2 * (1 + t**2) + c1**2)**1.5
    return num / den

def torsion(t):
    """Torsion of the conical helix."""
    num = np.pi**3 * (6 + t**2)
    den = 3 * (np.pi**2 * (4 + t**2) + 4*(2 + t**2)**2)
    return num / den

# ------------------------------------------------
# Example outputs
# ------------------------------------------------
if __name__ == "__main__":
    print("=== Difference sequence (major/minor steps) ===")
    for m in range(1, 6):
```

```
    maj = 16*m - 6
    mi  = 8*m + 1
    Um  = maj + mi
    print(f"m={m}: Major={maj}, Minor={mi}, U_m={Um}")

print("\n=== Arc-length comparison: parabola vs helix ===")
pairs = [(1,5),(5,7),(11,13),(17,19),(23,25),(35,37)]
for n1, n2 in pairs:
    t1 = theta_n(n1)
    t2 = theta_n(n2)
    Lp = arc_length_parabola(n1, n2)
    Lh = arc_length_helix(t1, t2)
    dev = (Lh - Lp) / Lp * 100
    print(f"n={n1}->{n2}: L_Parabola={Lp:.6f}, L_Helix={Lh:.6f}, "
          f"Dev={dev:+.2f}%")

print("\n=== Curvature and torsion at prime-number angles ===")
primes = [5, 7, 11, 13, 17, 19, 23]
for p in primes:
    t = theta_n(p)
    print(f"p={p:3d}: theta={t:.4f}, "
          f"kappa_2D={curvature(t):.6f}, "
          f"kappa_3D={curvature_3d(t):.6f}, "
          f"tau={torsion(t):.6f}")
```

C.2 Numerical Verifications

The following tables supplement the above Python code with independent comparisons that confirm the analytical formulae from the respective main chapters.

C.2.1 Torsion Formula (Theorem 13.3)

The torsion formula

$$\tau(t) = \frac{\pi^3(6+t^2)}{3[\pi^2(4+t^2)+4(2+t^2)^2]}$$

was verified by direct comparison with the numerical computation via the cross product $(\gamma' \times \gamma'') \cdot \gamma''' / |\gamma' \times \gamma''|^2$. The following values agree to at least 10 decimal places:

Table C.1: Verification of the torsion at selected points

t	τ_{formula}	$\tau_{\text{numerical}}$
5	$0.100055070\ldots$	$0.100055070\ldots$
10	$0.025691658\ldots$	$0.025691658\ldots$
20	$0.006451662\ldots$	$0.006451662\ldots$
47	$0.001169444\ldots$	$0.001169444\ldots$

C.2.2 Difference Formulae (Theorem 2.5)

The formulae $\Delta_{\text{major},m} = 16m - 6$ and $\Delta_{\text{minor},m} = 8m + 1$ were verified by direct computation from $k(6m-1) - k(6m-5)$ and $k(6m+1) - k(6m-1)$ for $m = 1, \ldots, 20$ (complete agreement).

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